

概率论作业

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前言

前言是最后写的.

Y-Problem 1.1.5.1

设 A, B, C, D 是某一实验中的 4 个事件. 试用他们的运算表示如下事件:

(1) 4 个事件恰好发生一个; (2) 恰好发生两个; (3) 至少发生三个; (4) 至多一个.

解:

$$(1) \quad (A \cap B^c \cap C^c \cap D^c) \cup (A^c \cap B \cap C^c \cap D^c) \cup (A^c \cap B^c \cap C \cap D^c) \cup (A^c \cap B^c \cap C^c \cap D);$$

$$(2) \quad (A \cap B \cap C^c \cap D^c) \cup (A \cap B^c \cap C \cap D^c) \cup (A \cap B^c \cap C^c \cap D) \cup (A^c \cap B \cap C \cap D^c) \cup (A^c \cap B \cap C^c \cap D^c) \cup (A^c \cap B^c \cap C \cap D);$$

$$(3) \quad (A \cap B \cap C \cap D^c) \cup (A \cap B \cap C^c \cap D) \cup (A \cap B^c \cap C \cap D) \cup (A^c \cap B \cap C \cap D) \cup (A \cap B \cap C \cap D);$$

$$(4) \quad (A \cap B^c \cap C^c \cap D^c) \cup (A^c \cap B \cap C^c \cap D^c) \cup (A^c \cap B^c \cap C \cap D^c) \cup (A^c \cap B^c \cap C^c \cap D) \cup (A^c \cap B^c \cap C^c \cap D^c).$$

Y-Problem 1.1.5.6

设事件列 $\{A_n\}$ 单调上升: 即对任意 $n \geq 1$ 有 $A_n \subset A_{n+1}$. 试用概率论语言证明

$$\overline{\lim}_n A_n = \bigcup_{n=1}^{\infty} A_n = \underline{\lim}_n A_n$$

对单调下降事件列, 相应的结论是什么?

证明: 由事件列的单增性可知

$$\bigcap_{n=k}^{\infty} A_n = A_k$$

从而由下极限的定义即知第二个等号成立, 对第一个等号, 由上极限的定义即知

$$\overline{\lim}_n A_n = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n \subset \bigcup_{n=1}^{\infty} A_n$$

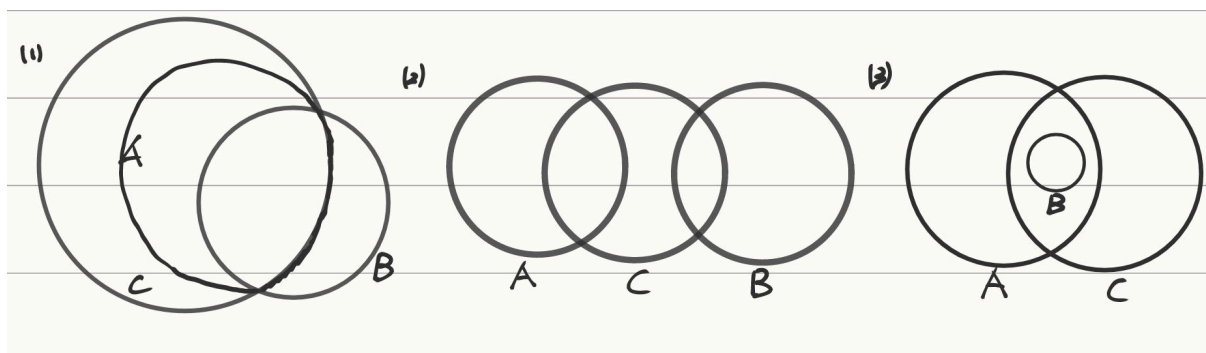
任取 $x \in \bigcup_{n=1}^{\infty} A_n$, 可知存在 $p \geq 1$ 使得 $x \in A_p \subset A_{p+1} \subset \dots$, 从而对任意 $k \geq 1$ 均有 $x \in \bigcup_{n=k}^{\infty} A_n$, 从而 $x \in \overline{\lim}_n A_n$. 于是第一个等号成立.

对单调下降事件列, 同理可证 $\overline{\lim}_n A_n = \bigcap_{n=1}^{\infty} A_n = \underline{\lim}_n A_n$.

Y-Problem 1.1.5.8

试画出表示 7 题中各事件关系的 Venn 图.

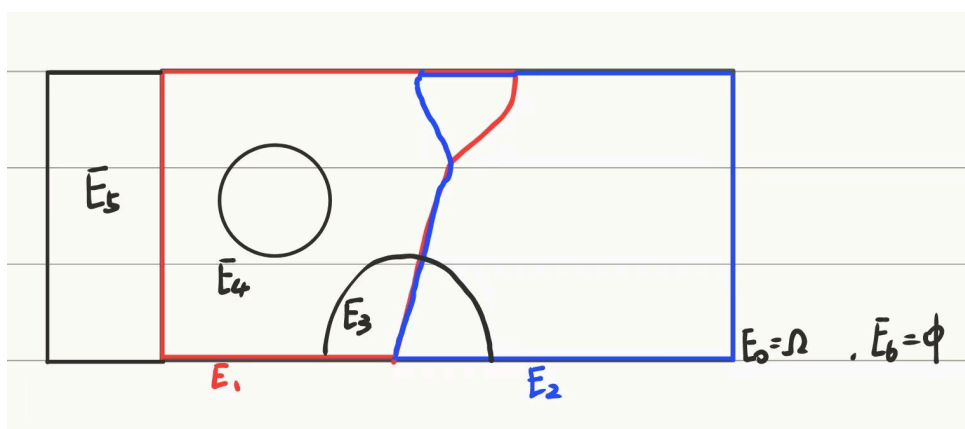
解: 如图所示



Y-Problem 1.1.5.9

从 0 到 9 这 10 个数字中有重复地取 5 个. 以 E_i 表示有某数字被取出 i 次. 试画出表示 $E_0, \dots, E_6$ 的 Venn 图.

解: 如图所示



K-Problem 2.1.1

Proof $\left(\bigcup_j A_j\right) \setminus \left(\bigcup_j B_j\right) \subset \bigcup_j (A_j \setminus B_j)$, $\left(\bigcap_j A_j\right) \setminus \left(\bigcap_j B_j\right) \subset \bigcup_j (A_j \setminus B_j)$.
When is there equality?

Pf:

任取 $x \in \left(\bigcup_j A_j\right) \setminus \left(\bigcup_j B_j\right)$, 我们有 $\forall j, x \notin B_j$ 且 $\exists k, x \in A_k$, 故 $x \in A_k \setminus B_k$, 故有 $x \in \bigcup_j (A_j \setminus B_j)$, 第一个式子得证.

考察取等条件, 即考察何时后一个集合属于前一个, 即 $\exists k, x \in A_k \setminus B_k \Rightarrow x \in \bigcup_j A_j, x \notin \bigcup_j B_j$, 前一个条件自然成立, 只需 $\bigcup_j (A_j \setminus B_j) \subset \left(\bigcup_j B_j\right)^c = \bigcap_j B_j^c$, 也即对任意 j, k , 均有 $A_j \setminus B_j \subset B_k^c$.

再看第二个式子: 任取 $x \in \left(\bigcap_j A_j\right) \setminus \left(\bigcap_j B_j\right)$, 有 $\forall j, x \in A_j, \exists k, x \notin B_k$, 因此 $x \in A_k \setminus B_k$, 故 $x \in \bigcup_j (A_j \setminus B_j)$, 从而第二个式子得证.

考察取等条件, 即考察何时后一个集合属于前一个, 即 $\exists k, x \in A_k \setminus B_k \Rightarrow x \in \bigcap_j A_j, x \notin \bigcap_j B_j$, 后一个条件自然成立, 因此只需 $\bigcup_j (A_j \setminus B_j) \subset \bigcap_j A_j$, 也即对任意 j, k , 均有 $A_j \setminus B_j \subset A_k$

K-Problem 2.1.2

The best way to define the symmetric difference is through indicators of sets as follows :

$$1_{A\Delta B} = 1_A + 1_B \pmod{2}$$

where we have arithmetical addition modulo 2 on the right side . All properties of Δ follow easily from this definition, some of which are rather tedious to verify otherwise . As examples :

$$\begin{aligned}(A\Delta B)\Delta C &= A\Delta(B\Delta C) \\ (A\Delta B)\Delta(B\Delta C) &= A\Delta C \\ (A\Delta B)\Delta(C\Delta D) &= (A\Delta C)\Delta(B\Delta D) \\ A\Delta B = C &\Leftrightarrow A = B\Delta C \\ A\Delta B = C\Delta D &\Leftrightarrow A\Delta C = B\Delta D\end{aligned}$$

Pf:

两个集合相等当且仅当二者的示性函数相同, 示性函数的加法为 $\mathbb{F}_2$ 上的加法, 故

$$\begin{aligned}1_{(A\Delta B)\Delta C} &= 1_{A\Delta B} + 1_C = 1_A + 1_B + 1_C = 1_A + 1_{B\Delta C} = 1_{A\Delta(B\Delta C)} \\ 1_{(A\Delta B)\Delta(B\Delta C)} &= 1_A + 2 \cdot 1_B + 1_C = 1_A + 1_C = 1_{A\Delta C} \\ 1_{(A\Delta B)\Delta(C\Delta D)} &= 1_A + 1_B + 1_C + 1_D = 1_{(A\Delta C)\Delta(B\Delta D)} \\ 1_{A\Delta B} = 1_C &\Leftrightarrow 1_A + 1_B = 1_C \Leftrightarrow 1_A = 1_C + 1_B \Leftrightarrow 1_A = 1_{B\Delta C} \\ 1_{A\Delta B} = 1_A + 1_B &= 1_C + 1_D = 1_{C\Delta D} \Leftrightarrow 1_{A\Delta C} = 1_A + 1_C = 1_B + 1_D = 1_{B\Delta D}\end{aligned}$$

因此题目中的性质得证.

K-Problem 2.1.3

If Ω has exactly n points, then $\mathcal{L} = 2^\Omega$ has 2^n members. The B.F generated by n given sets “without relations among them” has 2^{2^n} members.

Pf:

(问: “without relations among them” 是什么意思? 以及 2^{2^n} 在中文译本中是 2^{2^n})

由排列组合的知识即知 $|\mathcal{L}| = 2^n$.

这里将 “without relation among them” 理解为 $\forall A, B, A \not\subset B, A \not\subset B^c$.

则这 n 个集合将 Ω 划分为了 2^n 个两两不交的非空集合, 并且这些集合的并是 Ω . 对这 2^n 个集合取并即可得到题中的 B.F., 由于这些集合两两不交, 因此只需决定每个集合在取并时的状态, 因此最终可以得到 2^{2^n} 个元素.

K-Problem 2.1.5

The interior of any collection of B.F.'s $\{\mathcal{F}_\alpha, \alpha \in A\}$ is the maximal B.F. contained in all of them; it is indifferently denoted by $\bigcap_{\alpha \in A} \mathcal{F}_\alpha$ or $\bigwedge_{\alpha \in A} \mathcal{F}_\alpha$.

Pf:

先证明 $\mathcal{F} = \bigcap_{\alpha \in A} \mathcal{F}_\alpha$ 是 B.F.: 由 $\{\mathcal{F}_j\}$ 为 B.F. 构成的集合即知对 $A \in \mathcal{F}$, 有

$$\forall \alpha, A \in \mathcal{F}_\alpha \implies \forall \alpha, A^c \in \mathcal{F}_\alpha \implies A^c \in \bigcap_{\alpha \in A} \mathcal{F}_\alpha = \mathcal{F}$$

对 $\{A_i\} \subset \mathcal{F}$, 有

$$\forall \alpha, \{A_i\} \subset \mathcal{F}_\alpha \implies \forall \alpha, \bigcup_i A_i \in \mathcal{F}_\alpha \implies \bigcup_i A_i \in \mathcal{F}$$

从而 $\mathcal{F}$ 构成 B.F., 由 $\mathcal{F}$ 的定义即知其最大性.

K-Problem 2.1.6

The union of a countable collection of B.F.'s $\{\mathcal{F}_j\}$ such that $\mathcal{F}_j \subset \mathcal{F}_{j+1}$ need not be a B.F., but there exists a minimal B.F. containing all of them, denoted by $\bigvee_j \mathcal{F}_j$. In general, $\bigvee_{\alpha \in A} \mathcal{F}_\alpha$ denotes the minimal B.F. containing all $\mathcal{F}_\alpha, \alpha \in A$.

[HINT: $\Omega =$ the set of positive integers; $\mathcal{F}_j =$ the B.F. generated by those up to j .]

Pf:

首先 2^Ω 作为 B.F. 必然包含了所有 $\mathcal{F}_j$, 因此我们记

$$\mathcal{F} = \bigcap_{\forall j, \mathcal{F}_j \subset \mathcal{G} \text{ 且 } \mathcal{G} \text{ 为 B.F.}} \mathcal{G}$$

于是由 2.1.5 即知 $\mathcal{F}$ 为 B.F., 且由定义即知 $\mathcal{F}$ 的最小性.

下面以 HINT 中的例子说明 $\bigcup_j \mathcal{F}_j$ 未必是 B.F.: 对 $n \geq 1$, 有 $\{2n\} \in \mathcal{F}_{2n} \subset \bigcup_j \mathcal{F}_j$, 因此若 $\bigcup_j \mathcal{F}_j$ 为 B.F., 则必然包含 $A = \{2n : n \geq 1\}$, 因此 A 会包含在某个 $\mathcal{F}_j$ 中, 根据 $\mathcal{F}_j$ 的定义即得矛盾, 因此 $\bigcup_j \mathcal{F}_j$ 不是 B.F.

Y-Problem 1.2.4.4

从 n 对不同的鞋子中任取 $2m$ 只 ($2m < n$), 求其中恰有 k 双配对的鞋子的概率.

解: 记事件为 A , 样本空间为 Ω , 则 $|\Omega| = \binom{n}{2m}$, A 发生即先从 n 双鞋中选择 k 双, 剩余 $2m - 2k$ 个要选的鞋属于 $2m - 2k$ 对鞋, 因此先从 $n - k$ 对鞋中选择 $2m - 2k$ 对, 再从每对鞋中二选一, 即 $|A| = \binom{n}{k} \binom{n-k}{2m-2k} 2^{2m-2k}$, 从而

$$P(A) = \frac{\binom{n}{k} \binom{n-k}{2m-2k} 2^{2m-2k}}{\binom{n}{2m}}$$

Y-Problem 1.2.4.6

将三个不同的球放入编号为 1,2,3,4 的 4 个盒, 每球入各盒均为等可能的. 试求如下事件的概率: (1) 恰有两个空盒; (2) 有球的盒的最小编号为 2.

解:

(1) $|\Omega| = 4^3 = 64$, $|A_1| = \binom{4}{2} \binom{2}{1} \binom{3}{2} = 36$, 故 $P(A_1) = \frac{36}{64} = \frac{9}{16}$;

(2) 按“有球的盒的数量”讨论, 则 $|A| = 1 + \binom{2}{1} \binom{2}{1} \binom{3}{2} + 3! = 19$, 故 $P(A_2) = \frac{19}{64}$.

Y-Problem 1.2.4.9

从一副 52 张扑克牌中任取出 5 张, 求下列事件的概率: (1) 以 A 打头的同花顺次 5 张; (2) 其他同花顺次 5 张; (3) 有 4 张同点数; (4) 3 张同点数, 另外两张也同点数; (5) 5 张同花; (6) 异花顺次 5 张; (7) 3 张同点, 另外 2 张不同点; (8) 5 张中有两对; (9) 5 张中恰有 1 对; (10) 其他情形.

解:

$$(1) P(A_1) = \frac{|A_1|}{|\Omega|} = \frac{4}{\binom{52}{5}} = \frac{4}{2598960} = \frac{1}{649740};$$

$$(2) P(A_2) = \frac{|A_2|}{|\Omega|} = \frac{8 \times 4}{\binom{52}{5}} = \frac{2}{162435};$$

$$(3) P(A_3) = \frac{|A_3|}{|\Omega|} = \frac{48 \binom{13}{1}}{\binom{52}{5}} = \frac{1}{4165};$$

$$(4) P(A_4) = \frac{|A_4|}{|\Omega|} = \frac{\binom{13}{1} \binom{4}{3} \binom{12}{1} \binom{4}{2}}{\binom{52}{5}} = \frac{6}{4165};$$

$$(5) P(A_5) = \frac{|A_5|}{|\Omega|} = \frac{\binom{4}{1} \binom{13}{5}}{\binom{52}{5}} = \frac{33}{16660};$$

$$(6) P(A_6) = \frac{|A_6|}{|\Omega|} = \frac{9 \times (4^5 - 4)}{\binom{52}{5}} = \frac{9}{2548};$$

$$(7) P(A_7) = \frac{|A_7|}{|\Omega|} = \frac{\binom{13}{1} \binom{4}{3} \binom{12}{2} 4^2}{\binom{52}{5} \times 2!} = \frac{88}{4165};$$

$$(8) P(A_8) = \frac{|A_8|}{|\Omega|} = \frac{\binom{13}{2} \binom{4}{2}^2 \binom{11}{1} \binom{4}{1}}{\binom{52}{5}} = \frac{198}{4165};$$

$$(9) P(A_9) = \frac{|A_9|}{|\Omega|} = \frac{\binom{13}{1} \binom{4}{2} \binom{12}{3} 4^3}{\binom{52}{5}} = \frac{352}{833};$$

$$(10) P(A_{10}) = 1 - \sum_{i=1}^9 P(A_i) = \frac{639}{1274}.$$

Y-Problem 1.2.4.12

袋中装有编号 1 至 N 的 N 个球, 现在有放回地任取 n 个, 依次记下其号码, 求: (1) n 个号码按严格上升排列的概率; (2) n 个号码非降排序的概率.

解:

(1) $|\Omega| = N^n$, 事件 A_1 等价于从 N 个球中取出 n 个不同的球, 取完后再排序即对应一个严格上升的排列, 因此 $|A_1| = \binom{N}{n}$, 故有

$$P(A_1) = \frac{\binom{N}{n}}{N^n}$$

(2) 设 $\{a_i\}_{1 \leq i \leq n}$ 为一个非降排序, 考虑

$$b_i = a_i + i - 1$$

则可知 $\{b_i\}_{1 \leq i \leq n}$ 为 1 至 $N + n - 1$ 的一个严格上升排列. 另外, 对 1 至 $N + n - 1$ 的一个严格上升排列 $\{b_i\}_{i \leq i \leq n}$, 可知 $\{b_i - i + 1\}_{1 \leq i \leq n}$ 对应 1 至 N 的一个非降排序, 从而有 $|A_2| = \binom{N+n-1}{n}$, 故

$$P(A_2) = \frac{\binom{N+n-1}{n}}{N^n}$$

Y-Problem 1.3.3

甲、乙两船欲停靠在同一码头. 假设它们都有可能在某一天的一昼夜内任何时刻到达, 且甲船与乙船到达后各需在码头停留 3 与 4 小时. 求有船到达是需等待空出码头的概率.

解: 设甲、乙停靠的时间 (单位: 时) 分别为 x, y , 则 $\Omega = \{(x, y) : 0 \leq x, y \leq 24\}$, 有船到达时需要等待意味着

$$x < y < x + 3 \text{ 或 } y < x < y + 4$$

故作图可知

$$P(A) = 1 - \frac{\frac{1}{2} \times 20^2 + \frac{1}{2} \times 21^2}{24^2} = \frac{311}{1152}$$

Y-Problem 1.3.5

在区间 $(-1, 1)$ 上任取两数 ξ, η , 求二次方程 $x^2 + \xi x + \eta = 0$ 的两个根: (1) 都是实数的概率; (2) 都是正数的概率.

解:

(1) 易知 $\Omega = \{(\xi, \eta) : -1 < \xi, \eta < 1\}$, $A = \{(\xi, \eta) : \xi^2 \geq 4\eta\}$, 故有

$$P(A) = \frac{\int_{-1}^1 \left(\frac{\xi^2}{4} + 1 \right) d\xi}{4} = \frac{13}{24}$$

(2) 易知 $\Omega = \left\{ (\xi, \eta) : 0 < \eta \leq \frac{\xi^2}{4} \wedge \xi < 0 \right\}$, 故有

$$P(A_2) = \frac{\int_{-1}^0 \frac{\xi^2}{4} d\xi}{4} = \frac{1}{48}$$

M-Problem 1.7.9

设 $\mathcal{C}$ 是一个集族, 证明:

(1) $m(\mathcal{C}) = \sigma(\mathcal{C})$ 当且仅当

$$A \in \mathcal{C} \implies A^c \in m(\mathcal{C}); A, B \in \mathcal{C} \implies A \cap B \in m(\mathcal{C}).$$

(2) $\lambda(\mathcal{C}) = \sigma(\mathcal{C})$ 当且仅当

$$A, B \in \mathcal{C} \implies A \cap B \in \lambda(\mathcal{C}).$$

证明: 课上已经给出了如下的证明:

(1) $m(\mathcal{G}) = \sigma(\mathcal{G}) \iff m(\mathcal{G}) \text{ is an algebra} \iff$

- Fact 1: $A, B \in m(\mathcal{G}) \implies AB \in m(\mathcal{G})$
- Fact 2: $A \in m(\mathcal{G}) \implies A^c \in m(\mathcal{G})$

Consider $\mathcal{G}_1 := \{A \in m(\mathcal{G}) \mid \forall B \in m(\mathcal{G}), AB \in m(\mathcal{G})\}$ (HW: $\mathcal{G}_1$ is a M.C.)

Fact 1 $\iff \mathcal{G}_1 = m(\mathcal{G}) \iff m(\mathcal{G}) \subset \mathcal{G}_1 \iff \mathcal{G} \subset \mathcal{G}_1$

$\iff \forall A \in \mathcal{G}, B \in m(\mathcal{G}) \Rightarrow AB \in m(\mathcal{G})$

$\iff \mathcal{G}_2 = \{A \in m(\mathcal{G}) \mid \forall B \in \mathcal{G}, AB \in m(\mathcal{G})\} = m(\mathcal{G})$ (HW: $\mathcal{G}_2$ is a M.C.)

$\iff \mathcal{G} \subset \mathcal{G}_2$

$\iff \forall A, B \in \mathcal{G}, AB \in m(\mathcal{G})$

Fact 2 $\iff \mathcal{G}_3 = \{A \in m(\mathcal{G}) \mid A^c \in m(\mathcal{G})\} = m(\mathcal{G})$ (HW: $\mathcal{G}_3$ is a M.C.)

$\iff \mathcal{G} \subset \mathcal{G}_3$

$\iff \forall A \in \mathcal{G}, A^c \in m(\mathcal{G}).$

(2) $\lambda(\mathcal{G}) = \sigma(\mathcal{G}) \iff \lambda(\mathcal{G}) \text{ is a } \pi\text{-class}$

$\iff \mathcal{G}_4 = \{A \in \lambda(\mathcal{G}) \mid \forall B \in \lambda(\mathcal{G}), AB \in \lambda(\mathcal{G})\} = \lambda(\mathcal{G})$ (HW: $\mathcal{G}_4$ is a λ -class)

$\iff \mathcal{G} \subset \mathcal{G}_4$

$\iff \forall A \in \mathcal{G}, B \in \lambda(\mathcal{G}), AB \in \lambda(\mathcal{G})$

$$\Leftrightarrow \mathcal{G}_5 = \{A \in \lambda(\mathcal{G}) \mid \forall B \in \mathcal{G}, AB \in \lambda(\mathcal{G})\} = \lambda(\mathcal{G}) \text{ (HW: } \mathcal{G}_5 \text{ is a } \lambda\text{-class)}$$

$$\Leftrightarrow \mathcal{G} \subset \mathcal{G}_5$$

$$\Leftrightarrow \forall A, B \in \mathcal{G}, AB \in \lambda(\mathcal{G}).$$

因此只需证明上述 $\mathcal{G}_{1,2,3}$ 为 M.C., $\mathcal{G}_{4,5}$ 为 λ -class.

对 $\mathcal{G}_1$, 设 $\{A_i\} \subset \mathcal{G}_1$ 为单调递增的集合列, 任取 $B \in m(\mathcal{G})$, 则

$$\left(\bigcup_{i=1}^{\infty} A_i\right) \cap B = \bigcup_{i=1}^{\infty} (A_i \cap B)$$

由 $\mathcal{G}_1$ 定义即知 $A_i \cap B \in m(\mathcal{G})$, 从而由 $m(\mathcal{G})$ 为 M.C. 知 $\left(\bigcup_{i=1}^{\infty} A_i\right) \cap B \in m(\mathcal{G})$, 这就说明 $\bigcup_{i=1}^{\infty} A_i \in \mathcal{G}_1$, 对下降列 $\{A_i\}$ 同理可证 $\bigcap_{i=1}^{\infty} A_i \in \mathcal{G}_1$, 故 $\mathcal{G}_1$ 为 M.C. 同理可证 $\mathcal{G}_2$ 为 M.C. 对于 $\mathcal{G}_3$, 取其中的上升列 $\{A_i\}$, 则

$$\left(\bigcup_{i=1}^{\infty} A_i\right)^c = \bigcap_{i=1}^{\infty} A_i^c$$

而 $\{A_i^c\}$ 构成 $m(\mathcal{G})$ 中的下降列, 从而 $\bigcap_{i=1}^{\infty} A_i^c \in m(\mathcal{G})$, 即 $\bigcup_{i=1}^{\infty} A_i \in \mathcal{G}_3$, 上升列同理, 故 $\mathcal{G}_3$ 为 M.C.

对于 $\mathcal{G}_4$, 与 $\mathcal{G}_1$ 的证明方法相同, 我们有:

- $\Omega \in \mathcal{G}_4$
- $A, B \in \mathcal{G}_4, A \subset B \Rightarrow B - A \in \mathcal{G}_4$
- $\{A_i\} \in \mathcal{G}_4 \wedge \{A_i\}$ 为上升列 $\Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{G}_4$

从而 $\mathcal{G}_4$ 为 λ -class, 同理可证 $\mathcal{G}_5$ 也是 λ -class, 这就补全了课上的证明.

M-Problem 1.7.10

设 $\mathcal{C}$ 是一个集族, 证明:

(1) $m(\mathcal{C}) = \sigma(\mathcal{C})$ 当且仅当

$$A \in \mathcal{C} \Rightarrow A^c \in m(\mathcal{C}); A, B \in \mathcal{C} \Rightarrow A \cup B \in m(\mathcal{C}).$$

(2) $\lambda(\mathcal{C}) = \sigma(\mathcal{C})$ 当且仅当

$$A, B \in \mathcal{C} \Rightarrow A \cup B \in \lambda(\mathcal{C})$$

证明: 本题只是将上一题中的 $A \cap B$ 换为了 $A \cup B$.

(1) 由 De-Morgan 律, 可知

- Fact 1: $A, B \in m(\mathcal{G}) \Rightarrow AB \in m(\mathcal{G})$
- Fact 2: $A \in m(\mathcal{G}) \Rightarrow A^c \in m(\mathcal{G})$

等价于

- Fact 1': $A, B \in m(\mathcal{G}) \implies A \cup B \in m(\mathcal{G})$
- Fact 2': $A \in m(\mathcal{G}) \implies A^c \in m(\mathcal{G})$

于是只需要将上一题中 $\mathcal{G}_{1,2,3}$ 中的 AB 换为 $A \cup B$ 即可. 与上一题的证明方法类似, 容易证明 $\mathcal{G}_{1,2,3}$ 仍然是 M.C., 因此上一题中 (1) 的逻辑链条仍成立, 命题得证.

(2) 注意上一题的证明的第一行为: $\lambda(\mathcal{G}) = \sigma(\mathcal{G}) \iff \lambda(\mathcal{G}) \text{ is a } \pi\text{-class}$

由于 $\lambda\text{-class}$ 满足对集合的补封闭, 因此由 De-Morgan 律, 可知

$$\lambda(\mathcal{G}) \text{ is a } \pi \text{ class} \iff \forall A, B \in \lambda(\mathcal{G}), A \cup B \in \lambda(\mathcal{G})$$

因此只需要将上一题中 $\mathcal{G}_{4,5}$ 中的 AB 换为 $A \cup B$ 即可. 与上一题的证明方法类似, 容易证明 $\mathcal{G}_{4,5}$ 仍然是 $\lambda\text{-class}$, 因此上一题中 (2) 的逻辑链条仍成立, 命题得证.

M-Problem 1.7.11

设 $\mathcal{C}$ 是一个集族, $\mathcal{C}_\delta$ 和 $\mathcal{C}_\sigma$ 分别表示由 $\mathcal{C}$ 生成的对可列交和可列并运算封闭的类. 若下面两个条件之一成立, 则 $m(\mathcal{C}) = \sigma(\mathcal{C})$:

- (1) $A \in \mathcal{C} \implies A^c \in \mathcal{C}_\sigma$; $A, B \in \mathcal{C} \implies A \cup B \in m(\mathcal{C})$;
- (2) $A \in \mathcal{C} \implies A^c \in \mathcal{C}_\delta$; $A, B \in \mathcal{C} \implies A \cap B \in m(\mathcal{C})$.

证明:

(1) 由前几题可知, 只需在给定条件

$$A, B \in \mathcal{C} \implies A \cup B \in m(\mathcal{C}) \text{ 并且 } A \in \mathcal{C} \implies A^c \in \mathcal{C}_\sigma$$

的情况下证明 $A \in \mathcal{C} \implies A^c \in m(\mathcal{C})$

因此设 $A \in \mathcal{C}$, 于是 $A^c \in \mathcal{C}_\sigma = \left\{ \bigcup_{i=1}^{\infty} A_i \mid A_i \in \mathcal{C} \right\}$, 故存在 $\mathcal{C}$ 中集族 $\{A_i\}$ 使得

$$A^c = \bigcup_{i=1}^{\infty} A_i$$

再记

$$B_1 = A_1, B_2 = A_1 \cup A_2, \dots, B_n = \bigcup_{i=1}^n A_i$$

则 $\{B_i\}$ 为单调上升的集合列, 并且由条件知 $\{B_i\} \subset m(\mathcal{C})$, 因此有

$$A^c = \bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} B_i \in m(\mathcal{C})$$

这就证明了 $A \in \mathcal{C} \implies A^c \in m(\mathcal{C})$, 综上知 $m(\mathcal{C}) = \sigma(\mathcal{C})$.

(2) 由前几题可知, 只需在给定条件

$$A, B \in \mathcal{C} \implies A \cap B \in m(\mathcal{C}) \text{ 并且 } A \in \mathcal{C} \implies A^c \in \mathcal{C}_\delta$$

的情况下证明 $A \in \mathcal{C} \implies A^c \in m(\mathcal{C})$

因此设 $A \in \mathcal{C}$, 于是 $A^c \in \mathcal{C}_\delta = \left\{ \bigcap_{i=1}^{\infty} A_i \mid A_i \in \mathcal{C} \right\}$, 故存在 $\mathcal{C}$ 中集族 $\{A_i\}$ 使得

$$A^c = \bigcap_{i=1}^{\infty} A_i$$

再记

$$B_1 = A_1, B_2 = A_1 \cap A_2, \dots, B_n = \bigcap_{i=1}^n A_i$$

则 $\{B_i\}$ 为单调下降的集合列, 并且由条件知 $\{B_i\} \subset m(\mathcal{C})$, 因此有

$$A^c = \bigcap_{i=1}^{\infty} A_i = \bigcap_{i=1}^{\infty} B_i \in m(\mathcal{C})$$

这就证明了 $A \in \mathcal{C} \implies A^c \in m(\mathcal{C})$, 综上知 $m(\mathcal{C}) = \sigma(\mathcal{C})$.

Y-Problem 1.4.6.5

从装有红, 黑, 白球各一个的袋子中任意有放回地取球, 直至各色球都取出过为止, 求取球次数: (1) 大于 k 的概率; (2) 恰为 k 的概率.

解:

(1) 设事件 B_1, B_2, B_3 分别为前 k 次取出的球均为红或黑, 黑或白, 红或白的概率, 则 (1) 中事件 $A_{>k} = B_1 \cup B_2 \cup B_3$, 而 $B_1 \cap B_2, B_2 \cap B_3, B_3 \cap B_1$ 分别表示前 k 次取出的球均为黑, 白, 红的概率, $B_1 \cap B_2 \cap B_3$ 概率为 0, 因此有

$$\begin{aligned} P(A_{>k}) &= P(B_1 \cup B_2 \cup B_3) \\ &= \sum_{i=1}^3 P(B_i) - \sum_{1 \leq i < j \leq 3} P(B_i \cap B_j) + P(B_1 \cap B_2 \cap B_3) \\ &= 3 \times \left(\frac{2}{3}\right)^k - 3 \times \left(\frac{1}{3}\right)^k \\ &= \frac{2^k - 1}{3^{k-1}} \end{aligned}$$

(2) 易知 $P(A_1) = 0$, $k \geq 2$ 时有

$$P(A_k) = P(A_{>k-1}) - P(A_{>k}) = \frac{2^{k-1} - 1}{3^{k-2}} - \frac{2^k - 1}{3^{k-1}} = \frac{2^{k-1} - 2}{3^{k-1}}$$

Y-Problem 1.4.6.8

用概率想法求 n 阶行列式展开式中含主对角线元的项数.

解: 根据行列式的组合定义, 其展开式中每一项相当于从第 i 行选出第 a_i 个元素, 使得 $(a_1, a_2, \dots, a_n)$ 为 $1, 2, \dots, n$ 的一个排列, 因此 $|\Omega| = n!$. 而一项中含有主元等价于存在 $1 \leq i \leq n$ 使得 $a_i = i$, 那么事件 A^c 就代表 $\forall 1 \leq i \leq n, a_i \neq i$, 这就是经典的错排问题, 根据书中例题, 可知

$$P(A) = 1 - P(A^c) = \frac{1}{1!} - \frac{1}{2!} + \dots + (-1)^{n-1} \frac{1}{n!}$$

$$|A| = P(A)|\Omega| = n! \sum_{k=1}^n (-1)^{k-1} \frac{1}{k!}$$

Y-Problem 1.4.6.9

试证明: 对任意两个事件 A 与 B 有

$$P(A) \vee P(B) \leq P(A \cup B) \leq 2(P(A) \vee P(B))$$

证明: 我们有

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B) \leq 2(P(A) \vee P(B))$$

再由 $P(A) \geq P(A \cap B), P(B) \geq P(A \cap B)$ 即知

$$P(A \cup B) \geq P(A) \vee P(B)$$

Y-Problem 1.4.6.11

求证对任意 n 个事件 $A_1, A_2, \dots, A_n$, 有

$$P\left(\bigcap_{k=1}^n A_k\right) \geq \sum_{k=1}^n P(A_k) - n + 1$$

证明: 原式等价于

$$\sum_{k=1}^n P(A_k^c) = \sum_{k=1}^n (1 - P(A_k)) = n - \sum_{k=1}^n P(A_k) \geq 1 - P\left(\bigcap_{k=1}^n A_k\right) = P\left(\bigcup_{k=1}^n A_k^c\right)$$

这就是课上讲过的 Bool's equality.

Y-Problem 1.4.6.12

试证明:

$$P\left(\varliminf_n A_n\right) \leq \varliminf_n P(A_n), \quad P\left(\varlimsup_n A_n\right) \geq \varlimsup_n P(A_n)$$

证明: 根据定义, 有

$$P\left(\varliminf_n A_n\right) = P\left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k\right) = \lim_n P\left(\bigcap_{k=n}^{\infty} A_k\right)$$

令 $B_n = \bigcap_{k=n}^{\infty} A_k \subset A_n$, 于是 $P(B_n) \leq P(A_n)$, 对 n 取下极限即知

$$P\left(\varliminf_n A_n\right) = \lim_n P(B_n) \leq \varliminf_n P(A_n)$$

同理有

$$P\left(\varlimsup_n A_n\right) = P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) = \lim_n P\left(\bigcup_{k=n}^{\infty} A_k\right) \geq \varlimsup_n P(A_n)$$

Y-Problem 1.4.6.13

设 $\mathcal{O}$ 为 $\mathbb{R}$ 上的开集全体, $\mathcal{F}$ 为 $\mathbb{R}$ 的有理顶点开区间全体. 求证 $\mathbb{R}$ 中的 Borel 集类

$$\mathcal{B} = \sigma(\mathcal{O}) = \sigma(\mathcal{F})$$

(第一个等号正是 Borel 集名称的由来)

证明: 我们知道 $\mathbb{R}$ 上的 Borel 集定义为

$$\mathcal{P} = \{(-\infty, x) : x \in \mathbb{R}\}, \quad \mathcal{B} = \sigma(\mathcal{P})$$

可知 $\mathcal{O} \subset \mathcal{B}$, $\mathcal{P} \subset \sigma(\mathcal{O})$, 故 $\sigma(\mathcal{O}) \subset \mathcal{B}$, $\mathcal{B} \subset \sigma(\mathcal{O})$, 故有 $\mathcal{B} = \sigma(\mathcal{O})$.

对于 $\mathcal{F}$, 易知 $\sigma(\mathcal{F}) \subset \mathcal{B}$. 对于 $\mathcal{P}$ 中 x 为有理数的区间 P_x , 可知 $P_x \in \mathcal{F}$, 对 x 为无理数的情形, 取 $\{x_i\}$ 为 $\mathbb{R}$ 中一系列逼近 x 且满足 $x_i < x$ 的有理数列, 则

$$(-\infty, x) = \bigcup_{i=1}^{\infty} (-\infty, x_i)$$

故此时 $P_x \in \sigma(\mathcal{F})$, 从而 $\mathcal{P} \subset \sigma(\mathcal{F})$, 故 $\mathcal{B} \subset \sigma(\mathcal{F})$, 故 $\mathcal{B} = \sigma(\mathcal{O}) = \sigma(\mathcal{F})$.

K-Problem 2.2.6

Now let $\Delta \notin \mathcal{F}$ be such that

$$\Delta \subset F \in \mathcal{F} \implies P(F) = 1.$$

Such a set is called *thick* in $(\Omega, \mathcal{F}, P)$. If $E = \Delta \cap F, F \in \mathcal{F}$, define $P^*(E) = P(F)$. Then P^* is a well-defined (what does it mean?) p.m. on $(\Delta, \Delta \cap \mathcal{F})$. This procedure is called the *adjunction* of Δ to $(\Omega, \mathcal{F}, P)$.

Proof: 先证明 P^* 是良定义的: 设 F_1, F_2 满足 $\Delta \cap F_1 = \Delta \cap F_2$, 要验证

$$P^*(\Delta \cap F_1) = P^*(\Delta \cap F_2)$$

即证 $P(F_1) = P(F_2)$. 为此, 我们证明 $\Delta \subset F_1^c \cup F_2$, 设 $\delta \in \Delta$ 且 $\delta \notin F_1^c$, 则 $\delta \in \Delta \cap F_1 = \Delta \cap F_2$, 则 $\delta \in F_2$, 故 $\delta \in F_1^c \cup F_2$. 由 $\Delta \subset F_1^c \cup F_2$, 即知

$$\begin{aligned} 1 &= P(F_1^c \cup F_2) = 1 - P(F_1) + P(F_2) - P(F_2 \cap F_1^c) \\ \Rightarrow P(F_1) &= P(F_2) - P(F_2 \cap F_1^c) = P(F_2 - (F_2 \cap F_1^c)) = P(F_1 \cap F_2) \end{aligned}$$

同理可知

$$P(F_2) = P(F_2 \cap F_1)$$

于是 $P(F_1) = P(F_2)$, 故 P^* 为良定义的.

再证明 $\Delta \cap \mathcal{F} \subset 2^\Delta$ 为 Δ 上的 σ -代数: 首先 $\Delta \in \Delta \cap \mathcal{F}$, 取 $E = \Delta \cap F, F \in \mathcal{F}$, 则 $E^c = \Delta - (\Delta \cap F) = \Delta \cap F^c$, 由 $F^c \in \mathcal{F}$ 知 $E^c \in \Delta \cap \mathcal{F}$. 取 $\{E_i\} \subset \Delta \cap \mathcal{F}$, 其中 $E_i = \Delta \cap F_i$, 则

$$\bigcup_{i=1}^{\infty} E_i = \Delta \cap \left(\bigcup_{i=1}^{\infty} F_i \right) \wedge \left(\bigcup_{i=1}^{\infty} F_i \right) \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} E_i \in \Delta \cap \mathcal{F}$$

于是 $\Delta \cap \mathcal{F}$ 构成 Δ 上的 σ -代数.

最后证明 P^* 为概率测度: 由定义, 有

$$P^*(E) = P^*(\Delta \cap F) = P(F) \geq 0$$

$$P^*(\Delta) = P^*(\Delta \cap \Omega) = P(\Omega) = 1$$

$$P^*\left(\sum_{i=1}^{\infty} E_i\right) = P^*\left(\Delta \cap \left(\sum_{i=1}^{\infty} F_i\right)\right) = P\left(\sum_{i=1}^{\infty} F_i\right) = \sum_{i=1}^{\infty} P(F_i) = \sum_{i=1}^{\infty} P^*(E_i)$$

综上, $(\Delta, \Delta \cap \mathcal{F}, P^*)$ 构成概率空间.

Borel-Cantelli's Lemma:

Given probability space $(\Omega, \mathcal{F}, P), \forall \{A_n\} \subset \mathcal{F}, n \geq 1$

(1) If $\sum_{n=1}^{\infty} P(A_n) < +\infty$, then $P(\overline{\lim}_n A_n) = 0$;

(2) If $\{A_n\}$ are **independence**, i.e. $\forall \mathcal{J} \subset \mathbb{N}, P\left(\bigcap_{j \in \mathcal{J}} A_j\right) = \prod_{j \in \mathcal{J}} P(A_j)$, then

$$\sum_{n=1}^{\infty} P(A_n) = \infty \iff P\left(\overline{\lim}_n A_n\right) = 1$$

Proof:

(1) 根据定义可知

$$\begin{aligned} P\left(\overline{\lim}_n A_n\right) &= P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) = \lim_n P\left(\bigcup_{k=n}^{\infty} A_k\right) \leq \lim_n \sum_{k=n}^{\infty} P(A_k) = 0 \\ &\implies P\left(\overline{\lim}_n A_n\right) = 0 \end{aligned}$$

在证明 (2) 前, 我们先证明如下引理:

Lemma: 设 $\{A_i\}$ 为一列独立的事件, 则 $\{A_i^c\}$ 也是一列独立的事件.

Pf: 不妨取指标集 $\mathcal{J} = \mathbb{N}$, 利用容斥原理, 有

$$\begin{aligned} P\left(\bigcap_{i=1}^n A_i^c\right) &= 1 - P\left(\bigcup_{i=1}^n A_i\right) \\ &= 1 - \sum_{i=1}^n P(A_i) + \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) - \cdots + (-1)^n P(A_1 A_2 \cdots A_n) \\ &= 1 - \sum_{i=1}^n P(A_i) + \sum_{1 \leq i < j \leq n} P(A_i)P(A_j) - \cdots + (-1)^n P(A_1)P(A_2) \cdots P(A_n) \\ &= \prod_{i=1}^n (1 - P(A_i)) = \prod_{i=1}^n P(A_i^c) \end{aligned}$$

取 $n \rightarrow \infty$ 即知 $\{A_i^c\}$ 为一列独立的事件.

(2) 利用独立性的定义以及如上引理, 有

$$\begin{aligned} 1 - P\left(\overline{\lim}_n A_n\right) &= P\left(\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right)^c\right) = P\left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k^c\right) = \lim_n P\left(\bigcap_{k=n}^{\infty} A_k^c\right) \\ &= \lim_n \prod_{k=n}^{\infty} P(A_k^c) = \lim_n e^{\sum_{k=n}^{\infty} \ln(1-P(A_k))} \leq \lim_n e^{-\sum_{k=n}^{\infty} P(A_k)} \end{aligned}$$

于是若 $\sum_{k=1}^{\infty} P(A_k) = \infty$, 则 $\sum_{k=n}^{\infty} P(A_k) = \infty$ 对任意 n 成立, 于是上式右侧极限为 0, 故 $P\left(\overline{\lim}_n A_n\right) = 1$; 反之, 若 $P\left(\overline{\lim}_n A_n\right) = 1$, 则由 (1) 的证明过程即知

$$\lim_n \sum_{k=n}^{\infty} P(A_k) \geq 1 \implies \sum_{n=1}^{\infty} P(A_n) = \infty$$

Y-Problem 1.5.4.1

假定生男孩和生女孩是等可能的, 在一个有 3 个孩子的家庭中, 已知有男孩, 求至少有一个女孩的概率.

解: 记 A : 有女孩, B : 有男孩, 则

$$P(B) = 1 - \left(\frac{1}{2}\right)^3 = \frac{7}{8}, P(AB) = 1 - \left(\frac{1}{2}\right)^3 - \left(\frac{1}{2}\right)^3 = \frac{6}{8}$$

于是所求即为

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{6}{7}$$

Y-Problem 1.5.4.3

袋中有 $2n - 1$ 只白球与 $2n$ 只黑球, 现任取出 n 只, 发现它们是同色的, 求同为黑色的概率.

解: 记 A : 取 n 只球发现其同为黑色; B : 取 n 只球发现其同色, 则 $A \subset B$, 故 $AB = A$, 于是有

$$P(A|B) = \frac{P(A)}{P(B)} = \frac{|A|}{|B|} = \frac{\binom{2n}{n}}{\binom{2n}{n} + \binom{2n-1}{n}} = \frac{2}{3}$$

Y-Problem 1.5.4.6

假设一架坠毁的飞机掉在 3 个可能区域中的任何一个等可能的, 若飞机坠毁在区域 i 中 ($i = 1, 2, 3$), 则由于地理环境的影响, 经快速检查后发现其残骸的概率是 α_i ($0 < \alpha_i < 1$). 现快速检查区域 1 后未发现残骸, 求飞机坠毁在区域 i 的概率.

解: 设 A_i : 飞机坠毁在区域 i , B : 快速检查区域 1 后未发现残骸, 则

$$P(A_1|B) = 0$$

$$P(A_2|B) = \frac{P(A_2B)}{P(B)} = \frac{\alpha_2}{\alpha_2 + \alpha_3}$$

$$P(A_3|B) = \frac{P(A_3B)}{P(B)} = \frac{\alpha_3}{\alpha_2 + \alpha_3}$$

Y-Problem 1.5.4.9

甲乙两袋各装有 1 红 1 黑两个球, 每次从两袋中各任取 1 球交换后放回. 求经 n 次交换后, 甲袋中分别包含 2 个红球, 1 红 1 黑, 2 个黑球的概率 p_n, q_n, r_n .

解: 设对应的事件分别为 P_n, Q_n, R_n , 于是 $\{P_n, Q_n, R_n\}$ 对任意 $n \geq 0$ 均构成 Ω 的分割, 并且由对称性知

$$p_n = r_n, p_n + q_n + r_n = 1 \Rightarrow p_n = \frac{1 - q_n}{2}$$

以 $P_{n-1}, Q_{n-1}, R_{n-1}$ 作为 Ω 的分割对第 n 次摸球并交换的过程进行分析, 得到

$$\begin{aligned} p_n &= 0 \cdot p_{n-1} + \left(\frac{1}{2} \cdot \frac{1}{2}\right) q_{n-1} + 0 \cdot r_{n-1} \Rightarrow q_n = 1 - \frac{1}{2} q_{n-1} \\ &\Rightarrow q_n - \frac{2}{3} = -\frac{1}{2} \left(q_{n-1} - \frac{2}{3}\right) \end{aligned}$$

结合 $q_0 = 1$ 即知

$$q_n = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n} \Rightarrow p_n = r_n = \frac{2^n + (-1)^{n+1}}{3 \cdot 2^{n+1}}$$

Y-Problem 1.5.4.10

r 个人作传球游戏, 从某甲开始, 每次持球者均等可能地传给其余 $r - 1$ 个人中的任一个. 求下列事件的概率:

- (1) 传了 n 次, 球没有回到过甲手中;
- (2) 传了 $n(n < r - 1)$ 次, 没有人接到过两次球 (甲开始时持球算作已接球 1 次);
- (3) 第 n 次仍由甲传出.

解:

- (1) 记该事件为 A_n , 对应概率为 p_n , 则 $A_n \subset A_{n-1}$ 且 $p_1 = 1$, 于是

$$\begin{aligned} P(A_n) &= P(A_n A_{n-1}) = P(A_n | A_{n-1}) P(A_{n-1}) \Rightarrow p_n = \frac{r-2}{r-1} p_{n-1} \\ &\Rightarrow p_n = \left(\frac{r-2}{r-1}\right)^{n-1} \end{aligned}$$

- (2) 记该事件为 B_n , 对应概率为 q_n , 则 B_n 发生当且仅当传了 n 次后有 $n + 1$ 个不同的人接到过球, 再由 $B_n \subset B_{n-1}$ 以及 $q_1 = 1$ 可知

$$\begin{aligned} P(B_n) &= P(B_n | B_{n-1}) P(B_{n-1}) \Rightarrow q_n = \frac{r-n}{r-1} q_{n-1} \\ &\Rightarrow q_n = \frac{(r-2)(r-3)\cdots(r-n)}{(r-1)^{n-1}} (n \geq 2), q_1 = 1 \end{aligned}$$

事实上 (1), (2) 也可以使用乘法原理来解决.

(3) 记该事件为 C_n , 对应概率为 r_n , 则 C_n 发生需要 $\overline{C_{n-1}}$ 发生并且第 $n-1$ 个人传给甲, 于是由 $r_1 = 1$ 知

$$\begin{aligned} r_n &= (1 - r_{n-1}) \cdot \frac{1}{r-1} \Rightarrow r_n - \frac{1}{r} = \frac{-1}{r-1} \left(r_{n-1} - \frac{1}{r} \right) \\ &\Rightarrow r_n = \left(\frac{-1}{r-1} \right)^{n-1} \frac{r-1}{r} + \frac{1}{r} \end{aligned}$$

Y-Problem 1.6.5.1

假定生男孩和生女孩是等可能的, 证明: 在有 3 个孩子的家庭中, 事件 $E = \{\text{此家庭至多有 1 个女孩}\}$ 与事件 $F = \{\text{家庭中有男孩也有女孩}\}$ 相互独立. 如果考虑有 2 个孩子的家庭呢?

证明: 直接计算有

$$P(E) = \frac{1 + \binom{3}{1}}{2^3} = \frac{1}{2}, P(F) = 1 - \frac{1+1}{2^3} = \frac{3}{4}, P(EF) = \frac{3}{2^3} = \frac{3}{8} = P(E)P(F)$$

故 E, F 相互独立. 对于两个家庭的孩子, 有

$$P(E) = \frac{3}{4}, P(F) = \frac{1}{2}, P(EF) = \frac{1}{2} \neq P(E)P(F)$$

故此时 E, F 不独立.

Y-Problem 1.6.5.10

试证事件 $A_1, \dots, A_n$ 相互独立的充分必要条件是: 对每个 $\widehat{A}_k = A_k$ 或 A_k^c , 总有

$$P(\widehat{A}_1 \cdots \widehat{A}_n) = P(\widehat{A}_1) \cdots P(\widehat{A}_n)$$

证明: 必要性: 通过调整编号, 不妨设 $\widehat{A}_i = A_i^c (i = 1, \dots, k)$, 其余 $\widehat{A}_j = A_j$. 记 $B = A_{k+1} \cdots A_n$, 则有

$$\begin{aligned} &P(A_1^c) \cdots P(A_k^c) P(A_{k+1}) \cdots P(A_n) \\ &= P(B) \prod_{i=1}^k (1 - P(A_i)) \\ &= P(B) - \sum_{i=1}^k P(A_i B) + \sum_{1 \leq i < j \leq k} P(A_i A_j B) + \cdots + (-1)^k P(A_1 \cdots A_k B) \end{aligned}$$

再记 $B_i = B \cap A_i (i = 1, \dots, k)$, 于是

$$\begin{aligned}
& P(B) - \sum_{i=1}^k P(A_i B) + \sum_{1 \leq i < j \leq k} P(A_i A_j B) + \cdots + (-1)^k P(A_1 \cdots A_k B) \\
&= P(B) - \sum_{i=1}^k P(B_i) + \sum_{1 \leq i < j \leq k} P(B_1 B_2) + \cdots + (-1)^k P(B_1 \cdots B_k) \\
&= P(B) - P\left(\bigcup_{i=1}^k B_i\right) = P(B) - P\left(B \cap \left(\bigcup_{i=1}^k A_i\right)\right) \\
&= P\left(B \cap \left(\bigcup_{i=1}^k A_i\right)^c\right) = P(A_1^c \cdots A_k^c A_{k+1} \cdots A_n)
\end{aligned}$$

充分性: 显然有 $P(A_1 \cdots A_n) = P(A_1) \cdots P(A_n)$, 归纳证明: 对任意 $2 \leq k \leq n-1$, 以及 $1 \leq i_1 \leq \cdots \leq i_k \leq n$, 有

$$P(A_{i_1} \cdots A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$$

当 $k = n-1$ 时, 首先我们有

$$\begin{aligned}
P(A_1 \cdots A_{n-1}) &= P(A_1 \cdots A_{n-1} A_n) + P(A_1 \cdots A_{n-1} A_n^c) \\
&= P(A_1) \cdots P(A_{n-1}) P(A_n) + P(A_1) \cdots P(A_{n-1}) (1 - P(A_n)) \\
&= P(A_1) \cdots P(A_{n-1})
\end{aligned}$$

对于其他排列也是同理的. 现设命题对 k 成立, 在 $k-1$ 处, 取 $j \neq i_1, \dots, i_{k-1}$, 则由归纳假设即有

$$\begin{aligned}
P(A_{i_1} \cdots A_{i_{k-1}}) &= P(A_{i_1} \cdots A_{i_{k-1}} A_j) + P(A_{i_1} \cdots A_{i_{k-1}} A_j^c) \\
&= P(A_{i_1}) \cdots P(A_{i_{k-1}}) P(A_j) + P(A_{i_1}) \cdots P(A_{i_{k-1}}) (1 - P(A_j)) \\
&= P(A_{i_1}) \cdots P(A_{i_{k-1}})
\end{aligned}$$

归纳即知充分性成立, 因此命题得证.

Y-Problem 1.6.5.11

假定 $0 < P(B) < 1$, 证明: 事件 A 与 B 独立的一个充分必要条件是

$$P(A|B) = P(A|B^c)$$

证明: 我们有

$$P(A|B) = P(A|B^c) \iff \frac{P(AB)}{P(B)} = \frac{P(AB^c)}{P(B^c)}$$

易见 $P(AB) + P(AB^c) = P(A)$, 因此

$$P(A|B) = P(A|B^c) \Leftrightarrow \frac{P(AB)}{P(B)} = \frac{P(A) - P(AB)}{1 - P(B)} \Leftrightarrow P(AB) = P(A)P(B)$$

Y-Problem 1.6.5.13

若事件 A, B, C 满足

$$P(AB|C) = P(A|C) \cdot P(B|C)$$

其中 $P(C) > 0$, 则称事件 A 与 B 关于事件 C **条件独立**. 证明:

(1) 当 $P(BC) > 0$ 时, 上式条件独立性的充分必要条件是

$$P(A|BC) = P(A|C)$$

(2) 举例说明“独立”与“条件独立”两者没有蕴含关系.

证明:

(1) 我们有

$$\begin{aligned} P(AB|C) = P(A|C)P(B|C) &\Leftrightarrow \frac{P(ABC)}{P(C)} = \frac{P(AC)P(BC)}{P^2(C)} \\ &\Leftrightarrow \frac{P(ABC)}{P(BC)} = \frac{P(AC)}{P(C)} \\ &\Leftrightarrow P(A|BC) = P(A|C) \end{aligned}$$

(2) 条件独立未必独立: 取事件 $A \subset C \subset B$, 满足 $0 < P(A) < P(C) < P(B) < 1$, 于是 $ABC = AC = A$, $BC = C$, 于是 $P(A|BC) = P(A|C)$, 即 A, B 关于事件 C 条件独立. 但此时 $P(AB) = P(A) > P(A)P(B)$, A, B 不独立.

独立未必条件独立: 取 1.6.5.1 中的事件 E, F , 再记 $G = \{\text{家中至多有 2 个女孩}\}$, 计算知

$$P(E|G) = \frac{4}{7}, P(F|G) = \frac{6}{7}, P(EF|G) = \frac{3}{7} \neq P(E|G)P(F|G)$$

即 E, F 独立但关于事件 G 不独立.

M-Problem 1.7.21

设 $(\Omega, \mathcal{F}, P)$ 是一个概率空间, $\Omega_0 \subset \Omega$. 记 $\Omega_0 \cap \mathcal{F} = \{\Omega_0 \cap A : A \in \mathcal{F}\}$, 称之为 $\mathcal{F}$ 在 Ω_0 上的限制. 若 $P(\Omega_0) > 0$, $\Omega_0 \in \mathcal{F}$, 则定义集函

$$P'(B) = \frac{P(B)}{P(\Omega_0)}, B \in \Omega_0 \cap \mathcal{F}$$

证明: $\Omega_0 \cap \mathcal{F}$ 是 Ω_0 上的一个 σ -代数; P' 是 $\Omega_0 \cap \mathcal{F}$ 上的一个概率测度. 三重偶 $(\Omega_0, \Omega_0 \cap \mathcal{F}, P')$ 称为概率空间 $(\Omega, \mathcal{F}, P)$ 在 Ω_0 上的限制.

证明: 由 $\Omega_0 \in \mathcal{F}$ 即知 $\Omega_0 \in \Omega_0 \cap \mathcal{F}$. 任取 $\Omega_0 \cap A \in \Omega_0 \cap \mathcal{F}$, 由 $A^c \in \mathcal{F}$ 即知

$$\Omega_0 - (\Omega_0 \cap A) = \Omega_0 \cap A^c \in \Omega_0 \cap \mathcal{F}$$

再取 $\Omega_0 \cap \mathcal{F}$ 中的集合列 $\{\Omega_0 \cap A_i\}$, 由 $\bigcup_i A_i \in \mathcal{F}$ 即知

$$\bigcup_{i=1}^{\infty} (\Omega_0 \cap A_i) = \Omega_0 \cap \left(\bigcup_{i=1}^{\infty} A_i \right) \in \Omega_0 \cap \mathcal{F}$$

由此即知 $\Omega_0 \cap \mathcal{F}$ 构成 Ω_0 中的 σ -代数.

P' 显然满足 $P'(B) \geq 0, \forall B \in \Omega_0 \cap \mathcal{F}$, 以及 $P'(\Omega_0) = 1$. 设 $\{B_i = \Omega_0 \cap A_i\}$ 为 $\Omega_0 \cap \mathcal{F}$ 中不交的集合列, 于是

$$P' \left(\sum_{i=1}^{\infty} B_i \right) = \frac{P \left(\sum_{i=1}^{\infty} B_i \right)}{P(\Omega_0)} = \sum_{i=1}^{\infty} \frac{P(B_i)}{P(\Omega_0)} = \sum_{i=1}^{\infty} P'(B_i)$$

这就证明了 P' 构成 $\Omega_0 \cap \mathcal{F}$ 上的概率测度, $(\Omega_0, \Omega_0 \cap \mathcal{F}, P')$ 构成概率空间.

M-Problem 1.7.27

设 $\mathcal{F}$ 是一个代数, P_1 和 P_2 是 $\sigma(\mathcal{F})$ 上的两个概率测度, 用单调类定理证明: 若 P_1 和 P_2 在 $\mathcal{F}$ 上是一致的, 则在 $\sigma(\mathcal{F})$ 上也是一致的.

证明: 记 $\mathcal{G} := \{A \in \sigma(\mathcal{F}) \mid P_1(A) = P_2(A)\}$, 只需证 $\sigma(\mathcal{F}) \subset \mathcal{G}$. 我们有 $\mathcal{F} \subset \mathcal{G}$, 由 $\mathcal{F}$ 为代数即知 $\lambda(\mathcal{F}) = \sigma(\mathcal{F})$, 故只需证 $\mathcal{G}$ 为 λ 类.

- $P_1(\Omega) = P_2(\Omega) = 1 \implies \Omega \in \mathcal{G}$
- $A, B \in \mathcal{G}, A \subset B \implies P_1(B - A) = P_1(B) - P_1(A) = P_2(B) - P_2(A) = P_2(B - A) \implies B - A \in \mathcal{G}$
- $\{A_n\} \subset \mathcal{G}$ 单调递增, $A = \lim_n A_n \implies P_1(A) = \lim_n P_1(A_n) = \lim_n P_2(A_n) = P_2(A) \implies A \in \mathcal{G}$

这就验证了 $\mathcal{G}$ 为 λ 类, 从而 $\sigma(\mathcal{F}) \subset \mathcal{G}$, 即 $P_1|_{\sigma(\mathcal{F})} = P_2|_{\sigma(\mathcal{F})}$.

M-Problem 1.7.33

设 $\{A_n, n \geq 1\}$ 是独立的事件序列, 证明若对每个 $k \geq 1$,

$$\sum_{n \geq k} P(A_n | A_k^c \cap \cdots \cap A_{n-1}^c) = \infty$$

则 $P(\overline{\lim}_n A_n) = 1$. 若取消独立性条件, 则上述条件也能推出 $P(\overline{\lim}_n A_n) = 1$.

证明: 直接证明不独立的情况. 固定一个 $k \geq 1$, 考察级数的部分和

$$\begin{aligned} S_n &= P(A_{k+1}|A_k^c) + P(A_{k+2}|A_k^c A_{k+1}^c) + \cdots + P(A_n|A_k^c \cdots A_{n-1}^c) \\ &= \frac{P(A_{k+1}A_k^c)}{P(A_k^c)} + \frac{P(A_{k+2}A_k^c A_{k+1}^c)}{P(A_k^c A_{k+1}^c)} + \cdots + \frac{P(A_n A_k^c \cdots A_{n-1}^c)}{P(A_k^c \cdots A_{n-1}^c)} \\ &\leq \frac{P(A_{k+1}A_k^c) + P(A_{k+2}A_k^c A_{k+1}^c) + \cdots + P(A_n A_k^c \cdots A_{n-1}^c)}{P(A_k^c \cdots A_{n-1}^c)} \end{aligned}$$

注意到事件 $A_{k+1}A_k^c, \cdots, A_n A_k^c \cdots A_{n-1}^c$ 是两两不交的, 因此有

$$S_n \leq \frac{P((A_{k+1}A_k^c) \cup (A_{k+2}A_k^c A_{k+1}^c) \cup \cdots \cup (A_n A_k^c \cdots A_{n-1}^c))}{P(A_k^c \cdots A_{n-1}^c)} \leq \frac{1}{P(A_k^c \cdots A_{n-1}^c)}$$

于是

$$P(A_k^c \cdots A_{n-1}^c) \leq \frac{1}{S_n}$$

令 $n \rightarrow \infty$ 即知

$$P\left(\bigcap_{i=k}^{\infty} A_i^c\right) = 0$$

于是

$$P\left(\overline{\lim_n A_n}\right) = 1 - P\left(\lim_n A_n^c\right) = 1 - \lim_{k \rightarrow \infty} P\left(\bigcap_{i=k}^{\infty} A_i^c\right) = 1$$

Y-Problem 2.1.5.1

设随机变量 ξ 的分布函数为 $F(x)$, 证明 $\eta = e^\xi$ 也是随机变量, 并求 η 的分布函数.

证明: 由

$$\{\omega : \eta(\omega) < x\} = \begin{cases} \{\omega : \xi(\omega) < \log x\}, & x > 0 \\ \emptyset, & x \leq 0 \end{cases} \in \mathcal{B}(\mathbb{R})$$

即知 $\eta = e^\xi$ 为随机变量. 从而

$$F_\eta(x) = P \circ \eta^{-1}((-\infty, x)) = \begin{cases} F(\log x), & x > 0 \\ 0, & x \leq 0 \end{cases}$$

Y-Problem 2.1.5.2

若 $\xi_n, n \geq 1$ 为随机变量, 证明: $\sup_n \xi_n, \inf_n \xi_n, \overline{\lim}_n \xi_n, \underline{\lim}_n \xi_n$ 均为随机变量.

证明: 由上界与下界的性质即有

$$\begin{aligned} \left\{x : \sup_{n \geq k} \xi_n(x) < t\right\} &= \bigcup_{m=1}^{\infty} \bigcap_{n=k}^{\infty} \left\{x : \xi_n(x) < t - \frac{1}{m}\right\} \\ \left\{x : \inf_{n \geq k} \xi_n(x) > t\right\} &= \bigcup_{m=1}^{\infty} \bigcap_{n=k}^{\infty} \left\{x : \xi_n(x) > t + \frac{1}{m}\right\} \end{aligned}$$

于是由 $\{(-\infty, x) : x \in \mathbb{R}\}$ 与 $\{(x, +\infty) : x \in \mathbb{R}\}$ 均可生成 $\mathcal{B}(\mathbb{R})$ 即知 $\sup_{n \geq k} \xi_n$ 与 $\inf_{n \geq k} \xi_n$ 为随机变量, 取 $k = 1$ 即知 $\sup_n \xi_n$ 与 $\inf_n \xi_n$ 为随机变量. 由

$$\begin{aligned} \overline{\lim}_n \xi_n &= \inf_{k \geq 1} \sup_{n \geq k} \xi_n \\ \underline{\lim}_n \xi_n &= \sup_{k \geq 1} \inf_{n \geq k} \xi_n \end{aligned}$$

即知 $\overline{\lim}_n \xi_n$ 与 $\underline{\lim}_n \xi_n$ 为随机变量.

Y-Theorem 2.1.4

概率空间 $(\Omega, \mathcal{F}, P)$ 上的实值函数 $\xi(\omega)$ 为随机变量的充要条件为: 存在简单随机变量序列 $\{\xi_n, n \geq 1\}$, 使

$$\lim_n \xi_n(\omega) = \xi(\omega)$$

而且当 ξ 非负时, 还可取 $\{\xi_n\}$ 为非负简单随机变量的非降序列.

证明: 必要性书中已证, 充分性由 $\lim_n \xi_n = \overline{\lim}_n \xi_n = \underline{\lim}_n \xi_n$ 与上一题即知.

K-Problem 2.1.11

Take $\Omega = \mathbb{R}^n$ or a separable metric space in Exercise 10 and let $\mathcal{D}$ be the class of all open sets. Let $\mathcal{H}$ be a class of real-valued functions on Ω satisfying the following conditions:

- $1 \in \mathcal{H}$ and $1_D \in \mathcal{H}$ for each $D \in \mathcal{D}$;
- $\mathcal{H}$ is a vector space, namely: if $f_1, f_2 \in \mathcal{H}$ and $c_1, c_2 \in \mathbb{R}$, then $c_1 f_1 + c_2 f_2 \in \mathcal{H}$;
- $\mathcal{H}$ is closed with respect to increasing limits of positive functions, namely: if $f_n \in \mathcal{H}$, $0 \leq f_n \leq f_{n+1}$ for all n , and $f = \lim_n f_n < \infty$, then $f \in \mathcal{H}$.

Then $\mathcal{H}$ contains all Borel measurable functions on Ω , namely all finite-valued functions measurable with respect to the topological Borel field (= the minimal B.F. containing all open sets of Ω).

Proof: 记 $\mathcal{F} = \{A \subset \Omega : 1_A \in \mathcal{H}\}$, 显然有 $\mathcal{D} \subset \mathcal{F}$. 下证 $\sigma(\mathcal{D}) \subset \mathcal{F}$. 由 D 为 π 类即知 $\sigma(\mathcal{D}) = \lambda(\mathcal{D})$, 因此只需要证明 $\mathcal{F}$ 为 λ 类.

易见 $\Omega \in \mathcal{F}$, 取 $A, B \in \mathcal{F}$, $A \subset B$, 于是 $1_{B-A} = 1_B - 1_A \in \mathcal{H}$, 即 $B - A \in \mathcal{F}$. 再取 $\mathcal{F}$ 中单调上升的集合列 $\{A_j\}$, 则由 $1_{\lim_n A_n} = \lim_n 1_{A_n}$ 并且 $\{1_{A_n}\}$ 递增即知 $1_{\lim_n A_n} \in \mathcal{H}$, 即 $\lim_n A_n \in \mathcal{F}$. 综上 $\mathcal{F}$ 为 λ 类, 于是 $\sigma(\mathcal{D}) \subset \mathcal{F}$.

取 ξ 为 $E \in B(\mathcal{D})$ 上的有限集可测函数, 记

$$\xi^+ = \max(\xi, 0) = \frac{1}{2}(|\xi| + \xi), \xi^- = -\min(\xi, 0) = \frac{1}{2}(|\xi| - \xi)$$

于是 $\xi = \xi^+ - \xi^-$, 并且 ξ^+ 与 ξ^- 均为非负的有限集随机变量. 由上一题即知存在非负简单随机变量的非降序列 $\{\xi_n^+\}$ 与 $\{\xi_n^-\}$ 使得

$$\lim_n \xi_n^+ = \xi^+, \lim_n \xi_n^- = \xi^-$$

已证 $\mathcal{H}$ 包含所有 1_E 的线性组合, 因此其包含所有 E 上的简单随机变量, 于是由 $\mathcal{H}$ 对递增序列的有限集极限函数封闭即知 $\xi^+, \xi^- \in \mathcal{H}$, 从而 $\xi \in \mathcal{H}$.

K-Problem 1.2.2

Let F be a d.f. with points of jump $\{a_j\}$. Prove that the sum

$$\sum_{x-\varepsilon < a_j < x} [F(a_j) - F(a_j -)]$$

converges to zero as $\varepsilon \downarrow 0$, for every x . What if the summation above is extended to $x - \varepsilon < a_j \leq x$ instead? Give another proof of the continuity of F_c in Theorem 1.2.1 by using this problem.

Proof: 对于 $(x - \varepsilon, x)$ 中有限个递增的 a_i (不妨设为 $a_1, \dots, a_n$), 由 F 递增即知

$$F(x - \varepsilon) \leq F(a_1 -) < F(a_1) \leq \cdots \leq F(a_n -) < F(a_n) \leq F(x -)$$

从而

$$\begin{aligned} \sum_{i=1}^n [F(a_i) - F(a_i -)] &\leq F(x -) - F(x - \varepsilon) \\ \Rightarrow \sum_{i=1}^{\infty} [F(a_i) - F(a_i -)] &\leq F(x -) - F(x - \varepsilon) \end{aligned}$$

令 $\varepsilon \downarrow 0$, 则 $RHS \downarrow 0$, 于是

$$\lim_{\varepsilon \downarrow 0} \sum_{x-\varepsilon < a_j < x} [F(a_j) - F(a_j -)] = 0$$

若将下标变为 $x - \varepsilon < a_j \leq x$, 则

$$\begin{aligned} &\sum_{x-\varepsilon < a_j \leq x} [F(a_j) - F(a_j -)] \\ &= \sum_{x-\varepsilon < a_j < x} [F(a_j) - F(a_j -)] + F(x) - F(x -) \xrightarrow{\varepsilon \downarrow 0} F(x) - F(x -) \end{aligned}$$

在书中定义了

$$F_d(x) = \sum_j b_j \delta_{a_j}(x), \quad b_j = F(a_j) - F(a_j -)$$

$$F_c(x) = F(x) - F_d(x)$$

于是 $F_d(x)$ 表示 $(-\infty, x]$ 上所有 a_j 的 b_j 之和, 故有

$$F_d(x) - F_d(x - \varepsilon) = \sum_{x-\varepsilon < a_j \leq x} [F(a_j) - F(a_j -)]$$

于是

$$F_c(x) - F_c(x - \varepsilon) = [F(x) - F(x - \varepsilon)] - [F_d(x) - F_d(x - \varepsilon)]$$

根据上文所证, 可知 $\lim_{\varepsilon \downarrow 0} [F_c(x) - F_c(x - \varepsilon)] = 0$, 于是 F_c 左连续. 由于 F_c 为两个右连续函数之差, 从而其右连续, 因此 F_c 连续.

K-Problem 1.3.2

Prove Theorem 1.3.2.

Theorem 1.3.2. Every d.f. F can be written as the convex combination of a discrete, a singular continuous, and an absolutely continuous d.f. Such a decomposition is unique.

Proof: 存在性: 由书中 12 页的内容知, 记

$$F_{ac}(x) = \int_{-\infty}^x F'(t) dt, \quad F_s(x) = F(x) - F_{ac}(x)$$

则 F_{ac} 为绝对连续函数, F_s 为奇异函数, 并且二者均为递增函数, 右连续, 且满足 $F_{ac} \leq F, F_s \leq F$. 于是由定理 1.2.1 知 F_s 有分解

$$F_s = F_c + F_d$$

其中 F_c 为连续函数, 特别地, 其为奇异连续函数, F_d 为 $\delta_a(x)$ 的可列线性组合. 故

$$F = F_{ac} + F_c + F_d$$

容易知道, F_{ac}, F_d, F_c 在 $-\infty$ 处均取 0 值, 均为递增的右连续函数且 $\leq F$. 于是记

$$a_1 = F_{ac}(\infty), a_2 = F_c(\infty), a_3 = F_s(\infty)$$

$$F_1 = \frac{F_{ac}}{a_1}, F_2 = \frac{F_c}{a_2}, F_3 = \frac{F_d}{a_3}$$

若某个 ∞ 处函数值为 0, 则其对应的 $F \equiv 0$, 则相应的 a 为 0, 于是

$$F = a_1 F_1 + a_2 F_2 + a_3 F_3$$

其中 F_1 为绝对连续的分布函数, F_2 为奇异连续的分布函数, F_3 为离散分布函数, 并且 $a_1 + a_2 + a_3 = 1$, 这就证明了存在性.

唯一性: 设 $F = a_1 F_1 + a_2 F_2 + a_3 F_3 = a_1' F_1' + a_2' F_2' + a_3' F_3'$, 由定理 1.2.3 即知对于离散的部分有 $a_3' = a_3, F_3' = F_3$, 于是

$$a_1 F_1' - a_1' F_1' = a_3 F_3' - a_3' F_3' = 0 \text{ a.e.}$$

于是由绝对连续的性质即知 $a_1 F_1 = a_1' F_1'$, 于是 $a_3 F_3 = a_3' F_3'$, 再令 $x \rightarrow \infty$ 即知 $a_1 = a_1', a_3 = a_3'$, 从而 $F_1 = F_1', F_2 = F_2'$, 唯一性得证.

K-Problem 1.3.3

If the support of a d.f. is of measure zero, then F is singular. The converse is false.

Remark: A point x is said to belong to the support of the d.f. F iff for every $\varepsilon > 0$ we have $F(x + \varepsilon) - F(x - \varepsilon) > 0$. The set of all such x is called *the support* of F .

Proof: 记 f 的支集为 E , 从而对 $x \notin E$, 由定义知存在 $\varepsilon > 0$ 使得

$$F(x + \varepsilon) - F(x - \varepsilon) \leq 0$$

故由 F 递增知 $F|_{(x-\varepsilon, x+\varepsilon)} \equiv F(x)$, 故 $F'(x) = 0, \forall x \notin E$. 由 E 零测知 F 奇异.

反之, 考虑 $(0, 1)$ 上的 Cantor 函数 $g(x)$, 并将其延拓为 $g|_{(-\infty, 0]} \equiv 0, g|_{[1, \infty)} \equiv 1$, 则 g 为 $\mathbb{R}$ 上的连续单增函数且除去零测的 Cantor 集 C 后导数为 0.

取 $\mathbb{R}$ 中的可数稠密集 $\{a_1, a_2, \dots, a_n, \dots\}$, 定义函数

$$f(x) = \sum_{n=1}^{\infty} 2^{-n} g(x - a_n)$$

由 $0 \leq g(x) \leq 1, |2^{-n} g(x - a_n)| \leq 2^{-n}$ 知上述级数关于 x 一致收敛, 从而 f 连续, 并且 $f(-\infty) = 0, f(+\infty) = 1$. 设 $x_1 < x_2$, 则存在 i 使得 $x_1 < a_i < x_2$, 于是

$$\begin{aligned} g(x_1 - a_i) &= 0 < g(x_2 - a_i) \text{ 且 } g(x_1 - a_j) \leq g(x_2 - a_j) \quad \forall j \neq i \\ \implies f(x_1) &< f(x_2) \end{aligned}$$

即 f 严格单调递增. 最后, 由 Fubini 定理知

$$f'(x) = \sum_{n=1}^{\infty} 2^{-n} g'(x - a_n)$$

若 $f'(x) \neq 0$, 则存在某个 a_i 使得 $x - a_i \in C$, 即 $x \in \bigcup_{i=1}^{\infty} (C + a_i)$. 该集合为可列个零测集的并, 仍为零测集, 从而 $f' \equiv 0$ a.e. 于是 f 为严格单调递增的奇异分布函数, 其支集 $\mathbb{R}$ 并非零测集.

Y-Problem 2.1.5.6

已知随机变量 ξ 的密度函数为

$$p(x) = \begin{cases} x, & 0 < x \leq 1 \\ 2 - x, & 1 < x \leq 2 \end{cases}$$

试求: (1) ξ 的分布函数, (2) $P(\{0.2 < \xi < 1.2\})$.

解:

(1) 直接计算可知

$$F(x) = \int_{-\infty}^x p(t) dt = \begin{cases} 0, & x \leq 0 \\ \frac{1}{2}x^2, & 0 < x \leq 1 \\ -\frac{1}{2}x^2 + 2x - 1, & 1 < x \leq 2 \\ 1, & x > 2 \end{cases}$$

(2) 计算知

$$P(\{0.2 < \xi < 1.2\}) = F(1.2) - F(0.2) = 0.66$$

Y-Problem 2.1.5.8

设 $F(x)$ 是连续型分布函数, 试证对任意 $a < b$ 有

$$\int_{-\infty}^{+\infty} [F(x+b) - F(x+a)] dx = b - a$$

证明: 设密度函数为 $p(x)$, 则 $p \geq 0$ 且 $\int_{-\infty}^{\infty} p(x) dx = 1$, 故

$$\begin{aligned} \int_{-\infty}^{\infty} [F(x+b) - F(x+a)] dx &= \int_{-\infty}^{\infty} \int_{x+a}^{x+b} p(t) dt dx = \int_{-\infty}^{\infty} \int_a^b p(x+t) dt dx \\ &= \int_a^b \int_{-\infty}^{\infty} p(x+t) dx dt = \int_a^b 1 dt = b - a \end{aligned}$$

Y-Problem 2.1.5.9

设

$$F(x) = \begin{cases} \frac{1+2x}{3}, & 0 < x \leq 1 \\ 1, & x > 1 \end{cases}$$

试证明: (1) $F(x)$ 是一个分布函数; (2) $F(x)$ 既不是离散型也不是连续型的, 但可以写为这两种分布函数的线性组合.

证明:

(1) 我们有 $F(-\infty) = 0, F(\infty) = 1$, F 单调递增并且左连续, 于是 F 为分布函数.

(2) 显然 F 不是离散型的, 由 F 在 $x = 0$ 处不连续知 F 不是连续型的.

记

$$F_1(x) = \begin{cases} 0, & x \leq 0 \\ 1, & x > 0 \end{cases}, \quad F_2(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 < x \leq 1 \\ 1, & x > 1 \end{cases}$$

于是 F_1 与 F_2 分别为离散型与连续型分布函数, 并且 $F = \frac{1}{3}F_1 + \frac{2}{3}F_2$.

Y-Problem 2.2.5.2

同时掷两枚骰子, 直到某个骰子出现 6 点为止, 求恰好投掷 n 次的概率.

解: 记试验为: 同时掷两个骰子, 出现某个骰子为 6 即视为试验成功, 于是试验成功概率 $p = 1 - \left(\frac{5}{6}\right)^2 = \frac{11}{36}$, 所求即第 1 次成功发生在第 n 次试验的概率, 即

$$P = p(1-p)^{n-1} = \frac{11}{36} \times \left(\frac{25}{36}\right)^{n-1}$$

Y-Problem 2.2.5.4

甲、乙两队比赛篮球, 假定每一场甲、乙队获胜的概率分别为 0.4 和 0.6, 且各场胜负独立. 如果规定先胜 4 场者为冠军, 求甲队经过 i 场 ($i = 4, 5, 6, 7$) 比赛而成为冠军的概率 p_i , 再问与赛满三场的三场两胜制相比较, 采用那种赛制甲队最终夺冠的概率较小?

解: p_i 即第 4 次胜利发生在第 i 次试验的概率 (对于甲), 于是

$$p_i = \binom{i-1}{3} (0.4)^4 (0.6)^{i-4} \implies p_4 = 0.0256, p_5 = 0.06144$$

$$p_6 = 0.09216, p_7 = 0.110592$$

于是甲夺冠的概率 $P_1 = 0.289792$, 对于赛满三场的三局两胜制, 甲的获胜概率

$$P_2 = (0.4)^2 + 2 \times (0.4)^2 \times 0.6 = 0.352 > P_1$$

因此前一种赛制甲夺冠概率更小.

Y-Problem 2.2.5.5

求 n 重 Bernoulli 试验中成功偶数次的概率 P_n .

解: 我们有

$$P_n = \sum_{0 \leq k \leq n \wedge 2 \mid k} \binom{n}{k} p^k (1-p)^{n-k}$$

由二项式定理知

$$(1-2p)^n = \sum_{k=0}^n (-1)^k \binom{n}{k} p^k (1-p)^{n-k}, \quad 1 = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k}$$

两式相加即知

$$P_n = \frac{1}{2}(1 + (1-2p)^n)$$

Y-Problem 2.2.5.7

在可列重 Bernoulli 试验中, 以 ξ_i 表示第 i 次成功的等待时间, 求证 $\xi_2 - \xi_1$ 与 ξ_1 有相同的概率分布.

证明: 先给出 ξ_1 的分布:

$$P(\xi_1 = k) = p(1-p)^{k-1}$$

再利用全概率公式给出 $\xi_2 - \xi_1$ 的概率分布:

$$\begin{aligned} P(\xi_2 - \xi_1 = k) &= \sum_{i=1}^{\infty} P(\xi_2 - \xi_1 = k \mid \xi_1 = i) P(\xi_1 = i) \\ &= \sum_{i=1}^{\infty} p(1-p)^{k-1} \cdot p(1-p)^{i-1} = p^2(1-p)^{k-1} \sum_{i=1}^{\infty} (1-p)^{i-1} \\ &= p(1-p)^{k-1} = P(\xi_1 = k) \end{aligned}$$

故二者有相同的概率分布.

Y-Problem 2.3.4.4

假定每小时进入某商店的顾客数量服从 $\lambda = 200$ 的 Poisson 分布, 而进来的顾客将购买商品的概率均为 0.05, 且各顾客是否购物互相独立, 求在 1 小时中至少有 6 位顾客在此商店中购物的概率.

解: 由 Poisson 分布在随机选择下不变即知 1 小时内在超市进行购物的顾客数量 η 服从 Poisson 分布, 并且参数为 $200 \times 0.05 = 10$, 于是所求概率为

$$P = 1 - e^{-10} \sum_{k=0}^5 \frac{10^k}{k!}$$

Y-Problem 2.3.4.8

假定非负整值离散型分布的密度 $\{p_k\}$ 满足条件

$$\frac{p_k}{p_{k-1}} = \frac{\lambda}{k}, \quad k \geq 1$$

其中常数 $\lambda > 0$, 试证明此分布是以 λ 为参数的 Poisson 分布.

证明: 递推即知

$$p_1 = \lambda p_0, p_2 = \frac{\lambda^2}{2!} p_0, \dots, p_k = \frac{\lambda^k}{k!} p_0, \dots$$

于是由 $\{p_k\}$ 为密度函数即知

$$1 = \sum_{k=0}^{\infty} p_k = e^{\lambda} p_0 \Rightarrow p_0 = e^{-\lambda} \Rightarrow p_k = \frac{\lambda^k}{k!} e^{-\lambda}$$

即该分布为以 λ 为参数的 Poisson 分布.

Y-Problem 2.5.5.1

甲从 1,2,3,4 中任取一数 ξ , 乙再从 1, ..., ξ 中任取一整数 η , 试求 (ξ, η) 的联合分布与边缘分布.

解: 易知联合分布为

$$P(\xi = x, \eta = y) = \frac{1}{4} \times \frac{1}{x} = \frac{1}{4x}, 1 \leq y \leq x \leq 4 \wedge (x, y) \in \mathbb{Z}^2$$

边缘分布为

$$p(\xi = i) = \frac{1}{4}, i = 1, 2, 3, 4$$

$$p(\eta = 4) = \frac{1}{16}, p(\eta = 3) = \frac{7}{48}, p(\eta = 2) = \frac{13}{48}, p(\eta = 1) = \frac{25}{48}$$

Y-Problem 2.5.5.3

设 (ξ, η) 的联合密度函数为

$$p(x, y) = \frac{1}{2} \sin(x + y), \quad 0 < x, y < \frac{\pi}{2}$$

试求: (1) (ξ, η) 的联合分布函数; (2) η 的边缘密度函数.

解:

(1) 当 $0 < x, y < \frac{\pi}{2}$ 时, 我们有

$$F(x, y) = \int_0^x \int_0^y \frac{1}{2} \sin(x_1 + x_2) dx_2 dx_1 = \frac{1}{2} (\sin x + \sin y - \sin(x + y))$$

若 x, y 中有超过 $\frac{\pi}{2}$ 的, 则积分上限变为 $\frac{\pi}{2}$, 分类计算知

$$F(x, y) = \begin{cases} 1, & x, y > \frac{\pi}{2} \\ \frac{1}{2}(1 + \sin x - \cos y), & 0 < x < \frac{\pi}{2}, y \geq \frac{\pi}{2} \\ \frac{1}{2}(1 + \sin y - \cos x), & x \geq \frac{\pi}{2}, 0 < y < \frac{\pi}{2} \\ \frac{1}{2}(\sin x + \sin y - \sin(x + y)), & 0 < x, y < \frac{\pi}{2} \\ 0, & \text{else} \end{cases}$$

(2) 计算知

$$p_{\eta}(y) = \int_0^{\frac{\pi}{2}} p(x, y) dx = \frac{1}{2}(\sin y + \cos y), 0 < y < \frac{\pi}{2}$$

Y-Problem 2.5.5.5

设分布函数 $F_1(x)$ 与 $F_2(x)$ 对应的密度函数分别为 $p_1(x)$ 与 $p_2(x)$, 证明对于任何 $\alpha \in (-1, 1)$ 有

$$p_{\alpha}(x, y) = p_1(x)p_2(y)(1 + \alpha(2F_1(x) - 1)(2F_2(y) - 1))$$

是二维密度函数, 且以 $p_1(x)$ 和 $p_2(y)$ 为其边缘密度函数.

证明: 易知 $p_{\alpha}(x, y) \geq 0$, 积分即知

$$\begin{aligned} & \iint_{\mathbb{R}^2} p_{\alpha}(x, y) dx dy \\ &= \iint_{\mathbb{R}^2} p_1(x)p_2(y) dx dy + \alpha \iint_{\mathbb{R}^2} p_1(x)(2F_1(x) - 1)p_2(y)(2F_2(y) - 1) dx dy \\ &= 1 + \alpha \left(\int_{-\infty}^{\infty} p_1(x)(2F_1(x) - 1) dx \right) \left(\int_{-\infty}^{\infty} p_2(y)(2F_2(y) - 1) dy \right) \end{aligned}$$

我们有

$$\begin{aligned} \int_{-\infty}^{\infty} p_1(x)F_1(x) dx &= \int_{-\infty}^{\infty} \int_{-\infty}^x p_1(x)p_1(t) dt dx = \int_{-\infty}^{\infty} \int_t^{\infty} p_1(x)p_1(t) dx dt \\ &= \int_{-\infty}^{\infty} p_1(t)(1 - F_1(t)) dt = 1 - \int_{-\infty}^{\infty} p_1(t)F_1(t) dt \end{aligned}$$

于是

$$\begin{aligned} 2 \int_{-\infty}^{\infty} p_1(x)F_1(x) dx = 1 &\implies \int_{-\infty}^{\infty} p_1(x)(2F_1(x) - 1) dx = 0 \\ &\implies \iint_{\mathbb{R}^2} p_{\alpha}(x, y) dx dy = 1 \end{aligned}$$

故 $p_{\alpha}(x, y)$ 确为二维密度函数. 其边缘分布为

$$\begin{aligned} p_{\alpha 1}(x) &= \int_{-\infty}^{\infty} p_{\alpha}(x, y) dy \\ &= p_1(x) + \alpha p_1(x)(2F_1(x) - 1) \int_{-\infty}^{\infty} p_2(y)(2F_2(y) - 1) dy = p_1(x) \end{aligned}$$

同理 $p_{\alpha 2}(y) = p_2(y)$.

Y-Problem 2.5.5.8

设 (ξ, η) 服从区域 $D = \{(x, y) : 0 < x^2 < y < x < 1\}$ 上的均匀分布. 试求:

(1) ξ 的边缘密度函数; (2) $P(\eta > \frac{1}{2})$.

解:

(1) 我们有

$$p_{\xi}(x) = \int_{x^2}^x \frac{1}{m(D)} dy = \int_{x^2}^x 6 dy = -6x^2, 0 < x < 1$$

(2) 我们有

$$P\left(\eta > \frac{1}{2}\right) = \int_{\frac{1}{2}}^1 p_{\eta}(y) dy = \int_{\frac{1}{2}}^1 \int_y^{\sqrt{y}} 6 dx dy = \frac{7}{4} - \sqrt{2}$$

Y-Problem 2.5.5.9

设 (ξ, η) 为二维正态随机向量, 求 (ξ, η) 落入以下区域的概率.

$$D = \left\{ (x, y) : \frac{(x - a_1)^2}{\sigma_1^2} - \frac{2r(x - a_1)(y - a_2)}{\sigma_1 \sigma_2} + \frac{(y - a_2)^2}{\sigma_2^2} \leq \lambda^2 \right\}$$

解: 设该正态分布的密度函数为 $p(x, y)$, 于是所求即

$$\begin{aligned} P &= \iint_D p(x, y) dx dy \\ &\stackrel{\text{伸缩变换}}{=} \iint_{D'=\{x^2-2rxy+y^2<\lambda^2\}} \frac{1}{2\pi\sqrt{1-r^2}} \exp\left(-\frac{x^2-2rxy+y^2}{2(1-r^2)}\right) dx dy \\ &\stackrel{\text{顺时针旋转}\frac{\pi}{4}}{=} \iint_{D''=\{(1-r)x^2+(1+r)y^2<\lambda^2\}} \frac{1}{2\pi\sqrt{1-r^2}} \exp\left(-\frac{1}{2}\left(\frac{x^2}{1+r} + \frac{y^2}{1-r}\right)\right) dx dy \\ &\stackrel{\text{伸缩变换}}{=} \iint_{D'''=\{x^2+y^2<\frac{\lambda^2}{2(1-r^2)}\}} \frac{1}{\pi} \exp(-(x^2+y^2)) dx dy \\ &\stackrel{\text{三角换元}}{=} \frac{1}{\pi} \int_0^{2\pi} \int_0^{\sqrt{\frac{\lambda^2}{2(1-r^2)}}} r e^{-r^2} dr d\theta = 1 - e^{-\frac{\lambda^2}{2(1-r^2)}} \end{aligned}$$

Y-Problem 2.4.5.1

设 ξ 服从区间 $(0, 5)$ 上的均匀分布, 求方程 $4x^2 + 4\xi x + \xi + 2 = 0$ 有实根的概率.

解: 该方程有实根等价于 $\xi \geq 2$, 于是所求概率即为 $\frac{3}{5}$.

Y-Problem 2.4.5.3

假定随机变量 ξ 只取区间 $(0, 1)$ 中的值, 且对任何 $0 < x < y < 1$, ξ 落在子区间 (x, y) 内的概率仅与 $y - x$ 有关. 求证 ξ 服从区间 $(0, 1)$ 上的均匀分布.

证明: 先证明: $P(\xi = x)$ 对任意 $x \in (0, 1)$ 均为 0: 任取 $x_1, x_2 \in (0, 1)$, 取 $\varepsilon > 0$ 使得 $B(x_1, \varepsilon) \cap B(x_2, \varepsilon) = \emptyset$, 于是由条件即知

$$\begin{aligned} P(\xi = x_1) &= P(\xi \in B(x_1, \varepsilon)) - P(x_1 - \varepsilon < \xi < x_1) - P(x_1 < \xi < x_1 + \varepsilon) \\ &= P(\xi \in B(x_2, \varepsilon)) - P(x_2 - \varepsilon < \xi < x_2) - P(x_2 < \xi < x_2 + \varepsilon) \\ &= P(\xi = x_2) \end{aligned}$$

由任意性即知 ξ 在每点处的概率均相等, 因此只能为 0. 于是对任意 $n \geq 1$, 有

$$\begin{aligned} 1 &= \sum_{i=1}^n P\left(\frac{i-1}{n} < \xi < \frac{i}{n}\right) \Rightarrow P\left(\frac{i-1}{n} < \xi < \frac{i}{n}\right) = \frac{1}{n}, i = 1, \dots, n \\ &\Rightarrow P\left(\xi < \frac{k}{n}\right) = \frac{k}{n}, 0 \leq k \leq n \end{aligned}$$

故分布函数 $F(x)|_{(0,1) \cap \mathbb{Q}} \equiv x$, 由 F 左连续, 对无理数 $x \in (0, 1)$, 只需取有理数列 $\{x_n\}$ 从左侧逼近 x 即知 $F(x) = x$, 故 $(0, 1)$ 上有 $F(x) \equiv x$, 即 ξ 为均匀分布.

Y-Problem 2.4.5.7

随机变量 ξ 服从标准正态分布, $\eta = \xi$ 或 $-\xi$ 视 $|\xi| \leq 1$ 或 $|\xi| > 1$ 而定. 求 η 分布.

解: 设标准正态分布的分布为 $F(x)$. 利用全概率公式, $x \leq -1$ 时

$$\begin{aligned} P(\eta < x) &= P(\eta < x \mid |\xi| \leq 1)P(|\xi| \leq 1) + P(\eta < x \mid |\xi| > 1)P(|\xi| > 1) \\ &= P(\xi > -x) = 1 - F(-x) \end{aligned}$$

当 $-1 < x \leq 1$ 时, 有

$$P(\eta < x) = P(-1 \leq \xi < x) + P(\xi > 1) = F(x)$$

当 $x > 1$ 时

$$P(\eta < x) = P(-1 \leq \xi \leq 1) + P(-x < \xi < -1) + P(\xi > 1) = 1 - F(-x)$$

Y-Problem 2.4.5.9

设 ξ 服从参数为 λ 的指数分布, 求 $\eta = \lfloor \xi \rfloor + 1$ 的分布.

解: 易知 η 为离散型, 计算即知

$$P(\eta = x) = P(x-1 \leq \xi < x) = \begin{cases} e^{-\lambda(x-1)} - e^{-\lambda x}, & x = 1, 2, \dots \\ 0, & x \leq 0 \end{cases}$$

Y-Problem 2.6.3.7

设随机变量 (ξ, η) 的联合密度函数为

$$p(x, y) = \frac{1 + xy}{4}, \quad |x| < 1 \wedge |y| < 1$$

求证 ξ 与 η 不独立, 但 ξ^2 与 η^2 是独立的.

证明: 先求联合分布函数

$$F(x, y) = \begin{cases} 1, & x, y \geq 1 \\ \frac{x+1}{2}, & -1 < x < 1, y \geq 1 \\ \frac{y+1}{2}, & -1 < y < 1, x \geq 1 \\ \frac{1}{16}(x^2y^2 - x^2 - y^2 + 4xy + 4x + 4y + 5), & -1 < x, y < 1 \\ 0, & \text{else} \end{cases}$$

于是 ξ 和 η 的分布函数为

$$F_1(x) = \begin{cases} 0, & x \leq -1 \\ \frac{x+1}{2}, & -1 < x \leq 1 \\ 1, & x > 1 \end{cases}; F_2(y) = \begin{cases} 0, & y \leq -1 \\ \frac{y+1}{2}, & -1 < y \leq 1 \\ 1, & y > 1 \end{cases}$$

显然 $F(x, y) \neq F_1(x)F_2(y)$. 而 ξ^2, η^2 的分布函数分别为

$$G_1(x) = \begin{cases} 0, & x \leq 0 \\ \sqrt{x}, & 0 < x \leq 1 \\ 1, & x > 1 \end{cases}; G_2(y) = \begin{cases} 0, & y \leq 0 \\ \sqrt{y}, & 0 < y \leq 1 \\ 1, & y > 1 \end{cases}$$

而 (ξ^2, η^2) 的联合分布函数为

$$G(x, y) = \begin{cases} 0, & x \leq 0 \vee y \leq 0 \\ \sqrt{x}, & 0 < x \leq 1, y > 1 \\ \sqrt{y}, & 0 < y \leq 1, x > 1 \\ \sqrt{xy}, & 0 < x, y \leq 1 \\ 1, & x > 1 \wedge y > 1 \end{cases}$$

其中 $0 < x, y \leq 1$ 时的 $G(x, y)$ 由

$$F(\sqrt{x}, \sqrt{y}) + F(-\sqrt{x}, -\sqrt{y}) - F(\sqrt{x}, -\sqrt{y}) - F(-\sqrt{x}, \sqrt{y})$$

得到. 由此即知 $G(x, y) = G_1(x)G_2(y)$, 即 ξ^2 与 η^2 独立.

L-Problem 1

Assume that $\{\xi_n, n \geq 1\}$ are iid (independent and identical distributed) random variables and $\xi_i \sim \text{Exp}(1)$. Let $M_n = \max\{\xi_1, \dots, \xi_n\}$. Proof:

$$P\left(\omega : \lim_{n \rightarrow \infty} \frac{M_n(\omega)}{\log n} = 1\right) = 1$$

(Hint: Borel-Cantelli lemma)

Proof: 任取 $0 < \varepsilon < 1$, 由独立性知

$$\begin{aligned} P(M_n < (1 - \varepsilon) \log n) &= P\left(\bigcap_{i=1}^n \{\xi_i < (1 - \varepsilon) \log n\}\right) = \prod_{i=1}^n P(\xi_i < (1 - \varepsilon) \log n) \\ &= \left(1 - \frac{1}{n^{1-\varepsilon}}\right)^n = e^{n \log(1 - \frac{1}{n^{1-\varepsilon}})} < \frac{1}{e^{n^\varepsilon}} < \frac{1}{n^2} (n \rightarrow \infty) \end{aligned}$$

于是由 Borel-Cantelli 引理即知

$$\sum_{n=1}^{\infty} P(M_n < (1 - \varepsilon) \log n) < +\infty \implies P\left(\overline{\lim}_n \left\{ \frac{M_n}{\log n} < (1 - \varepsilon) \right\}\right) = 0$$

取 $\omega \in \left\{ \omega : \underline{\lim}_{n \rightarrow \infty} \frac{M_n(\omega)}{\log n} < 1 \right\}$, 于是存在足够小的 ε 与无穷个 n 满足

$$\frac{M_n(\omega)}{\log n} < 1 - \varepsilon$$

根据集合上极限的等价定义即知 $\omega \in \overline{\lim}_n \left\{ \omega : \frac{M_n(\omega)}{\log n} < 1 - \varepsilon \right\}$, 于是

$$P\left(\omega : \underline{\lim}_n \frac{M_n(\omega)}{\log n} < 1\right) = 0 \implies P\left(A := \left\{ \omega : \underline{\lim}_n \frac{M_n(\omega)}{\log n} \geq 1 \right\}\right) = 1$$

同理可得

$$\begin{aligned} P\left(\overline{\lim}_n \left\{ \frac{M_n}{\log n} > 1 + \varepsilon \right\}\right) &= 0 \implies P\left(\omega : \overline{\lim}_n \frac{M_n(\omega)}{\log n} > 1\right) = 0 \\ &\implies P\left(B := \left\{ \omega : \overline{\lim}_n \frac{M_n(\omega)}{\log n} \leq 1 \right\}\right) = 1 \end{aligned}$$

于是即有

$$\begin{aligned} P\left(\omega : \lim_{n \rightarrow \infty} \frac{M_n(\omega)}{\log n} = 1\right) &= P(A \cap B) = 2 - P(A \cup B) \geq 1 \\ &\implies P\left(\omega : \lim_{n \rightarrow \infty} \frac{M_n(\omega)}{\log n} = 1\right) = 1 \end{aligned}$$

L-Problem 2

Let $\{\xi_n, n \geq 1\}$ be a sequence of r.v.'s on $(\Omega, \mathcal{F}, P)$. Define

$$\mathcal{G}_n := \sigma(\xi_{n+1}, \xi_{n+2}, \dots) := \sigma\left(\bigcup_{k=1}^{\infty} \xi_{n+k}^{-1}(\mathcal{B}(\mathbb{R}))\right)$$

$$\mathcal{F}_n := \sigma(\xi_1, \dots, \xi_n) := \sigma\left(\bigcup_{k=1}^n \xi_k^{-1}(\mathcal{B}(\mathbb{R}))\right)$$

$$\mathcal{F}_{\infty} := \sigma(\xi_1, \dots, \xi_n, \dots) := \sigma\left(\bigcup_{k=1}^{\infty} \xi_k^{-1}(\mathcal{B}(\mathbb{R}))\right)$$

where $\xi_i^{-1}(\mathcal{B}(\mathbb{R})) = \{\xi_i^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$ is a σ -algebra on Ω .

Proof:

(1) $\mathcal{G}_n = \sigma(\mathcal{E}_n)$, $\mathcal{F}_n = \sigma(\mathcal{C}_n)$, $\mathcal{F}_{\infty} = \sigma(\bigcup_{n=1}^{\infty} \mathcal{F}_n)$, where

$$\mathcal{E}_n = \left\{ \bigcap_{i=1}^k \{\xi_{n+i} < x_{n+i}\} : k \geq 1, x_{n+1}, \dots, x_{n+k} \in \overline{\mathbb{R}} \right\}$$

$$\mathcal{C}_n = \left\{ \bigcap_{k=1}^n \{\xi_k < x_k\} : x_1, \dots, x_n \in \overline{\mathbb{R}} \right\}, \quad \overline{\mathbb{R}} = [-\infty, +\infty]$$

(2) (Kolmogorov's 0-1 law): If $\{\xi_n, n \geq 1\}$ are independent, then the *tail* σ -algebra

$$\mathcal{G} := \bigcap_{n=1}^{\infty} \mathcal{G}_n, \quad \mathcal{G}_n \text{ is defined as (1)}$$

contains only events of probability 0 or 1, i.e., if $A \in \mathcal{G}$, then $P(A) = 0$ or 1.

Proof:

(1) 易见 $\mathcal{E}_n \subset \mathcal{G}_n$, 于是 $\sigma(\mathcal{E}_n) \subset \mathcal{G}_n$, 下证 $\mathcal{G}_n \subset \sigma(\mathcal{E}_n)$, 这只需证

$$\bigcup_{k=1}^{\infty} \xi_{n+k}^{-1}(\mathcal{B}(\mathbb{R})) \subset \sigma(\mathcal{E}_n)$$

只需证对任意 $k \geq 1$ 与 $B \in \mathcal{B}(\mathbb{R})$, 有 $\xi_{n+k}^{-1}(B) \in \sigma(\mathcal{E}_n)$

由随机变量的性质, 有

$$\begin{aligned} \xi_{n+k}^{-1}(B) &\in \xi_{n+k}^{-1}(\mathcal{B}(\mathbb{R})) = \xi_{n+k}^{-1}(\sigma(\{(-\infty, x) : x \in \mathbb{R}\})) \\ &= \sigma(\xi_{n+k}^{-1}(\{(-\infty, x) : x \in \mathbb{R}\})) \\ &\subset \sigma(\mathcal{E}_n) \end{aligned}$$

其中最后一个包含关系是令 $\mathcal{E}_n$ 中除 x_{n+k} 外的所有 $x = +\infty$ 得到的.

于是得到 $\mathcal{G}_n = \sigma(\mathcal{E}_n)$, 完全一致的证明给出 $\mathcal{F}_n = \sigma(\mathcal{C}_n)$.

对于 $\mathcal{F}_n$, 首先易知 $\mathcal{F}_n \subset \mathcal{F}_\infty$, 从而 $\bigcup_{n=1}^{\infty} \mathcal{F}_n \subset \mathcal{F}_\infty$, 故 $\sigma\left(\bigcup_{n=1}^{\infty} \mathcal{F}_n\right) \subset \mathcal{F}_\infty$. 下证 $\mathcal{F}_\infty \subset \sigma\left(\bigcup_{n=1}^{\infty} \mathcal{F}_n\right)$, 由定义即知这只需证对任意 $k \geq 1$ 与 $B \in \mathcal{B}(\mathbb{R})$, 均有

$$\xi_k^{-1}(B) \in \bigcup_{n=1}^{\infty} \mathcal{F}_n$$

而这是显然的, 故 $\mathcal{F}_\infty = \sigma\left(\bigcup_{n=1}^{\infty} \mathcal{F}_n\right)$.

(2) 使用如下引理:

Lemma: 设 $(\Omega, \mathcal{F}, P)$ 是概率空间, $\{\mathcal{G}_i\}_{i \in I}$ 为一列 σ 代数, 且每个 $\mathcal{G}_i$ 均由 π 类 $\mathcal{A}_i$ 生成, 即 $\mathcal{G}_i = \sigma(\mathcal{A}_i), i \in I$. 于是 $\{\mathcal{G}_i\}_{i \in I}$ 独立当且仅当

$$P\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} P(A_i), \forall A_i \in \mathcal{A}_i \text{ and finite subset } J \subset I$$

回到本题, 由随机变量的独立性即知 $\mathcal{E}_n$ 与 $\mathcal{C}_n$ 独立, 且二者均为 π 类. 于是由引理得 $\mathcal{G}_n = \sigma(\mathcal{E}_n)$ 与 $\mathcal{F}_n = \sigma(\mathcal{C}_n)$ 独立, 再由 $\mathcal{G} \subset \mathcal{G}_n$ 知 $\mathcal{G}$ 与 $\mathcal{F}_n$ 独立, 对任意 $n \geq 1$, 于是由 $\mathcal{F}_n$ 为递增的 σ 代数列即知 $\bigcup_{n=1}^{\infty} \mathcal{F}_n$ 与 $\mathcal{G}$ 独立. 且二者均为 π 类, 故由引理即知 $\mathcal{F}_\infty = \sigma\left(\bigcup_{n=1}^{\infty} \mathcal{F}_n\right)$ 与 $\mathcal{G} = \sigma(\mathcal{G})$ 独立. 而 $\mathcal{G} \subset \mathcal{F}_\infty$, 从而 $\mathcal{G}$ 与 $\mathcal{G}$ 独立.

于是, 由独立性的定义, 对任意 $A \in \mathcal{G}$, 有

$$P(A) = P(A \cap A) = P^2(A) \implies P(A) = 0 \vee P(A) = 1$$

证毕.

Y-Problem 2.3.4.7

试证明: 函数方程 $F(x+y) = F(x)F(y)$ 的单调 (或连续) 解必为指数函数 $F(x) = a^x$. 从而函数方程 $G(x+y) = G(x) + G(y)$ 的单调 (或连续) 解必为线性函数 $G(x) = kx$.

证明: 首先有 $F(0) = F^2(0)$, 若 $F(0) = 0$, 令 $y = 0$ 即知这对应平凡解 $F \equiv 0$. 设 $F(0) = 1$, 记 $a := F(1)$, 并不妨设 F 是单增的. 令 $y = -x$, 得 $F(x)F(-x) = 1$, 由于 $x > 0 \Rightarrow F(x) > 1$, 因此知 F 在 $\mathbb{R}$ 上恒正. 对任意正整数 k , 有

$$F(k) = F(1+1+\cdots+1) = F(1) \cdot F(1) \cdots F(1) = a^k$$

再由 $F(x)F(-x) = 1$ 即知 $F(x) = a^x, \forall x \in \mathbb{Z}$, 设 $x = \frac{p}{q} \in \mathbb{Q}$, 于是

$$a^p = F(p) = F(x+x+\cdots+x) = F(x)^q \Rightarrow F(x) = a^{\frac{p}{q}} = a^x$$

从而 $F(x) = a^x, \forall x \in \mathbb{Q}$. 设 $x \in \mathbb{R} \setminus \mathbb{Q}$, 取有理递增列 $\{p_n\}$ 与有理递减列 $\{q_n\}$ 使得 $p_n < x < q_n \forall n$ 且 $\lim_{n \rightarrow \infty} p_n = \lim_{n \rightarrow \infty} q_n = x$. 于是根据单调函数的性质, 有

$$F(x) \geq \sup_{p < x} F(p) \geq \sup_n F(p_n) = a^x$$

$$F(x) \leq \inf_{q > x} F(q) \leq \inf_n F(q_n) = a^x$$

从而 $F(x) = a^x, \forall x \in \mathbb{R}$, 其中 $a = F(1) > 0$.

对于 $G(x+y) = G(x) + G(y)$, 令 $F(x) = e^{G(x)} > 0$, 从而 G 单调给出 F 单调, 且 $F(x+y) = F(x)F(y)$, 于是 $F(x) = a^x$, 故 $G(x) = (\ln a)x$ 为线性函数.

Y-Problem 2.6.3.2

在可列重 Bernoulli 试验中, 以 ξ_i 表第 i 次成功的等待时间, 求已知 $\xi_2 = n (n \geq 2)$ 时 ξ_1 的条件分布.

解: 设每次试验成功率为 p . 在 $\xi_2 = n$ 时, 易知 ξ_1 只能取 $1, 2, \dots, n-1$, 于是

$$P(\xi_1 = k | \xi_2 = n) = \frac{P(\xi_1 = k, \xi_2 = n)}{P(\xi_2 = n)} = \frac{p^2(1-p)^{n-2}}{\binom{n-1}{1} p^2(1-p)^{n-2}} = \frac{1}{n-1}$$

这体现了 Bernoulli 试验的独立性.

Y-Problem 2.6.3.5

设 (ξ, η) 是连续型随机变量, ξ 有密度函数

$$p_1(x) = \lambda^2 x e^{-\lambda x}, x > 0$$

而 η 服从 $(0, \xi)$ 上的均匀分布, 试求 η 的密度函数.

证明: 易知 $\eta > 0$, 从而对 $x > 0$, 利用全概率公式, 有

$$\begin{aligned} F_{\eta}(x) &= \int_{\mathbb{R}} P(\eta < x | \xi = t) dF_{\xi}(t) = \int_{\mathbb{R}} P(\eta < x | \xi = t) p_1(t) dt \\ &= \int_0^x p_1(t) dt + \int_x^{\infty} \frac{x}{t} p_1(t) dt = (-\lambda x e^{-\lambda x} - e^{-\lambda x} + 1) + (\lambda x e^{-\lambda x}) \\ &= 1 - e^{-\lambda x}, x > 0 \end{aligned}$$

从而知 η 满足指数分布, 其密度函数为 $p_{\eta}(x) = \lambda e^{-\lambda x}, x > 0$.

Y-Problem 2.6.3.8

若 (ξ, η) 独立, 都服从 -1 与 1 上的等可能分布, 而 $\zeta = \xi\eta$, 求证 ξ, η, ζ 两两独立但整体不独立.

证明: 直接计算, 有

$$P(\zeta = 1) = P(\xi = 1, \eta = 1) + P(\xi = -1, \eta = -1) = \frac{1}{2}, P(\zeta = -1) = \frac{1}{2}$$

并且

$$\begin{aligned} P(\xi = i, \eta = j) &= \frac{1}{4} = P(\xi = i)P(\eta = j), i, j \in \{1, -1\} \\ P(\zeta = i, \eta = j) &= P\left(\xi = \frac{i}{j}, \eta = j\right) = \frac{1}{4} = P(\zeta = i)P(\eta = j), i, j \in \{1, -1\} \\ P(\zeta = i, \xi = j) &= P\left(\xi = j, \eta = \frac{i}{j}\right) = \frac{1}{4} = P(\zeta = i)P(\xi = j), i, j \in \{1, -1\} \end{aligned}$$

于是 ξ, η, ζ 两两独立, 但是

$$P(\zeta = 1, \xi = 1, \eta = -1) = 0 \neq P(\zeta = 1)P(\xi = 1)P(\eta = -1)$$

即 ξ, η, ζ 整体不独立.

Y-Problem 2.7.8.2

假定随机变量 ξ_1, ξ_2 相互独立, 对 $i = 1, 2, \xi_i \sim \text{Poisson}(\lambda_i)$, 试求:

(1) $\xi_1 + \xi_2$ 的分布; (2) 已知 $\xi_1 + \xi_2 = n$ 时 ξ_1 的条件分布.

解: (1) 由独立性, 我们有

$$P(\xi_1 + \xi_2 = k) = \sum_{i=0}^k p_i q_{k-i} = e^{-(\lambda_1 + \lambda_2)} \sum_{i=0}^k \frac{\lambda_1^i \lambda_2^{k-i}}{i!(k-i)!} = \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} (\lambda_1 + \lambda_2)^k$$

即 $\xi_1 + \xi_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

(2) 此时 ξ_1 只取 $0, 1, \dots, n$ 这些值, 由独立性, 我们有

$$\begin{aligned} p(\xi_1 = k | \xi_1 + \xi_2 = n) &= \frac{P(\xi_1 = k, \xi_2 = n - k)}{P(\xi_1 + \xi_2 = n)} = \frac{e^{-(\lambda_1 + \lambda_2)} \frac{\lambda_1^k \lambda_2^{n-k}}{k!(n-k)!}}{\frac{e^{-(\lambda_1 + \lambda_2)}}{n!} (\lambda_1 + \lambda_2)^n} \\ &= \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}, k = 0, 1, \dots, n \end{aligned}$$

即 $\xi_1 |_{\xi_1 + \xi_2 = n} \sim B\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right)$

Y-Problem 2.7.8.8

设随机变量 $\xi_1, \dots, \xi_r$ 互相独立, 都服从参数为 λ 的指数分布. 试证: $\xi_1 + \dots + \xi_r$ 服从 Γ 分布 $\Gamma(\lambda, r)$.

证明: 我们先给出数学分析中一个熟知的结论:

Lemma:

$$\int_{\substack{t_1 + \dots + t_n < x \\ t_1, \dots, t_n > 0}} 1 \, dt_n \dots dt_1 = \frac{x^n}{n!}$$

回到本题, 使用归纳法. $r = 1$ 时显然, 设命题对 $r = k - 1$ 成立, 即:

$$F_{\xi_1 + \dots + \xi_{k-1}}(x) = \int_{t_1 + \dots + t_{k-1} < x} p(t_1, \dots, t_{k-1}) \, dt_{k-1} \dots dt_1 = \int_0^x \frac{\lambda^{k-1}}{\Gamma(k-1)} t^{k-2} e^{-\lambda t} \, dt$$

当 $r = k$ 时, 由独立性, 我们有

$$\begin{aligned} F_{\xi_1 + \dots + \xi_k}(x) &= \int_{t_1 + \dots + t_k < x} p(t_1, \dots, t_k) \, dt_k \dots dt_1 \\ &= \int_0^x \int_0^{x-t_1} \dots \int_0^{x-t_1-\dots-t_{k-1}} \lambda^k e^{-\lambda(t_1 + \dots + t_k)} \, dt_k \dots dt_1 \\ &= -\lambda^{k-1} e^{-\lambda x} \frac{x^{k-1}}{(k-1)!} + \int_0^x \dots \int_0^{x-t_1-\dots-t_{k-2}} \lambda^{k-1} e^{-\lambda(t_1 + \dots + t_{k-1})} \, dt_{k-1} \dots dt_1 \\ &= -\lambda^{k-1} e^{-\lambda x} \frac{x^{k-1}}{(k-1)!} + F_{\xi_1 + \dots + \xi_{k-1}}(x) \\ &= -\lambda^{k-1} e^{-\lambda x} \frac{x^{k-1}}{(k-1)!} + \int_0^x \frac{\lambda^{k-1}}{\Gamma(k-1)} t^{k-2} e^{-\lambda t} \, dt \\ &\stackrel{\text{分部积分}}{=} \int_0^x \frac{\lambda^k}{\Gamma(k)} t^{k-1} e^{-\lambda t} \, dt \end{aligned}$$

归纳即知 $\xi_1 + \dots + \xi_r \sim \Gamma(\lambda, r)$.

Y-Problem 2.7.8.11

设随机变量 ξ, η 独立, 同服从标准正态分布, 试证: $\frac{\xi}{\eta}$ 服从 Cauchy 分布.

证明: 由定理 2.7.5, $\frac{\xi}{\eta}$ 具有密度函数

$$\begin{aligned} p_{\frac{\xi}{\eta}}(z) &= \int_{\mathbb{R}} |y| p(z y, y) dy = \int_{\mathbb{R}} |y| p_{\xi}(z y) p_{\eta}(y) dy \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} |y| e^{-\frac{z^2 y^2}{2} - \frac{y^2}{2}} dy = \frac{1}{\pi} \int_0^{\infty} y e^{-\frac{z^2+1}{2} y^2} dy \\ &= \frac{1}{\pi(z^2 + 1)} \end{aligned}$$

从而 $\frac{\xi}{\eta} \sim C(1, 0)$.

Y-Problem 2.7.8.12

设随机变量 ξ, η 独立, 同服从 $\lambda = 1$ 的指数分布, 试证: $\xi + \eta$ 与 $\frac{\xi}{\xi + \eta}$ 互相独立.

证明: 记 $\zeta = \xi + \eta, \varphi = \frac{\xi}{\xi + \eta}$, 由 2.7.8.8 即知 $\zeta \sim \Gamma(1, 2)$, 对于 φ , 易知其取值范围是 $(0, 1)$, 故对 $z \in (0, 1)$, 有

$$F_{\varphi}(z) = \iint_{\frac{x}{x+y} < z} p(x, y) dy dx = \int_0^{\infty} \int_{(\frac{1}{z}-1)x}^{\infty} e^{-(x+y)} dy dx = z$$

于是可以写出

$$F_{\zeta}(x) = \int_0^x \frac{1}{2} t e^{-t} dt = \frac{1}{2}(1 - e^{-x} - x e^{-x}), x > 0; F_{\varphi}(y) = \begin{cases} 1, & y \geq 1 \\ y, & 0 < y < 1 \end{cases}$$

计算 ζ 与 φ 的联合分布, 对 $x > 0, 0 < y < 1$, 有

$$\begin{aligned} F_{(\zeta, \varphi)}(x, y) &= P\left(\xi + \eta < x, \frac{\xi}{\xi + \eta} < y\right) = \iint_{t_1+t_2 < x, \frac{t_1}{t_1+t_2} < y} e^{-(t_1+t_2)} dt_2 dt_1 \\ &= \int_0^{xy} \int_{(\frac{1}{y}-1)t_1}^{x-t_1} e^{-(t_1+t_2)} dt_2 dt_1 = y(1 - e^{-x} - x e^{-x}) \\ &= F_{\zeta}(x) F_{\varphi}(y) \end{aligned}$$

对于 $x > 0, y \geq 1$ 时上式显然成立, 这就验证了 $\xi + \eta$ 与 $\frac{\xi}{\xi + \eta}$ 的独立性.

Y-Problem 2.7.8.16

设随机变量 ξ, η 独立, ξ 服从 $p = \frac{1}{2}$ 的 Bernoulli 分布, η 服从区间 $(0, 1)$ 上的均匀分布. 试求: (1) $\xi + \eta$ 的分布函数; (2) $\frac{\xi}{2} + \eta$ 的分布函数; (3) $\xi \eta$ 的分布函数.

解:

(1) 利用全概率公式以及独立性, 有

$$P(\xi + \eta < x) = \frac{1}{2}P(\eta < x) + \frac{1}{2}P(\eta < x - 1) = \begin{cases} \frac{1}{2}x, & 0 < x \leq 2 \\ 1, & x > 2 \end{cases}$$

(2) 利用全概率公式以及独立性, 有

$$P\left(\frac{\xi}{2} + \eta < x\right) = \frac{1}{2}P(\eta < x) + \frac{1}{2}P\left(\eta < x - \frac{1}{2}\right) = \begin{cases} \frac{1}{2}x, & 0 < x \leq \frac{1}{2} \\ x - \frac{1}{4}, & \frac{1}{2} < x \leq 1 \\ \frac{1}{2}x + \frac{1}{4}, & 1 < x \leq \frac{3}{2} \\ 1, & x > \frac{3}{2} \end{cases}$$

(3) 利用全概率公式以及独立性, 有

$$P(\xi\eta < x) = \frac{1}{2} + \frac{1}{2}P(\eta < x) = \begin{cases} \frac{1}{2}x + \frac{1}{2}, & 0 < x \leq 1 \\ 1, & x > 1 \end{cases}$$

L-Problem 1

Assume that $\xi \sim \Gamma\left(\frac{1}{2}, \frac{n}{2}\right)$, $\eta \sim \Gamma\left(\frac{1}{2}, \frac{m}{2}\right)$ are independent, show that

(1) $\xi + \eta$ and $\frac{\xi}{\xi + \eta}$ are independent;

(2) $\xi + \eta \sim \Gamma\left(\frac{1}{2}, \frac{m+n}{2}\right)$, $\frac{\xi}{\xi + \eta} \sim \text{Beta}\left(\frac{n}{2}, \frac{m}{2}\right)$.

Proof: 先计算 $\xi + \eta$ 的密度函数

$$\begin{aligned} p_{\xi+\eta}(z) &= \int_{\mathbb{R}} p_{\xi}(x)p_{\eta}(z-x) dx = \int_0^z \left(\frac{x^{\frac{n}{2}-1} e^{-\frac{1}{2}x}}{\sqrt{2^n} \Gamma(\frac{n}{2})} \right) \left(\frac{(z-x)^{\frac{m}{2}-1} e^{-\frac{1}{2}(z-x)}}{\sqrt{2^m} \Gamma(\frac{m}{2})} \right) dx \\ &= \frac{e^{-\frac{1}{2}z} z^{\frac{n+m}{2}-1}}{\sqrt{2^{n+m}} \Gamma(\frac{n}{2}) \Gamma(\frac{m}{2})} \int_0^1 t^{\frac{n}{2}-1} (1-t)^{\frac{m}{2}-1} dt \\ &= \frac{\left(\frac{1}{2}\right)^{\frac{m+n}{2}}}{\Gamma(\frac{n}{2}) \Gamma(\frac{m}{2})} B\left(\frac{n}{2}, \frac{m}{2}\right) z^{\frac{m+n}{2}-1} e^{-\frac{1}{2}z} \\ &= \frac{\left(\frac{1}{2}\right)^{\frac{m+n}{2}}}{\Gamma(\frac{m+n}{2})} z^{\frac{m+n}{2}-1} e^{-\frac{1}{2}z}, z > 0 \end{aligned}$$

于是 $\xi + \eta \sim \Gamma\left(\frac{1}{2}, \frac{m+n}{2}\right)$. 易知 $\frac{\xi}{\xi + \eta}$ 取值范围是 $(0, 1)$, 从而其分布函数为

$$F_{\frac{\xi}{\xi+\eta}}(z) = \iint_{\frac{x}{x+y} < z} p_{\xi}(x)p_{\eta}(y) dy dx = \int_0^{\infty} \int_{x(\frac{1}{z}-1)}^{\infty} \frac{e^{-\frac{1}{2}(x+y)} x^{\frac{n}{2}-1} y^{\frac{m}{2}-1}}{\sqrt{2^{n+m}} \Gamma(\frac{n}{2}) \Gamma(\frac{m}{2})} dy dx$$

从而求导得到其密度函数为

$$\begin{aligned}
 p_{\frac{\xi}{\xi+\eta}}(z) &= \frac{d\left(F_{\frac{\xi}{\xi+\eta}}\right)}{dz}(z) = \int_0^\infty \frac{x}{z^2} \frac{e^{-\frac{1}{2}(x+x(\frac{1}{z}-1))} x^{\frac{n}{2}-1} \left(x(\frac{1}{z}-1)\right)^{\frac{m}{2}-1}}{\sqrt{2^{n+m}} \Gamma(\frac{n}{2}) \Gamma(\frac{m}{2})} dx \\
 &= \frac{\left(\frac{1}{z}-1\right)^{\frac{m}{2}-1}}{z^2 \sqrt{2^{n+m}} \Gamma(\frac{n}{2}) \Gamma(\frac{m}{2})} \int_0^\infty e^{-\frac{x}{2z}} x^{\frac{n+m}{2}-1} dx \\
 &\stackrel{\text{换元}}{=} \frac{\left(\frac{1}{z}-1\right)^{\frac{m}{2}-1} (2z)^{\frac{m+n}{2}}}{z^2 \sqrt{2^{n+m}} \Gamma(\frac{n}{2}) \Gamma(\frac{m}{2})} \int_0^\infty t^{\frac{m+n}{2}-1} e^{-t} dt \\
 &= \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{n}{2}) \Gamma(\frac{m}{2})} z^{\frac{n}{2}-1} (1-z)^{\frac{m}{2}-1} \\
 &= \frac{z^{\frac{n}{2}-1} (1-z)^{\frac{m}{2}-1}}{B(\frac{n}{2}, \frac{m}{2})}
 \end{aligned}$$

于是 $\frac{\xi}{\xi+\eta} \sim B(\frac{n}{2}, \frac{m}{2})$.

最后验证独立性, 记 $\zeta = \xi + \eta, \varphi = \frac{\xi}{\xi+\eta}$, 则对 $z > 0, 0 < w < 1$

$$\begin{aligned}
 F_{(\zeta, \varphi)}(z, w) &= P\left(\xi + \eta < z, \frac{\xi}{\xi + \eta} < w\right) = \iint_{x+y < z, \frac{x}{x+y} < w} p_\xi(x) p_\eta(y) dy dx \\
 &= \int_0^{zw} \int_{(\frac{1}{w}-1)x}^{z-x} p_\xi(x) p_\eta(y) dy dx
 \end{aligned}$$

求导可知

$$\begin{aligned}
 \frac{\partial F}{\partial w} &= \int_0^{zw} \frac{x}{w^2} p_\xi(x) p_\eta\left(x\left(1 - \frac{1}{w}\right)\right) dx \\
 p_{(\zeta, \varphi)}(z, w) &= \frac{\partial^2 F}{\partial z \partial w} = z p_\xi(zw) p_\eta(z(1-w)) \\
 &= z \frac{(zw)^{\frac{n}{2}-1} e^{-\frac{1}{2}zw}}{\sqrt{2^n} \Gamma(\frac{n}{2})} \cdot \frac{(z(1-w))^{\frac{m}{2}-1} e^{-\frac{1}{2}z(1-w)}}{\sqrt{2^m} \Gamma(\frac{m}{2})} \\
 &= \frac{\left(\frac{1}{2}\right)^{\frac{m+n}{2}}}{\Gamma(\frac{m+n}{2})} z^{\frac{n+m}{2}-1} e^{-\frac{1}{2}z} \cdot \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{n}{2}) \Gamma(\frac{m}{2})} w^{\frac{n}{2}-1} (1-w)^{\frac{m}{2}-1} \\
 &= p_\zeta(z) p_\varphi(w)
 \end{aligned}$$

对于 $z > 0, w > 1$ 的情况, 独立性是显然的.

于是我们验证了独立性成立.

L-Problem 2

Assume that $\xi \sim N(0, 1), \eta \sim N(0, 1)$ are independent, show that $U := \frac{\xi}{\eta}$ and $V = \xi^2 + \eta^2$ are independent, $U \sim C(1, 0), V \sim \Gamma(\frac{1}{2}, 1)$.

Proof: 对于 $\frac{\xi}{\eta}$, 可知其密度函数为

$$\begin{aligned} p_{\frac{\xi}{\eta}}(x) &= \int_{\mathbb{R}} |y| p(\xi y, y) dy = \int_{\mathbb{R}} |y| \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2 y^2}{2}} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy \\ &= \frac{1}{\pi} \int_0^\infty y e^{-\frac{z^2+1}{2} y^2} dy \\ &= \frac{1}{\pi} \frac{1}{1+z^2}, z > 0 \end{aligned}$$

从而 $\frac{\xi}{\eta} \sim C(0, 1)$. 对 $\xi^2 + \eta^2$, 先计算分布函数, 有

$$\begin{aligned} F_{\xi^2+\eta^2}(z) &= \iint_{x^2+y^2 < z} p_\xi(x) p_\eta(y) dx dy = \frac{1}{2\pi} \iint_{x^2+y^2 < z} e^{-\frac{1}{2}(x^2+y^2)} dy dx \\ &= \frac{1}{2\pi} \int_0^{\sqrt{z}} \int_0^{2\pi} e^{-\frac{r^2}{2}} r d\theta dr = \int_0^{\sqrt{z}} r e^{-\frac{r^2}{2}} dr \end{aligned}$$

于是求导可得密度函数

$$p_{\xi^2+\eta^2}(z) = \frac{1}{2} e^{-\frac{z}{2}}$$

从而 $\xi^2 + \eta^2 \sim \Gamma(\frac{1}{2}, 1)$. 最后验证独立性, 设 $\zeta = \frac{\xi}{\eta}, \varphi = \xi^2 + \eta^2$, 计算联合分布有

$$\begin{aligned} F_{(\zeta, \varphi)}(z, w) &= P\left(\frac{\xi}{\eta} < z, \xi^2 + \eta^2 < w\right) = \iint_{\frac{y}{x} < z, x^2+y^2 < w} \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)} dy dx \\ &= \frac{1}{2\pi} \int_0^{\sqrt{w}} \int_0^{\arctan z} e^{-\frac{1}{2}r^2} r d\theta dr + \frac{1}{2\pi} \int_0^{\sqrt{w}} \int_\pi^{\pi+\arctan z} r e^{-\frac{1}{2}r^2} d\theta dr \\ &= \frac{\arctan z}{\pi} \int_0^{\sqrt{w}} r e^{-\frac{1}{2}r^2} dr \end{aligned}$$

于是求导可得联合密度函数

$$p_{(\zeta, \varphi)}(z, w) = \frac{\partial^2 F}{\partial z \partial w} = \frac{e^{-\frac{1}{2}w}}{2\pi(1+z^2)} = p_\zeta(z) p_\varphi(w)$$

从而 $\frac{\xi}{\eta}$ 与 $\xi^2 + \eta^2$ 独立.

L-Problem 3

Assume that $\{\xi_n\}$ are iid r.v with $\xi_1 \sim N(0, 1)$, $\{\eta_k\}$ are iid r.v. with $\eta_1 \sim U_{[0,1]}$, $\{\zeta_k\}$ are iid r.v. with $\zeta_1 \sim \exp(1)$. Define

$$A = \frac{\xi_1}{\xi_2} + \frac{\xi_3}{\xi_4} + \dots + \frac{\xi_{2n-1}}{\xi_{2n}}; B = \frac{\xi_1\xi_2 + \xi_3\xi_4 + \xi_5\xi_6}{\sqrt{\xi_1^2 + \xi_3^2 + \xi_5^2}}; C = \min_{1 \leq k \leq n} \{\eta_k\};$$

$$D = \max_{1 \leq k \leq n} \{\eta_k\}; E = \frac{\zeta_1 + \dots + \zeta_m}{\zeta_1 + \dots + \zeta_n} (m < n); G = \frac{(n-m)E}{m(1-E)}$$

计算下列问题:

- (1) A 和 B 的概率分布函数;
- (2) $(\eta_1, C), (\eta_1, D), (C, D)$ 的概率分布函数或密度函数;
- (3) $P(\omega : \eta_1 = C), P(\omega : \eta_1 = D), F_{\eta_1|D=y}(x)$;
- (4) E, G 的密度函数.

Solution:

(1) 由 Problem 2, 可知每个 $\varphi_n := \frac{\xi_{2n-1}}{\xi_{2n}} \sim C(1, 0)$, 并且整体独立. 我们先证明: 两个独立的服从 $C(1, 0)$ 的分布相加后服从 $C(2, 0)$, 直接计算其密度函数即有

$$p_{\varphi_1+\varphi_2}(z) = \frac{1}{\pi^2} \int_{\mathbb{R}} \frac{1}{(1+x^2)(1+(z-x)^2)} dx = \frac{2}{\pi(z^2+4)}$$

故 $\varphi_1 + \varphi_2 \sim C(2, 0)$, 同理若 $\varphi_n \sim C(n, 0), \varphi_m \sim C(m, 0)$ 且二者独立, 则

$$\varphi_n + \varphi_m \sim C(n+m, 0)$$

回到本题, 可知 $\varphi_1 + \varphi_2 \sim C(2, 0)$, 再由 $\varphi_1 + \varphi_2$ 与 φ_3 独立知 $\varphi_1 + \varphi_2 + \varphi_3 \sim C(3, 0)$, 递推下去, 即知 $A = \frac{\xi_1}{\xi_2} + \frac{\xi_3}{\xi_4} + \dots + \frac{\xi_{2n-1}}{\xi_{2n}} = \varphi_1 + \dots + \varphi_n \sim C(n, 0)$.

对于 B , 利用课上所讲的方法即知

$$\begin{aligned} F_B(z) &= \iiint_{\mathbb{R}^3} P\left(\frac{x_1\xi_2 + x_3\xi_4 + x_5\xi_6}{\sqrt{x_1^2 + x_3^2 + x_5^2}} < z \middle| \xi_{1,3,5} = x_{1,3,5}\right) dF_{(\xi_1, \xi_3, \xi_5)}(x_1, x_3, x_5) \\ &= \iiint_{\mathbb{R}^3} P\left(\frac{x_1\xi_2 + x_3\xi_4 + x_5\xi_6}{\sqrt{x_1^2 + x_3^2 + x_5^2}} < z\right) dF_{(\xi_1, \xi_3, \xi_5)}(x_1, x_3, x_5) \\ &= \iiint_{\mathbb{R}^3} \Phi(z) dF_{(\xi_1, \xi_3, \xi_5)}(x_1, x_3, x_5) = \Phi(z) \end{aligned}$$

于是 $B \sim N(0, 1)$.

(2) 设 $C_1 = (\eta_1, C)$. 易知 $x, y > 0$ 时 $F_{C_1}(x, y) > 0$. $x, y > 1$ 时 $F_{C_1}(x, y) = 1$. 而 $F_{C_1}(x, y)$ 即事件 $\{\eta_1 < x, \min \eta_i < y\}$ 的概率. 若 $y \geq x$, 则该事件等价于 $\{\eta_1 < x\}$, 从而 $F_{C_1}(x, y) = F_{\eta_1}(x)$. 若 $y < x$, 则有

$$\begin{aligned} P\left\{\eta_1 < x, \min_{1 \leq k \leq n} \eta_k < y\right\} &= P(\eta_1 < y) + P\left(\min_{2 \leq k \leq n} \eta_k < y\right)P(y < \eta_1 < x) \\ &= F_{\eta_1}(y) + P\left(\left(\bigcap_{k=2}^n \{\eta_k \geq y\}\right)^c\right)(F_{\eta_1}(x) - F_{\eta_1}(y)) \\ &= F_{\eta_1}(y) + (F_{\eta_1}(x) - F_{\eta_1}(y))\left(1 - \prod_{k=2}^n (1 - F_{\eta_k}(y))\right) \end{aligned}$$

综上, 可以写出

$$F_{C_1}(x, y) = \begin{cases} 1, & x, y > 1 \\ x, & 0 < x \leq 1, y \geq x \\ x - (x - y)(1 - y)^{n-1}, & 0 < y < x \leq 1 \\ 1 - (1 - y)^n, & 0 < y < 1, x > 1 \end{cases}$$

设 $D_1 = (\eta_1, D)$. 易知 $x, y > 0$ 时 $F_{D_1}(x, y) > 0$. $x, y > 1$ 时 $F_{D_1}(x, y) = 1$. 而 $F_{D_1}(x, y)$ 即事件 $\{\eta_1 < x, \max \eta_i < y\}$ 的概率. 若 $y \leq x$, 则概率等于 $F_D(y)$, 若 $y > x$, 则

$$\begin{aligned} &P\left(\eta_1 < x, \max_{1 \leq k \leq n} \eta_k < y\right) \\ &= P\left(\max_{1 \leq k \leq n} \eta_k < x\right) + P(\eta_1 < x)P\left(x \leq \max_{2 \leq i \leq k} \eta_k < y\right) \\ &= P\left(\bigcap_{k=1}^n \{\eta_k < x\}\right) + F_{\eta_1}(x)\left(P\left(\bigcap_{k=2}^n \{\eta_k < y\}\right) - P\left(\bigcap_{k=2}^n \{\eta_k < x\}\right)\right) \\ &= \prod_{k=1}^n F_{\eta_k}(x) + F_{\eta_1}(x)\left(\prod_{k=2}^n F_{\eta_k}(y) - \prod_{k=2}^n F_{\eta_k}(x)\right) \end{aligned}$$

综上, 可以写出

$$F_{D_1}(x, y) = \begin{cases} 1, & x, y \geq 1 \\ y^n, & 0 < y < 1, y \leq x \\ xy^{n-1}, & 0 < x < y < 1 \\ x, & 0 < x < 1, y > 1 \end{cases}$$

对于 (C, D) , 由定理 2.7.6 即知其密度函数为

$$p_{(C,D)}(x, y) = n(n-1)[F(y) - F(x)]^{n-2}p(x)p(y), x < y$$

(3) 我们有

$$\begin{aligned}
 P(\omega : \eta_1 = C) &= P(\omega : \eta_1 \leq \eta_2, \dots, \eta_1 \leq \eta_n) \\
 &= \int_0^1 \int_{x_1}^1 \cdots \int_{x_1}^1 p_{\eta_1}(x_1) \cdots p_{\eta_n}(x_n) dx_n \cdots dx_1 \\
 &= \int_0^1 \left(\int_{x_1}^1 1 dx \right)^{n-1} dx_1 = \int_0^1 (1 - x_1)^{n-1} dx_1 \\
 &= \frac{1}{n} \\
 P(\omega : \eta_1 = D) &= P(\omega : \eta_1 \geq \eta_2, \dots, \eta_1 \geq \eta_n) \\
 &= \int_0^1 \int_0^{x_1} \cdots \int_0^{x_1} 1 dx_n \cdots dx_1 \\
 &= \int_0^1 x_1^{n-1} dx_1 \\
 &= \frac{1}{n}
 \end{aligned}$$

对于 $F_{\eta_1|D=y}(x)$, 其只在 $y \in [0, 1]$ 时有定义. 若 $x \geq 1$, 则显然 $F_{\eta_1|D=y}(x) = 1$. 考虑 $0 < x, y < 1$ 的情况, 若 $x \geq y$ 时, 显然 $F_{\eta_1|D=y}(x) = 1$, 故考虑 $0 < x < y < 1$ 的情况, 此时由 (2) 的分布函数可知

$$\begin{aligned}
 p(x, y) &= \frac{\partial^2 F_{D_1}}{\partial x \partial y} = (n-1)y^{n-2}, 0 < x < y < 1 \\
 p(x, y) &= 0, 0 < y < 1, y < x; \\
 p_D(y) &= \int_0^\infty p(x, y) dx = \int_0^y (n-1)y^{n-2} dx = (n-1)y^{n-1}, 0 < y < 1
 \end{aligned}$$

从而在 $0 < x < y < 1$ 时有

$$F_{\eta_1|D=y}(x) = \frac{\int_0^x (n-1)y^{n-2} ds}{p_D(y)} = \frac{x}{y}$$

(4) 记 $\alpha = \zeta_1 + \cdots + \zeta_m, \beta = \zeta_{m+1} + \cdots + \zeta_n$, 由 2.7.8.8 即知 $\alpha \sim \Gamma(1, m), \beta \sim \Gamma(1, n-m)$, α 与 β 独立并且 $E = \frac{\alpha}{\alpha+\beta}$, 于是可以计算

$$F_E(z) = \iint_{\frac{x}{x+y} < z} p_\alpha(x) p_\beta(y) dy dx = \int_0^\infty \int_{(\frac{1}{z}-1)x}^\infty \frac{x^{m-1} e^{-x}}{(m-1)!} \frac{y^{n-1} e^{-y}}{(n-1)!} dy dx$$

求导可得

$$\begin{aligned}
p_E(z) = F'(z) &= \int_0^\infty \frac{x}{z^2} \frac{x^{m-1} e^{-x}}{(m-1)!} \frac{\left(\left(\frac{1}{z}-1\right)x\right)^{n-1} e^{-\left(\frac{1}{z}-1\right)x}}{(n-1)!} dx \\
&= \frac{\left(\frac{1}{z}-1\right)^{n-1}}{z^2(m-1)!(n-1)!} \int_0^\infty x^{n+m-1} e^{-\frac{x}{z}} dx \\
&= \frac{(1-z)^{n-1}}{z^{n+1}(m-1)!(n-1)!} z^{n+m} \Gamma(n+m) \\
&= \frac{z^{m-1}(1-z)^{n-1}}{B(n,m)}, z > 0
\end{aligned}$$

从而 $E \sim \text{Beta}(n, m)$, 且 $E \in (0, 1)$. 对于 G , 显然有

$$\begin{aligned}
F_G(z) &= P\left(\frac{(n-m)E}{m(1-E)} < z\right) = P\left(\frac{E}{1-E} < \frac{zm}{n-m}\right) \\
&= \begin{cases} P\left(0 < E < \frac{mz}{mz+n-m}\right), & z \geq 0 \\ 0, & z < 0 \end{cases} = \begin{cases} F_E\left(\frac{mz}{mz+n-m}\right) - F_E(0), & z \geq 0 \\ 0, & z < 0 \end{cases}
\end{aligned}$$

从而求导得

$$p_G(z) = \begin{cases} \frac{m(n-m)}{(mz+n-m)^2} p_E\left(\frac{mz}{mz+n-m}\right), & z \geq 0 \\ 0, & z < 0 \end{cases}$$

L-Problem 4

Assume that $\{\xi_k\}$ are iid and $\xi_1 \sim U_{[0,1]}$, 计算 $(\xi_1 \xi_2)^{\xi_3}$ 的分布.

Solution: 易知其取值范围是 $(0, 1)$, 计算其分布函数, 有

$$\begin{aligned}
F_{(\xi_1 \xi_2)^{\xi_3}}(w) &= \iiint_{(xy)^z < w} p(x, y, z) dz dy dx \\
&\stackrel{x \leftarrow e^x, y \leftarrow e^y}{=} \iiint_{e^{z(x+y)} < w} e^{x+y} dx dy dz = \int_0^1 F(z, w) dz
\end{aligned}$$

其中 $F(x, w)$ 为区域 $\{(x, y) : x + y < \frac{\log w}{z}, x < 0, y < 0\}$ 上的积分, 即

$$\begin{aligned}
F(z, w) &= \int_{-\infty}^{\frac{\log w}{z}} \int_{-\infty}^0 e^{x+y} dx dy + \int_{\frac{\log w}{z}}^0 \int_{-\infty}^{\frac{\log w}{z}-y} e^{x+y} dx dy \\
&= e^{\frac{\log w}{z}} - \frac{\log w}{z} e^{\frac{\log w}{z}}
\end{aligned}$$

于是

$$F(w) = \int_0^1 \left(1 - \frac{\log w}{z}\right) e^{\frac{\log w}{z}} dz \stackrel{t=\frac{\log w}{z}}{=} -\log w \int_{-\infty}^{\log w} \left(\frac{1}{t^2} - \frac{1}{t}\right) e^t dt$$

从而由分部积分知

$$\int_{-\infty}^a \frac{e^t}{t} dt = \frac{e^t}{t} \Big|_{-\infty}^a + \int_{-\infty}^a \frac{e^t}{t^2} dt \Rightarrow \int_{-\infty}^a \left(\frac{1}{t^2} - \frac{1}{t}\right) e^t dt = -\frac{e^a}{a}$$

于是

$$F(w) - \log w \left(-\frac{e^{\log w}}{\log w} \right) = w$$

从而 $(\xi_1 \xi_2)^{\xi_3} \sim U_{[0,1]}$.

K-Problem 3.2.2

If $E(|X|) < \infty$ and $\lim_{n \rightarrow \infty} \mathcal{P}(\Lambda_n) = 0$, then $\lim_{n \rightarrow \infty} \int_{\Lambda_n} X \, d\mathcal{P} = 0$. In particular

$$\lim_{n \rightarrow \infty} \int_{\{|X| > n\}} X \, d\mathcal{P} = 0$$

Proof: $\lim_n \int_{\Lambda_n} X \, d\mathcal{P} = 0$ 的证明见 3.2.8. 特别地, 取 $\Lambda_n = \{\omega : |X| > n\}$, 则由 $E(|X|) < \infty$ 即知 $\lim_{n \rightarrow \infty} \mathcal{P}(\Lambda_n) = 0$, 于是 $\lim_{n \rightarrow \infty} \int_{\{|X| > n\}} X \, d\mathcal{P} = 0$.

K-Problem 3.2.4

Let c be a fixed constant, $c > 0$. Then $E(|X|) < \infty$ if and only if

$$\sum_{n=1}^{\infty} \mathcal{P}(|X| \geq cn) < \infty$$

In particular, if the series converges for one value of c , it converges for all values of c .

Proof: 记 $\Lambda_n = \{nc \leq |X| < (n+1)c\}$, $n = 0, 1, \dots$, 于是

$$E(|X|) = \sum_{n=0}^{\infty} \int_{\Lambda_n} |X| \, d\mathcal{P} \Rightarrow c \sum_{n=0}^{\infty} n\mathcal{P}(\Lambda_n) \leq E(|X|) \leq c + c \sum_{n=0}^{\infty} n\mathcal{P}(\Lambda_n)$$

由 Abel 求和公式可知

$$\begin{aligned} \sum_{n=0}^N n\mathcal{P}(\Lambda_n) &= \sum_{n=1}^N \mathcal{P}(|X| \geq nc) - N\mathcal{P}(|X| \geq (N+1)c) \\ \Rightarrow \sum_{n=0}^N n\mathcal{P}(\Lambda_n) &\leq \sum_{n=1}^N \mathcal{P}(|X| \geq nc) = \sum_{n=0}^N n\mathcal{P}(\Lambda_n) + \frac{1}{c} \int_{\{|X| \geq (N+1)c\}} X \, d\mathcal{P} \end{aligned}$$

于是若 $E(|X|) < \infty$, 则 $\sum_{n=0}^{\infty} n\mathcal{P}(\Lambda_n) < \infty$, 且在上式中令 $N \rightarrow \infty$ 即得

$$\sum_{n=0}^{\infty} n\mathcal{P}(\Lambda_n) = \sum_{n=1}^{\infty} \mathcal{P}(|X| \geq nc)$$

若 $E(|X|) = \infty$, 则上述两级数均为 ∞ , 仍然相等, 于是可得

$$c \sum_{n=0}^{\infty} \mathcal{P}(|X| \geq nc) \leq E(|X|) \leq c + c \sum_{n=0}^{\infty} \mathcal{P}(|X| \geq nc)$$

即定理 3.2.1 的一般形式, 由上式立得题目成立.

L-Problem 3.2.6

Suppose that $\sup_n |X_n| \leq Y$ on Λ with $\int_{\Lambda} Y d\mathcal{P} < \infty$, Deduce from Fatou's Lemma:

$$\int_{\Lambda} \left(\overline{\lim}_{n \rightarrow \infty} X_n \right) d\mathcal{P} \geq \overline{\lim}_{n \rightarrow \infty} \int_{\Lambda} X_n d\mathcal{P}$$

Show this false if the condition involving Y is omitted.

Proof: $Y - X_n$ 满足 Fatou 引理的条件, 代入即知

$$\int_{\Lambda} Y d\mathcal{P} - \int_{\Lambda} \overline{\lim}_n X_n d\mathcal{P} \leq \int_{\Lambda} Y d\mathcal{P} - \overline{\lim}_n \int_{\Lambda} X_n d\mathcal{P}$$

于是由 $\int_{\Lambda} Y d\mathcal{P} < \infty$ 即知

$$\int_{\Lambda} \overline{\lim}_n X_n d\mathcal{P} \geq \overline{\lim}_n \int_{\Lambda} X_n d\mathcal{P}$$

若取消对控制函数的限制, 可取概率空间为 $(\Omega, \mathcal{F}, \mathcal{P}) = ([0, 1], \mathcal{B}([0, 1]), m)$, 其中 $\mathcal{P} = m$ 为 Lebesgue 测度. 取 $\Lambda = \Omega$, $X_n = n \cdot 1_{(0, \frac{1}{n})}$, 于是 $\overline{\lim}_n X_n = 0$, 且

$$\int_{\Lambda} X_n d\mathcal{P} = 1 \implies 0 = \int_{\Lambda} \overline{\lim}_{n \rightarrow \infty} X_n d\mathcal{P} < \overline{\lim}_{n \rightarrow \infty} \int_{\Lambda} X_n d\mathcal{P} = 1$$

即此时上述结论不再成立.

K-Problem 3.2.8

For any two sets Λ_1 and Λ_2 in $\mathcal{F}$, define

$$\rho(\Lambda_1, \Lambda_2) = \mathcal{P}(\Lambda_1 \Delta \Lambda_2)$$

then ρ is a pseudo-metric in the space of sets in $\mathcal{F}$; call the resulting metric space $M(\mathcal{F}, \mathcal{P})$. Prove that for each integrable r.v. X , the mapping of $M(\mathcal{F}, \mathcal{P})$ to $\mathbb{R}^1$ given $\Lambda \rightarrow \int_{\Lambda} X d\mathcal{P}$ is continuous. Similarly, the mappings on $M(\mathcal{F}, \mathcal{P}) \times M(\mathcal{F}, \mathcal{P})$ to $M(\mathcal{F}, \mathcal{P})$ given by

$$(\Lambda_1, \Lambda_2) \rightarrow \Lambda_1 \cup \Lambda_2, \Lambda_1 \cap \Lambda_2, \Lambda_1 \setminus \Lambda_2, \Lambda_1 \Delta \Lambda_2$$

are all continuous. If (see Sec. 4.2 below)

$$\overline{\lim}_n \Lambda_n = \underline{\lim}_n \Lambda_n$$

modulo a null set, we denote the common equivalence class of these two sets by $\lim_n \Lambda_n$. Prove that in this case $\{\Lambda_n\}$ converges to $\lim_n \Lambda_n$ in the metric ρ . Deduce Exercise 2 above as a special case.

Proof: 先验证 ρ 是伪度量: 对称性和规范性显然, 三角不等式验证如下:

$$\rho(\Lambda_1, \Lambda_2) \leq \mathcal{P}((\Lambda_1 \Delta \Lambda_3) \cup (\Lambda_3 \Delta \Lambda_2)) \leq \rho(\Lambda_1, \Lambda_3) + \rho(\Lambda_3, \Lambda_2)$$

验证积分映射的连续性: 固定一个 Λ_1 , 则对任意 $\Lambda_2 \in \mathcal{F}$, 有

$$\left| \int_{\Lambda_1} X d\mathcal{P} - \int_{\Lambda_2} X d\mathcal{P} \right| = \left| \int_{\Lambda_1 \setminus \Lambda_2} X d\mathcal{P} - \int_{\Lambda_2 \setminus \Lambda_1} X d\mathcal{P} \right| \leq \int_{\Lambda_1 \Delta \Lambda_2} |X| d\mathcal{P}$$

由于 X 可积, 由积分的绝对连续性即知对任意 $\varepsilon > 0$, 存在 $\delta > 0$, 使得 $\mathcal{P}(\Lambda_1 \Delta \Lambda_2) < \delta$ 时, 上述积分小于 ε , 于是可知映射连续.

验证集合运算的连续性: 设 $\Lambda_1, \Lambda_2, \Lambda'_1, \Lambda'_2 \in \mathcal{F}$, 有

$$\begin{aligned} \rho(\Lambda_1 \cup \Lambda_2, \Lambda'_1 \cup \Lambda'_2) &= \mathcal{P}((\Lambda_1 \cup \Lambda_2) \Delta (\Lambda'_1 \cup \Lambda'_2)) \leq \mathcal{P}((\Lambda_1 \Delta \Lambda'_1) \cup (\Lambda_2 \Delta \Lambda'_2)) \\ &\leq \mathcal{P}(\Lambda_1 \Delta \Lambda'_1) + \mathcal{P}(\Lambda_2 \Delta \Lambda'_2) = \rho(\Lambda_1, \Lambda'_1) + \rho(\Lambda_2, \Lambda'_2) \end{aligned}$$

$$\begin{aligned} \rho(\Lambda_1 \cap \Lambda_2, \Lambda'_1 \cap \Lambda'_2) &= \mathcal{P}((\Lambda_1 \cap \Lambda_2) \Delta (\Lambda'_1 \cap \Lambda'_2)) \leq \mathcal{P}((\Lambda_1 \Delta \Lambda'_1) \cup (\Lambda_2 \Delta \Lambda'_2)) \\ &\leq \mathcal{P}(\Lambda_1 \Delta \Lambda'_1) + \mathcal{P}(\Lambda_2 \Delta \Lambda'_2) = \rho(\Lambda_1, \Lambda'_1) + \rho(\Lambda_2, \Lambda'_2) \end{aligned}$$

$$\begin{aligned} \rho(\Lambda_1 \setminus \Lambda_2, \Lambda'_1 \setminus \Lambda'_2) &= \mathcal{P}((\Lambda_1 \setminus \Lambda_2) \Delta (\Lambda'_1 \setminus \Lambda'_2)) \leq \mathcal{P}((\Lambda_1 \Delta \Lambda'_1) \cup (\Lambda_2 \Delta \Lambda'_2)) \\ &\leq \mathcal{P}(\Lambda_1 \Delta \Lambda'_1) + \mathcal{P}(\Lambda_2 \Delta \Lambda'_2) = \rho(\Lambda_1, \Lambda'_1) + \rho(\Lambda_2, \Lambda'_2) \end{aligned}$$

$$\begin{aligned} \rho(\Lambda_1 \Delta \Lambda_2, \Lambda'_1 \Delta \Lambda'_2) &= \mathcal{P}((\Lambda_1 \Delta \Lambda_2) \Delta (\Lambda'_1 \Delta \Lambda'_2)) \leq \mathcal{P}((\Lambda_1 \Delta \Lambda'_1) \cup (\Lambda_2 \Delta \Lambda'_2)) \\ &\leq \mathcal{P}(\Lambda_1 \Delta \Lambda'_1) + \mathcal{P}(\Lambda_2 \Delta \Lambda'_2) = \rho(\Lambda_1, \Lambda'_1) + \rho(\Lambda_2, \Lambda'_2) \end{aligned}$$

综上即知这四个集合运算均连续.

再验证收敛性: 存在零测集 N 使 $\overline{\lim}_n \Lambda_n \setminus N = \underline{\lim}_n \Lambda_n \setminus N$, 记 $\Lambda = \lim_n \Lambda_n$, 先证:

$$1_{\Lambda_n} \xrightarrow{a.e.} 1_{\Lambda}$$

对 $\omega \notin N$, 由集合上下极限的等价定义, 若 $\omega \in \Lambda$, 则 ω 最终属于所有 Λ_n . 即 $1_{\Lambda_n}(\omega) \rightarrow 1 = 1_{\Lambda}(\omega)$; 若 $\omega \notin \Lambda$, 则 ω 至多属于有限个 Λ_n , 即 $1_{\Lambda_n}(\omega) \rightarrow 0 = 1_{\Lambda}(\omega)$. 这就验证了上式成立. 另一方面, 我们有

$$\rho(\Lambda_n, \Lambda) = \mathcal{P}(\Lambda_n \Delta \Lambda) = \int_{\Omega} 1_{\Lambda_n \Delta \Lambda} d\mathcal{P} = \int_{\Omega} |1_{\Lambda_n} - 1_{\Lambda}| d\mathcal{P}$$

易见 $|1_{\Lambda_n} - 1_{\Lambda}| < 2$ 且 $\int_{\Omega} 2 d\mathcal{P} = 2 < \infty$, 因此由控制收敛定理知

$$\lim_{n \rightarrow \infty} \rho(\Lambda_n, \Lambda) = \lim_{n \rightarrow \infty} \int_{\Omega} |1_{\Lambda_n} - 1_{\Lambda}| d\mathcal{P} = \int_{\Omega} \lim_{n \rightarrow \infty} |1_{\Lambda_n} - 1_{\Lambda}| d\mathcal{P} = 0 \implies \Lambda_n \xrightarrow{\rho} \Lambda$$

对 3.2.2, X 可积且 $\Lambda \xrightarrow{\rho} \emptyset$, 由积分映射的连续性立得 $\int_{\Lambda_n} X d\mathcal{P} \rightarrow \int_{\emptyset} X d\mathcal{P} = 0$.

K-Problem 3.1.6.5

设 ξ 为非负整值随机变量, 其数学期望存在, 求证: $E(\xi) = \sum_{i=1}^{\infty} P(\xi \geq i)$

证明: 由于 ξ 只取整值, 于是 $\xi|_{\Lambda_n} = n$. 利用钟开莱的定理 3.2.1 即知

$$E(\xi) = \sum_{n=0}^{\infty} \int_{\Lambda_n} \xi \, dP = \sum_{n=0}^{\infty} nP(\Lambda_n) = \sum_{n=1}^{\infty} P(\xi \geq n)$$

K-Problem 3.1.6.6

设 $F(x)$ 为非负随机变量的分布函数, 试证对任意 $s > 0$ 有

$$\int_0^{\infty} x^s \, dF(x) = \int_0^{\infty} s x^{s-1} (1 - F(x)) \, dx$$

证明: 利用二重积分, 我们有

$$\begin{aligned} \int_0^{\infty} s x^{s-1} (1 - F(x)) \, dx &= \int_0^{\infty} s y^{s-1} \int_y^{\infty} dF(x) \, dy \\ &= \int_0^{\infty} \int_0^x s y^{s-1} \, dy \, dF(x) = \int_0^{\infty} x^s \, dF(x) \end{aligned}$$

K-Problem 3.1.6.7

设随机变量 ξ 的分布函数为 $F(x)$, 且 ξ 的期望存在, 求证:

$$E(\xi) = \int_0^{\infty} (1 - F(x)) \, dx - \int_{-\infty}^0 F(x) \, dx$$

证明: 利用上一题的过程即知

$$\int_0^{\infty} (1 - F(x)) \, dx = \int_0^{\infty} x \, dF(x)$$

同理有

$$\int_{-\infty}^0 F(x) \, dx = \int_{-\infty}^0 \int_{-\infty}^y dF(x) \, dy = \int_{-\infty}^0 \int_x^0 dy \, dF(x) = - \int_{-\infty}^0 x \, dF(x)$$

于是

$$E(\xi) = \int_0^{\infty} x \, dF(x) + \int_{-\infty}^0 x \, dF(x) = \int_0^{\infty} (1 - F(x)) \, dx - \int_{-\infty}^0 F(x) \, dx$$

L-Problem 1

Assume that ξ and η are iid r.v.'s, proof that

$$P(|\xi + \eta| \leq 1) \leq 3P(|\xi - \eta| \leq 1)$$

Proof: 我们证明更强的: $P(|\xi + \eta| < 1) \leq 2P(|\xi - \eta| < 1)$.

Lemma 1: 设 $x_1, \dots, x_n \in \mathbb{R}$, 定义 $F(A) = \{i \mid x_i \in A\}$ 以及 $f(A) = |F(A)|$, 则

$$\exists k \in [1, n], \text{ s.t. } f[-x_k - 1, -x_k + 1] < 2f[x_k - 1, x_k + 1]$$

Proof: 令 $M = \max_{1 \leq k \leq n} f[x_k - 1, x_k + 1] > 0$, k 为使得 $f[x_k - 1, x_k + 1] = M$ 且让 $|x_k|$ 尽可能大的数. 于是有

$$f[-x_k - 1, -x_k + 1] \leq f[-x_k - 1, -x_k] + f[-x_k, -x_k + 1] \leq 2M$$

其中第二个不等号是因为若 $[-x_k - 1, -x_k]$ 中包含 x_j , 则 $F[x_j - 1, x_j + 1]$ 会包含 $F[-x_k - 1, -x_k]$, 于是 $f([-x_k - 1, -x_k]) \leq f([x_j - 1, x_j + 1]) \leq M$. 故只需证明上式的等号无法取到. 若取到, 则 $f[-x_k - 1, -x_k] = M$, 于是

- 若这 M 个数均等于 $-x_k$, 于是由 $f[-x_k, -x_k + 1] \leq M$, 即知 $F[-x_k, -x_k + 1]$ 中也只能是这 M 个 $-x_k$, 从而 $F[-x_k - 1, -x_k + 1]$ 中也是这 M 个 x_k , 于是

$$f[-x_k - 1, -x_k + 1] = M < 2M$$

- 存在某个 $-x_k - 1 \leq x_j < -x_k$, 则 $f[x_j - 1, x_j + 1] \geq f[-x_k - 1, -x_k] = M$ 且 $|x_j| > |x_k|$, 与 k 的定义矛盾.

综上知引理 1 成立.

Lemma 2: 设 $x_1, x_2, \dots, x_n \in \mathbb{R}$, 定义

$$A = \{(i, j) \mid |x_i + x_j| < 1\}, B = \{(i, j) \mid |x_i - x_j| < 1\}$$

则 $|A| < 2|B|$

Proof: 归纳, 当 $n = 1$ 时有 $|A| \leq 1 < 2 = 2|B|$. 设命题对 $n - 1$ 成立, 则在 n 处, 按照引理 1 取 k 使得 $a := f[-x_k - 1, -x_k + 1] < 2f[x_k - 1, x_k + 1] =: 2b$. 考察 x_k 对 $|A|, |B|$ 的贡献: x_k 对 $|A|$ 的贡献不超过 $2a$, 对 $|B|$ 的贡献为 $2b - 1$. 由 $a \leq 2b - 1$, 故 $2a \leq 2(2b - 1)$, 剩余部分由归纳假设知 $|A'| < 2|B'|$, 归纳即知引理 2 成立.

最后回到原题. 对该分布独立采样 n 次, 得到独立同分布的 $X_1, \dots, X_n$, 按引理 2 定义随机变量 $\mathcal{A}, \mathcal{B}$, 则

$$E(\mathcal{A}) = nP\left(|\xi| \leq \frac{1}{2}\right) + n(n-1)P(|\xi + \eta| \leq 1)$$

$$E(\mathcal{B}) = n + n(n-1)P(|\xi - \eta| \leq 1)$$

由于 $|A| < 2|B|$, 由期望的线性性即知 $E(\mathcal{A}) \leq 2E(\mathcal{B})$, 从而

$$nP\left(|\xi| \leq \frac{1}{2}\right) + n(n-1)P(|\xi + \eta| \leq 1) \leq 2n + 2n(n-1)P(|\xi - \eta| \leq 1)$$

两边同除 $n(n-1)$, 令 $n \rightarrow \infty$ 即得 $P(|\xi + \eta| \leq 1) \leq 2P(|\xi - \eta| \leq 1)$.

注: 本做法来自校答疑坊同学, 在此对他表示衷心的感谢!

L-Problem 2

Assume that $S = \sum_{k=1}^N X_k$, where N and $\{X_k, k \geq 1\}$ are independent, $N \sim G(p)$, $\{X_k, k \geq 1\}$ are iid r.v.'s with $X_1 \sim \text{Exp}(1)$. Find the distribution of S .

Solution: 由上一次作业的结论可知 $X_1 + \dots + X_N \sim \Gamma(1, N)$, 由全概率公式

$$\begin{aligned} F_S(x) &= \sum_{k=1}^{\infty} P(X_1 + \dots + X_N < x | N = k) P(N = k) \\ &= \sum_{k=1}^{\infty} \left((1-p)^{k-1} p \int_{-\infty}^x \frac{t^{k-1} e^{-t}}{(k-1)!} dt \right) \\ &= p \sum_{k=1}^{\infty} \int_{-\infty}^x \frac{((1-p)t)^{k-1} e^{-t}}{(k-1)!} dt \\ &= p \int_{-\infty}^x \sum_{k=1}^{\infty} \frac{((1-p)t)^{k-1} e^{-t}}{(k-1)!} dt \\ &= p \int_{-\infty}^x e^{-pt} dt = 1 - e^{-px} \end{aligned}$$

由幂级数的性质易知上述函数项级数在 $(-\infty, x]$ 上内闭一致收敛, 从而上述积分与求和换序是合法的, 由此即知 $S \sim \text{Exp}(p)$.

L-Problem 3

Take two point ξ and η randomly and independently with respect to the uniform distribution from the interval $[0, 1]$. Then the two points divide the $[0, 1]$ into three subintervals with lengths X, Y and Z , respectively. Calculate:

- (1) the probability that X, Y, Z constitute the lengths of three sides of a triangle;
- (2) the pdf of X, Y, Z .

Solution:

(1) 由三角不等式即知 X, Y, Z 三边构成三角形当且仅当 $\max(X, Y, Z) \geq \frac{1}{2}$, 从而若 ξ, η 落在 $\frac{1}{2}$ 同侧, 则必然无法构成三角形, 否则二者落在 $\frac{1}{2}$ 的两侧, 此时能构成三角形当且仅当 $|\xi - \eta| < \frac{1}{2}$, 综上即知

$$P = \int_0^{\frac{1}{2}} \int_{\frac{1}{2}}^{\xi + \frac{1}{2}} d\eta d\xi + \int_{\frac{1}{2}}^1 \int_{\xi - \frac{1}{2}}^{\frac{1}{2}} d\eta d\xi = \frac{1}{4}$$

(2) 易知

$$P(X < t) = P(\{\xi < t\} \cup \{\eta < t\}) = 1 - (1 - t)^2 = 2t - t^2$$

$$P(Y < t) = P(|\xi - \eta| < t) = \iint_{|x-y|<t, 0<x,y<1} dx dy = 2t - t^2$$

$$P(Z < t) = P(\{\xi > 1 - t\} \cup \{\eta > 1 - t\}) = 1 - (1 - t)^2 = 2t - t^2$$

其中 $t \in (0, 1)$.

Y-Problem 3.1.6.3

试求 Γ 分布 $\Gamma(\lambda, r)$ 的数学期望与方差.

解: 由积分变换定理, 我们有

$$\begin{aligned} E(\xi) &= \int_0^\infty x \cdot \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x} dx = \frac{\lambda^r}{\Gamma(r)} \frac{1}{\lambda^{r+1}} \int_0^\infty x^r e^{-x} dx = \frac{\Gamma(r+1)}{\Gamma(r)\lambda} = \frac{r}{\lambda} \\ D(\xi) &= \int_0^\infty \left(x - \frac{r}{\lambda}\right)^2 \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x} dx \\ &= \frac{\lambda^r}{\Gamma(r)} \left(\int_0^\infty x^{r+1} e^{-\lambda x} dx - \frac{2r}{\lambda} \int_0^\infty x^r e^{-\lambda x} dx + \frac{r^2}{\lambda^2} \int_0^\infty x^{r-1} e^{-\lambda x} dx \right) \\ &= \frac{\lambda^r}{\Gamma(r)} \left(\frac{\Gamma(r+2)}{\lambda^{r+2}} - \frac{2r}{\lambda} \frac{\Gamma(r+1)}{\lambda^{r+1}} + \frac{r^2}{\lambda^2} \frac{\Gamma(r)}{\lambda^r} \right) = \frac{r}{\lambda^2} \end{aligned}$$

Y-Problem 3.1.6.10

设随机变量 ξ, η 独立同 $N(a, \sigma^2)$ 分布, 求证:

$$E(\xi \vee \eta) = a + \frac{\sigma}{\sqrt{\pi}}$$

证明: 利用积分变换公式, 有

$$\begin{aligned} E(\xi \vee \eta) &= \iint_{\mathbb{R}^2} \max(x, y) p_\xi(x) p_\eta(y) dx dy \\ &= \frac{1}{2\pi\sigma^2} \iint_{x < y} y e^{-\frac{(x-a)^2 + (y-a)^2}{2\sigma^2}} dx dy + \frac{1}{2\pi\sigma^2} \iint_{y < x} x e^{-\frac{(x-a)^2 + (y-a)^2}{2\sigma^2}} dx dy \\ &= \frac{1}{2\pi\sigma^2} \int_{-\infty}^\infty \int_x^\infty y e^{-\frac{x^2+y^2}{2\sigma^2}} dy dx + \frac{1}{2\pi\sigma^2} \int_{-\infty}^\infty \int_y^\infty x e^{-\frac{x^2+y^2}{2\sigma^2}} dx dy \\ &\quad + \frac{a}{2\pi\sigma^2} \iint_{\mathbb{R}^2} e^{-\frac{x^2+y^2}{2\sigma^2}} dx dy \\ &= a + \frac{1}{4\pi\sigma^2} \left(\int_{\mathbb{R}} e^{-\frac{x^2}{2\sigma^2}} \cdot 2\sigma^2 e^{-\frac{x^2}{2\sigma^2}} dx + \int_{\mathbb{R}} e^{-\frac{y^2}{2\sigma^2}} \cdot 2\sigma^2 e^{-\frac{y^2}{2\sigma^2}} dy \right) \\ &= a + \frac{1}{\pi} \left(\int_{\mathbb{R}} e^{-\frac{x^2}{\sigma^2}} dx \right) \\ &= a + \frac{\sigma}{\sqrt{\pi}} \end{aligned}$$

Y-Problem 3.1.6.11

设随机变量 ξ, η 相互独立, 服从几何分布 $G(p)$, 求 $E(\xi \vee \eta)$.

解: 先求概率密度

$$\begin{aligned} P(\xi \vee \eta = k) &= P(\xi < k)P(\eta = k) + P(\eta < k)P(\xi = k) + P(\xi = k)P(\eta = k) \\ &= 2(1 - q^{k-1})q^{k-1}p + (q^{k-1}p)^2 = p^2q^{2k-2} - 2pq^{2k-2} + 2pq^{k-1} \end{aligned}$$

我们有

$$\begin{aligned} E(\xi \vee \eta) &= \sum_{k=1}^{\infty} kP(\xi \vee \eta = k) = (p^2 - 2p) \sum_{k=1}^{\infty} kq^{2k-2} + 2p \sum_{k=1}^{\infty} kq^{k-1} \\ &= \frac{p^2 - 2p}{(1 - q^2)^2} + \frac{2p}{(1 - q)^2} \stackrel{q=1-p}{=} \frac{2p - 3}{p(p - 2)} \end{aligned}$$

Y-Problem 3.1.6.15

袋中有 r 个红球与 b 个黑球, 任意一一不放回取出, 直到取出红球, 求取球次数的数学期望与方差.

解: 设取球次数为 X , 于是

$$\begin{aligned} P(X = k) &= \frac{\binom{b}{k-1}(k-1)!r}{\binom{r+b}{k}k!} = \frac{\binom{b}{k-1}r}{\binom{r+b}{k}k} \\ E(X) &= \sum_{k=1}^{r+b} kP(X = k) = r \sum_{k=1}^{b+1} \frac{\binom{b}{k-1}}{\binom{r+b}{k}} = \frac{b!r!}{(r+b)!} \sum_{k=1}^{b+1} k \binom{r+b-k}{r-1} \end{aligned}$$

我们有

$$\begin{aligned} \sum_{k=1}^{b+1} k \binom{r+b-k}{r-1} &= (r+b+1) \sum_{k=1}^{b+1} \binom{r+b-k}{r-1} - \sum_{k=1}^{b+1} (r+b+1-k) \binom{r+b-k}{r-1} \\ &= (r+b+1) \sum_{k=1}^{b+1} \binom{r+b-k}{r-1} - r \sum_{k=1}^{b+1} \binom{r+b+1-k}{r} \\ &= (r+b+1) \sum_{k=0}^b \binom{k+r-1}{r-1} - r \sum_{k=0}^b \binom{k+r}{r} \\ &= (r+b+1) \binom{r+b}{r} - r \binom{r+1+b}{r+1} = \binom{r+1+b}{r+1} \\ \implies E(X) &= \frac{r+b+1}{r+1} \end{aligned}$$

对方差, 记 $A = (r + b + 1)^2, B = -(2r + 2b + 3)$, 则

$$\begin{aligned}
 E(X^2) &= \sum_{k=1}^{r+b} k^2 P(X = k) = r \sum_{k=1}^{b+1} k \frac{\binom{b}{k-1}}{\binom{r+b}{k}} = \frac{b!r!}{(r+b)!} \sum_{k=1}^{b+1} k^2 \binom{r+b-k}{r-1} \\
 \sum_{k=1}^{b+1} k^2 \binom{r+b-k}{r-1} &= A \sum_{k=1}^{b+1} \binom{r+b-k}{r-1} + B \sum_{k=1}^{b+1} (r+b+1-k) \binom{r+b-k}{r-1} \\
 &\quad + \sum_{k=1}^{b+1} (r+b+1-k)(r+b+2-k) \binom{r+b-k}{r-1} \\
 &= A \sum_{k=1}^{b+1} \binom{r+b-k}{r-1} + Br \sum_{k=1}^{b+1} \binom{r+b+1-k}{r} \\
 &\quad + (r+1) \sum_{k=1}^{b+1} \binom{r+b+2-k}{r+1} \\
 &= A \sum_{k=0}^b \binom{k+r-1}{r-1} + Br \sum_{k=0}^b \binom{k+r}{r} \\
 &\quad + (r+1) \sum_{k=0}^b \binom{k+r+1}{r+1} \\
 &= A \binom{r+b}{r} + Br \binom{r+1+b}{r+1} + (r+1) \binom{r+2+b}{r+2} \\
 \Rightarrow D(X) &= E(X^2) - E^2(X) = \frac{rb(r+b+1)}{(r+1)^2(r+2)}
 \end{aligned}$$

Y-Problem 3.2.7.3

甲从 1,2,3,4 中任取一数 ξ , 乙再从 1, ..., ξ 中任取一数 η , 求 $\text{cov}(\xi, \eta)$

解: 易知 $E(\xi) = 2.5$, 对 η , 有

$$p(\eta = 4) = \frac{1}{16}, p(\eta = 3) = \frac{7}{48}, p(\eta = 2) = \frac{13}{48}, p(\eta = 1) = \frac{25}{48} \Rightarrow E(\eta) = \frac{7}{4}$$

对 $\xi\eta$, 有

$$p_1 = \frac{1}{16}, p_2 = \frac{1}{8}, p_3 = \frac{1}{12}, p_4 = \frac{3}{16}, p_6 = \frac{1}{12}, p_8 = \frac{1}{16}, p_9 = \frac{1}{12}, p_{12} = \frac{1}{16}, p_{16} = \frac{1}{16}$$

$$E(\xi\eta) = (1 + 8 + 12 + 16) \times \frac{1}{16} + (3 + 6 + 9) \times \frac{1}{12} + 2 \times \frac{1}{8} + 4 \times \frac{3}{16} = \frac{77}{16}$$

从而 $\text{cov}(\xi, \eta) = E(\xi\eta) - E(\xi)E(\eta) = \frac{7}{16}$.

Y-Problem 3.2.7.5

参加集会的 n 个人将他们的帽子混放在一起, 会后每人任取一顶帽子戴上, 以 ξ 表示他们中戴对自己帽子的人的个数, 求 $D(\xi)$.

解: 定义随机变量 ξ_i 为: $\xi_i = 1$ 当且仅当第 i 个人戴对了帽子. 于是

$$\xi = \sum_{i=1}^n \xi_i \Rightarrow E(\xi) = \sum_{i=1}^n E(\xi_i) = n \times \frac{1}{n} = 1$$

同理, 易知 $E(\xi_i^2) = \frac{1}{n}$, $E(\xi_i \xi_j) = \frac{1}{n(n-1)}$, $i \neq j$, 于是

$$\begin{aligned} E(\xi^2) &= \sum_{i=1}^n E(\xi_i^2) + 2 \sum_{1 \leq i < j \leq n} E(\xi_i \xi_j) = 1 + 2 \binom{n}{2} \frac{1}{n(n-1)} = 2 \\ &\Rightarrow D(\xi) = E(\xi^2) - E^2(\xi) = 1 \end{aligned}$$

Y-Problem 3.2.7.11

袋中有 N 个球, 其中白球数 τ 为随机变量, $E(\tau) = n$. 现从袋中有放回地任取 m 个球, 求取出的白球个数 ξ 的数学期望.

解: 固定白球数 τ , 则 $\xi \sim B(m, \frac{\tau}{N})$, 此时 $E_\tau(\xi) = \frac{m}{N}\tau$, 于是

$$E(\xi) = E(E_\tau(\xi)) = \frac{m}{N}E(\tau) = \frac{mn}{N}$$

Y-Problem 3.2.7.12

设随机变量 (ξ, η) 有联合密度函数

$$p(x, y) = \frac{1}{y} e^{-\frac{x}{y} - y}, \quad x, y > 0$$

求 $E(\xi|\eta)$ 与 $D(\xi|\eta)$.

解: 先求边缘密度与条件分布

$$p_\eta(y) = \int_0^\infty p(x, y) dx = e^{-y}, \quad p_{\xi|\eta=y}(x) = \frac{p(x, y)}{p_\eta(y)} = \frac{1}{y} e^{-\frac{x}{y}}$$

于是

$$\begin{aligned} E(\xi|\eta=y) &= \int_0^\infty x \cdot \frac{1}{y} e^{-\frac{x}{y}} dx = y \\ D(\xi|\eta=y) &= \int_0^\infty (x-y)^2 \cdot \frac{1}{y} e^{-\frac{x}{y}} dx = y^2 \end{aligned}$$

Y-Problem 3.2.7.16

在可列重 Bernoulli 试验中, 以 ξ_i 表第 i 次成功的等待时间, 求 $E(\xi_1|\xi_2)$ 与 $E(\xi_2|\xi_1)$.

解: 先求条件分布

$$p_{\xi_1|\xi_2=y}(x) = \frac{1}{y-1}, 1 \leq x \leq y-1, \quad p_{\xi_2|\xi_1=y}(x) = (1-p)^{x-y-1}p, x \geq y+1$$

于是

$$E(\xi_1|\xi_2=y) = \sum_{x=1}^{y-1} \frac{x}{y-1} = \frac{y}{2}, \quad E(\xi_2|\xi_1=y) = \sum_{x=y+1}^{\infty} xp(1-p)^{x-y-1} = y + \frac{1}{p}$$

Y-Problem 3.2.7.8

设 ξ, η 都是只有两个可能值的随机变量, 证明: 若 ξ, η 不相关, 则 ξ 与 η 独立.

证明: 设 ξ, η 分别满足, $\xi \sim \begin{pmatrix} x_1 & x_2 \\ p & 1-p \end{pmatrix}, \eta \sim \begin{pmatrix} x_3 & x_4 \\ q & 1-q \end{pmatrix}$ 于是

$$E(\xi) = x_1p + x_2(1-p), E(\eta) = x_3q + x_4(1-q)$$

$$E(\xi\eta) = x_1x_3p_{1,3} + x_1x_4p_{1,4} + x_2x_3p_{2,3} + x_2x_4p_{2,4}$$

其中

$$p_{1,3} + p_{2,3} = q, p_{1,3} + p_{1,4} = p, p_{1,4} + p_{2,4} = 1 - q, p_{2,3} + p_{2,4} = 1 - p$$

从而

$$\text{cov}(\xi, \eta) = 0 \iff$$

$$x_1x_3(p_{1,3} - pq) + x_1x_4(p - p_{1,3} - p(1-q)) + x_2x_3(q - p_{1,3} - q(1-p))$$

$$+ x_2x_4(1-q-p+p_{1,3} - (1-p)(1-q)) = 0$$

$$\iff (p_{1,3} - pq)(x_1 - x_2)(x_3 - x_4) = 0 \iff p_{1,3} = pq$$

由 $p_{1,3} = pq$ 即可推出其他 $p_{1,4}, p_{2,3}, p_{2,4}$ 满足独立性, 于是 ξ, η 独立.

Y-Problem 3.2.7.9

设 $\xi \sim N(0, 1)$, 证明: ξ 与 ξ^2 不相关, 但是不独立.

证明: 容易计算

$$E(\xi) = 0, E(\xi^2) = D(\xi) + (E(\xi))^2 = 1$$

$$E(\xi^3) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^3 e^{-\frac{x^2}{2}} dx \stackrel{\text{奇函数}}{=} 0 \implies \text{cov}(\xi, \xi^2) = 0$$

故 ξ, ξ^2 不相关. 设标准正态分布的密度为 $\varphi(x)$, 分布为 $\Phi(x)$, 对 $y > 0, x > \sqrt{y}$ 有

$$F_{\xi^2}(x) = \begin{cases} 0, & x < 0 \\ \Phi(\sqrt{x}) - \Phi(-\sqrt{x}), & x > 0 \end{cases}$$

$$F_{\xi, \xi^2}(x, y) = \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) \neq F_{\xi}(x)F_{\xi^2}(y) = \Phi(x)\Phi(\sqrt{y}) - \Phi(x)\Phi(-\sqrt{y})$$

于是 ξ, ξ^2 不独立且不相关.

Y-Problem 3.2.7.10

设平方可积随机变量 ξ 的密度函数为偶函数, 证明: ξ 与 $|\xi|$ 不相关, 但是不独立.

证明: 利用积分变换公式以及 $p(x)$ 为偶函数, 有

$$E(\xi|\xi|) = \int_{\mathbb{R}} x|x|p(x) dx = 0, E(\xi) = \int_{\mathbb{R}} xp(x) dx = 0 \implies \text{cov}(\xi, |\xi|) = 0$$

考察联合分布, 若 $\xi, |\xi|$ 独立, 对 $x > y > 0$, 有

$$F_{\xi, |\xi|}(x, y) = F_{|\xi|}(y) = F_{\xi}(x)F_{|\xi|}(y) \implies F_{\xi}(x) = 1, \forall x > 0$$

从而由 $F_{\xi}(x) + F_{\xi}(-x) = 1$ 以及 F_{ξ} 的左连续性知只能是

$$F_{\xi}(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases} \implies \xi \sim \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

此时 ξ 是平凡的且不具有密度函数, 因此对一般的 ξ, ξ 与 $|\xi|$ 不独立且不相关.

K-Problem 3.2.11

If $E(X^2) = 1$ and $E(|X|) \geq a > 0$, then $P(|X| \geq \lambda a) \geq (1 - \lambda)^2 a^2$ for $0 \leq \lambda \leq 1$.

Proof: $\lambda = 1$ 时显然, $\lambda = 0$ 时只需证 $a \leq 1$: 在 Hölder 不等式中取 $Y \equiv 1$, 得到

$$E(|X|) \leq E(|X|^p)^{\frac{1}{p}} \xrightarrow{p=2} 0 < a \leq E(|X|) \leq 1$$

现设 $0 < \lambda < 1$, 记 $\Lambda = \{\omega : |X| \geq \lambda a\}$, 先证明如下关系:

$$I_{\Lambda} \geq \frac{1}{t^2}(|X| - \lambda a)(\lambda a + 2t - |X|), \forall t > 0$$

这是因为若 $\omega \notin \Lambda$, 则 $|X| < \lambda a \implies \text{RHS} < 0 = \text{LHS}$; 若 $\omega \in \Lambda$, 则有

$$\text{RHS} \leq \frac{1}{t^2} \left(\frac{|X| - \lambda a + \lambda a + 2t - |X|}{2} \right)^2 = 1 = \text{LHS}$$

由上述不等式即知

$$\begin{aligned}
P(\Lambda) &= E(I_\Lambda) \geq E\left(\frac{1}{t^2}(|X| - \lambda a)(\lambda a + 2t - |X|)\right) \\
&= E\left(\frac{1}{t^2}(-\lambda a(\lambda a + 2t) + 2(\lambda a + t)|X| - |X|^2)\right) \\
&\geq \frac{1}{t^2}(-\lambda a(\lambda a + 2t) + 2(\lambda a + t)a - 1) \\
&= -(a^2\lambda^2 - 2\lambda a^2 + 1)\frac{1}{t^2} + 2a(1 - \lambda)\frac{1}{t}, \forall t > 0
\end{aligned}$$

上式为关于 $\frac{1}{t}$ 的二次函数, 且有

$$0 < 2\lambda a(1 - a) \leq a^2\lambda^2 - 2\lambda a^2 + 1 = 1 + \lambda a^2(\lambda - 2) < 1$$

于是上式在 $\frac{1}{t} = \frac{a(1-\lambda)}{a^2\lambda^2 - 2\lambda a^2 + 1}$ 处取到最大值, 代入可知

$$P(\Lambda) \geq \frac{(1 - \lambda)^2 a^2}{a^2\lambda^2 - 2\lambda a^2 + 1} > (1 - \lambda)^2 a^2$$

K-Problem 3.2.12

If $X \geq 0$ and $Y \geq 0$, $p \geq 0$, then $E((X + Y)^p) \leq 2^p(E(X^p) + E(Y^p))$. If $p > 1$, the factor 2^p can be replaced by 2^{p-1} . If $0 \leq p \leq 1$, it can be replaced by 1.

Proof: 对 $a, b \in \mathbb{R}, p \geq 0$, 记 $C_p = \max(1, 2^{p-1})$, 由课上所讲知

$$|a + b|^p \leq C_p(|a|^p + |b|^p)$$

对随机变量的情况, 当 $p > 1$ 时 $C_p = 2^{p-1}$, 故

$$(X + Y)^p \leq 2^{p-1}(X^p + Y^p) \implies E((X + Y)^p) \leq 2^{p-1}(E(X^p) + E(Y^p))$$

当 $0 \leq p < 1$ 时 $C_p = 1$, 故

$$(X + Y)^p \leq X^p + Y^p \implies E((X + Y)^p) \leq E(X^p) + E(Y^p)$$

K-Problem 3.2.13

If $X_j \geq 0$, then

$$E\left(\left(\sum_{j=1}^n X_j\right)^p\right) \leq \text{or} \geq \sum_{j=1}^n E(X_j^p)$$

according as $0 \leq p \leq 1$ or $p \geq 1$.

Proof: 当 $0 \leq p \leq 1$ 时, 容易将 C_r 不等式推广至 n 元情形, 即

$$(x_1 + \cdots + x_n)^p \leq x_1^p + \cdots + x_n^p, \quad x_1, \cdots, x_n \geq 0, 0 \leq p \leq 1$$

于是

$$\left(\sum_{j=1}^n X_j\right)^p \leq \sum_{j=1}^n X_j^p \implies E\left(\left(\sum_{j=1}^n X_j\right)^p\right) \leq \sum_{j=1}^n E(X_j^p)$$

当 $p \geq 1$ 时, 容易证明 $(a+b)^p \geq a^p + b^p, a, b \geq 0$, 递推即知

$$(x_1 + \cdots + x_n)^p \geq x_1^p + \cdots + x_n^p, \quad x_1, \cdots, x_n \geq 0, p \geq 1$$

于是

$$\left(\sum_{j=1}^n X_j\right)^p \geq \sum_{j=1}^n X_j^p \implies E\left(\left(\sum_{j=1}^n X_j\right)^p\right) \geq \sum_{j=1}^n E(X_j^p)$$

K-Problem 3.2.15

If $p > 0$, $E(|X|^p) < \infty$, then $x^p P(|X| > x) = o(1)$ as $x \rightarrow \infty$. Conversely, if $x^p P(|X| > x) = o(1)$ as $x \rightarrow \infty$, then $E(|X|^{p-\varepsilon}) < \infty$ for $0 < \varepsilon < p$.

Lemma: 设 $p > 0$, 则有

$$E(|X|^p) = p \int_0^\infty x^{p-1} P(|X| > x) dx$$

上式的含义是: 若一边有限, 则另一边有限且相等; 若一边无限, 则两边均无限.

Proof: 由 Toenelli 定理 (只要求非负可测, 不要求可积性) 可知

$$\begin{aligned} p \int_0^\infty x^{p-1} P(|X| > x) dx &= p \int_0^\infty \int_{-\infty}^{-x} x^{p-1} dF(t) dx + p \int_0^\infty \int_x^\infty x^{p-1} dF(t) dx \\ &= \int_{-\infty}^0 \int_0^{-t} p x^{p-1} dx dF(t) + \int_0^\infty \int_0^t p x^{p-1} dx dF(t) \\ &= \int_{\mathbb{R}} |t|^p dF(t) = E(|X|^p) \end{aligned}$$

Proof: 我们有

$$E(|X|^p) < \infty \iff \int_{\mathbb{R}} t^p dF(t) < \infty$$

于是可知

$$x^p P(|X| > x) = x^p \int_{\{|X|>x\}} dP = x^p \int_x^\infty dF(t) < \int_x^\infty t^p dF(t) \rightarrow 0 (x \rightarrow \infty)$$

反之, 由引理可得

$$E(|X|^{p-\varepsilon}) = (p-\varepsilon) \int_0^\infty x^{p-\varepsilon-1} P(|X| > x) dx$$

对任意 $\varepsilon' > 0$, 存在 $M > 0$ 使得 $x > M \implies x^p P(|X| > x) < \varepsilon'$, 于是

$$\begin{aligned} E(|X|^{p-\varepsilon}) &= (p-\varepsilon) \left(\int_0^M x^{p-\varepsilon-1} P(|X| > x) dx + \int_M^\infty x^{p-\varepsilon-1} P(|X| > x) dx \right) \\ &\leq (p-\varepsilon) \left(\int_0^M x^{p-\varepsilon-1} dx + \int_M^\infty \frac{\varepsilon'}{x^{\varepsilon+1}} dx \right) \\ &= M^{p-\varepsilon} + \varepsilon' \frac{(p-\varepsilon)M^\varepsilon}{\varepsilon} < +\infty \end{aligned}$$

综上命题得证.

K-Problem 3.2.20

For $r > 1$, we have

$$\int_0^\infty \frac{1}{u^r} E(X \wedge u^r) du = \frac{r}{r-1} E\left(X^{\frac{1}{r}}\right)$$

Proof: 由 $X^{\frac{1}{r}}$ 的存在性即知 $X \geq 0$, 从而

$$\begin{aligned} \int_0^\infty \frac{1}{u^r} E(X \wedge u^r) du &= \int_0^\infty \frac{1}{u^r} \int_0^\infty \min(x, u^r) dF_X(x) du \\ &= \int_0^\infty \int_0^{u^r} \frac{x}{u^r} dF_X(x) du + \int_0^\infty \int_{u^r}^\infty dF_X(x) du \\ &= \int_0^\infty \int_{x^{\frac{1}{r}}}^\infty \frac{x}{u^r} du dF_X(x) + \int_0^\infty \int_0^{x^{\frac{1}{r}}} du dF_X(x) \\ &= \frac{1}{r-1} \int_0^\infty x^{\frac{1}{r}} dF_X(x) + \int_0^\infty x^{\frac{1}{r}} dF_X(x) \\ &= \frac{r}{r-1} E\left(X^{\frac{1}{r}}\right) \end{aligned}$$

K-Problem 3.3.9

If X and Y are independent and $E(X)$ exists, then for any Borel set B , we have

$$\int_{\{Y \in B\}} X dP = E(X) P(Y \in B)$$

Proof: 我们有

$$\begin{aligned}\int_{\{Y \in B\}} X \, dP &= \int_{\Omega} X I_{\{Y \in B\}} \, dP_X \cdot \int_{\Omega} dP_Y = \int_{\Omega} \int_{\Omega} X I_{\{Y \in B\}} \, dP_X \, dP_Y \\ &\stackrel{\text{Fubini}}{=} \int_{\Omega} \int_{\Omega} X I_{\{Y \in B\}} \, dP_Y \, dP_X \stackrel{\text{indep}}{=} \int_{\Omega} X \int_{\Omega} I_{\{Y \in B\}} \, dP_Y \, dP_X \\ &= E(X)P(Y \in B)\end{aligned}$$

K-Problem 3.3.10

If X and Y are independent and for some $p > 0$: $E(|X + Y|^p) < \infty$, then $E(|X|^p) < \infty$ and $E(|Y|^p) < \infty$.

Proof: 我们有

$$\begin{aligned}\infty &> \iint_{\mathbb{R}^2} |x + y|^p \, dF_{(X,Y)}(x, y) \stackrel{\text{indep}}{=} \iint_{\mathbb{R}^2} |x + y|^p \, dF_X(x) \, dF_Y(y) \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |x + y|^p \, dF_X(x) \right) dF_Y(y) = \int_{\mathbb{R}} E(|X + y|^p) \, dF_Y(y)\end{aligned}$$

于是必然存在 $y_0 \in \mathbb{R}$ 使得 $E(|X + y|^p) < \infty$, 从而由 C_r 不等式知

$$E(|X|^p) = E(|X + y - y|^p) \leq C_p(E(|X + y|^p) + |y|^p) < \infty$$

同理可知 $E(|Y|^p) < \infty$.

K-Problem 3.3.11

If X and Y are independent, $E(|X|^p) < \infty$ for some $p \geq 1$ and $E(|Y|) = 0$, then $E(|X + Y|^p) \geq E(|X|^p)$.

Proof: 由积分变换定理, 有

$$\begin{aligned}E(|\xi + \eta|^p) &= \iint_{\mathbb{R}^2} |x + y|^p \, dF_{(\xi, \eta)}(x, y) \stackrel{\text{indep}}{=} \iint_{\mathbb{R}^2} |x + y|^p \, dF_{\xi}(x) \, dF_{\eta}(y) \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |x + y|^p \, dF_{\eta}(y) \right) dF_{\xi}(x) = \int_{\mathbb{R}} E(|x + \eta|^p) \, dF_{\xi}(x) \\ &\geq \int_{\mathbb{R}} |x|^p \, dF_{\xi}(x) = E(|\xi|^p)\end{aligned}$$

其中最后一步不等式是因为 $f(x) = |x|^p$ 在 $p \geq 1$ 时为凸函数, 由 Jensen 不等式知

$$E(|x + \eta|^p) = E(f(x + \eta)) \geq f(E(x + \eta)) = f(x + E(\eta)) = f(x) = |x|^p$$

Y-Problem 3.2.7.12

设随机变量 (ξ, η) 有联合密度函数

$$p(x, y) = \frac{1}{y} e^{-\frac{x}{y} - y}, \quad x, y > 0$$

求 $E(\xi|\eta)$ 与 $D(\xi|\eta)$.

解: 先求边缘密度与条件分布

$$p_{\eta}(y) = \int_0^{\infty} p(x, y) dx = e^{-y}, p_{\xi|\eta=y}(x) = \frac{p(x, y)}{p_{\eta}(y)} = \frac{1}{y} e^{-\frac{x}{y}}$$

于是

$$E(\xi|\eta = y) = \int_0^{\infty} x \cdot \frac{1}{y} e^{-\frac{x}{y}} dx = y, D(\xi|\eta = y) = \int_0^{\infty} (x - y)^2 \cdot \frac{1}{y} e^{-\frac{x}{y}} dx = y^2$$

Y-Problem 3.2.7.14

设随机变量 ξ_1 服从 $(0, 1)$ 上的均匀分布, 对 $k \geq 1$, 若已知 $\xi_k = x_k$, 则 ξ_{k+1} 服从 $(x_k, x_k + 1)$ 上的均匀分布, 求 $E(\xi_n)$.

解: 由题可知, $\xi_{k+1} - \xi_k \sim U_{(0,1)} \implies E(\xi_{k+1} - \xi_k) = \frac{1}{2}, k = 1, \dots, n-1$, 从而

$$E(\xi_n) = E(\xi_n - \xi_{n-1} + \dots + \xi_2 - \xi_1 + \xi_1) = E(\xi_1) + \sum_{k=1}^{n-1} E(\xi_{k+1} - \xi_k) = \frac{n}{2}$$

Y-Problem 3.2.7.15

试证对任何平方可积的随机变量 (ξ, η) 有

$$D(\xi) = E(D(\xi|\eta)) + D(E(\xi|\eta))$$

证明: 由积分变换公式可知

$$\begin{aligned} & E(D(\xi|\eta)) + D(E(\xi|\eta)) \\ &= \int_{\mathbb{R}} D(\xi|\eta = y) F_{\eta}(dy) + \int_{\mathbb{R}} (E(\xi|\eta = y) - E(E(\xi|\eta)))^2 F_{\eta}(dy) \\ &= \int_{\mathbb{R}} D(\xi|\eta = y) + E^2(\xi|\eta = y) - 2E(\xi)E(\xi|\eta = y) F_{\eta}(dy) + E^2(\xi) \\ &= \int_{\mathbb{R}} E(\xi^2|\eta = y) F_{\eta}(dy) - E^2(\xi) \\ &= E(E(\xi^2|\eta)) - E^2(\xi) = E(\xi^2) - E^2(\xi) = D(\xi) \end{aligned}$$

Y-Problem 3.2.7.16

在可列重 Bernoulli 试验中, 以 ξ_i 表第 i 次成功的等待时间, 求 $E(\xi_1|\xi_2)$ 与 $E(\xi_2|\xi_1)$.

解: 先求条件分布

$$p_{\xi_1|\xi_2=y}(x) = \frac{1}{y-1}, 1 \leq x \leq y-1, \quad p_{\xi_2|\xi_1=y}(x) = (1-p)^{x-y-1}p, x \geq y+1$$

于是

$$E(\xi_1|\xi_2=y) = \sum_{x=1}^{y-1} \frac{x}{y-1} = \frac{y}{2}, \quad E(\xi_2|\xi_1=y) = \sum_{x=y+1}^{\infty} xp(1-p)^{x-y-1} = y + \frac{1}{p}$$

K-Problem 9.1.3

If X is an integrable r.v., Y is a bounded r.v., and $\mathcal{G}$ a Borel subfield, then we have

$$E(E(X|\mathcal{G})Y) = E(E(Y|\mathcal{G})X)$$

Proof:

$E(X|\mathcal{G})$ 与 $E(Y|\mathcal{G})$ 均为 $\mathcal{G}$ 上的随机变量, 故有

$$\begin{aligned} E(E(X|\mathcal{G})Y) &= E(E(E(X|\mathcal{G})Y|\mathcal{G})) = E(E(X|\mathcal{G})E(Y|\mathcal{G})) \\ E(E(Y|\mathcal{G})X) &= E(E(E(Y|\mathcal{G})X|\mathcal{G})) = E(E(Y|\mathcal{G})E(X|\mathcal{G})) \\ &\Rightarrow E(E(X|\mathcal{G})Y) = E(E(Y|\mathcal{G})X) \end{aligned}$$

K-Problem 9.1.9

Let the $p(\cdot, \cdot)$ above be the 2-dimensional normal density:

$$\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\frac{x^2}{\sigma_1^2} - \frac{2\rho xy}{\sigma_1\sigma_2} + \frac{y^2}{\sigma_2^2}\right)\right)$$

where $\sigma_1 > 0, \sigma_2 > 0, 0 < \rho < 1$. Find the φ mentioned in Exercise 8 (i.e. the conditional density $p_{Y|X=x}(y)$) and

$$\int_{\mathbb{R}} y\varphi(x, y) dy$$

The latter should be a version of $E(Y|X=x)$; verify it.

Solution: 先求边缘分布

$$p_X(x) = \int_{-\infty}^{\infty} p(x, y) dy = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{x^2}{2\sigma_1^2}}$$

于是

$$\varphi(x, y) = p_{Y|X=x}(y) = \frac{p(x, y)}{p_\xi(x)} = \frac{1}{\sqrt{2\pi(1-\rho^2)}\sigma_2} \exp\left(-\frac{\left(\frac{y}{\sigma_2} - \frac{\rho x}{\sigma_1}\right)^2}{2(1-\rho^2)}\right)$$

从而有

$$\begin{aligned} \int_{\mathbb{R}} y\varphi(x, y) dy &= \int_{\mathbb{R}} \frac{y}{\sqrt{2\pi(1-\rho^2)}\sigma_2} \exp\left(-\frac{\left(\frac{y}{\sigma_2} - \frac{\rho x}{\sigma_1}\right)^2}{2(1-\rho^2)}\right) dy \\ &= \frac{\sigma_2}{\sqrt{2\pi(1-\rho^2)}} \int_{\mathbb{R}} y \exp\left(-\frac{\left(y - \frac{\rho x}{\sigma_1}\right)^2}{2(1-\rho^2)}\right) dy = \frac{\rho\sigma_2}{\sigma_1} x \end{aligned}$$

这就是条件期望 $E(Y|X=x)$ 的计算方法, 其中 (X, Y) 具有题中密度函数 $p(x, y)$.

K-Problem 9.1.10

Let $\mathcal{G}$ be a B.F., X and Y are two r.v.'s such that

$$E(Y^2|\mathcal{G}) = X^2, E(Y|\mathcal{G}) = X$$

Then $Y \stackrel{\text{a.s.}}{=} X$.

Proof: 首先, 我们有

$$\begin{aligned} E(X^2) = E(E(Y^2|\mathcal{G})) &= E(Y^2) < +\infty \implies E(XY) \leq \frac{1}{2}E(X^2 + Y^2) < \infty \\ &\implies E((X - Y)^2) < \infty \end{aligned}$$

故 $(X - Y)^2$ 可积. 易知此时 9.1.3 的证明仍成立, 即其结论在 XY 可积时成立, 故

$$\begin{aligned} E(X^2) &= E(E(Y|\mathcal{G})X) = E(E(X|\mathcal{G})Y) \stackrel{X \text{ is } \mathcal{G}\text{-meas}}{=} E(XY) \\ &\implies E((X - Y)^2) = 2(E(X^2) - E(XY)) = 0 \iff X \stackrel{\text{a.s.}}{=} Y \end{aligned}$$

K-Problem 9.1.11

As in Exercise 10 but suppose now for any $f \in C_K$

$$E(X^2|f(X)) = E(Y^2|f(X)); \quad E(X|f(X)) = E(Y|f(X))$$

Then $Y \stackrel{\text{a.e.}}{=} X$.

[HINT: By a monotone class theorem, the equations hold for $f = I_B, B \in \mathcal{B}^1$; now apply Exercise 10 with $\mathcal{G} = \mathcal{F}(X)$.]

Proof: 考虑集合

$$\mathcal{G} = \left\{ B \in \mathcal{B}^1 : \int_{X^{-1}(B)} XP(d\omega) = \int_{X^{-1}(X)} YP(d\omega), \right. \\ \left. \int_{X^{-1}(B)} X^2 P(d\omega) = \int_{X^{-1}(X)} Y^2 P(d\omega) \right\}$$

记 $\mathcal{K}$ 为 $\mathbb{R}$ 中紧集构成的集合, 易知 $\mathcal{K}$ 为 π 类, 先证明: $\mathcal{K} \subset \mathcal{G}$. 取 $B \in \mathcal{K}$, 由题知

$$E(X^2|I_B(X)) = E(Y^2|I_B(X)); \quad E(X|I_B(X)) = E(Y|I_B(X))$$

其中, $E(X|I_B(X)) = E(X|\sigma(I_B(X))) = E(X|\sigma(I_B(X)^{-1}(\mathcal{B}^1)))$, 取 $\{1\} \in \mathcal{B}^1$, 可得 $X^{-1}(B) = I_B(X)^{-1}(\{1\}) \in \sigma(I_B(X))$, 于是

$$\begin{aligned} \int_{X^{-1}(B)} XP(d\omega) &= \int_{X^{-1}(B)} E(X|I_B(X))P(d\omega) \\ &= \int_{X^{-1}(B)} E(Y|I_B(X))P(d\omega) = \int_{X^{-1}(B)} YP(d\omega) \end{aligned}$$

对 X^2, Y^2 同理, 因此可知 $B \in \mathcal{G} \implies \mathcal{K} \subset \mathcal{G}$.

再证明 $\mathcal{G}$ 构成 λ 类. 显然有 $\mathbb{R} \in \mathcal{G}$. 设 $A, B \in \mathcal{G}$ 且 $A \subset B$, 于是

$$\begin{aligned} \int_{X^{-1}(B \setminus A)} XP(d\omega) &= \int_{X^{-1}(B)} XP(d\omega) - \int_{X^{-1}(A)} XP(d\omega) \\ &= \int_{X^{-1}(B)} YP(d\omega) - \int_{X^{-1}(A)} YP(d\omega) = \int_{X^{-1}(B \setminus A)} YP(d\omega) \end{aligned}$$

对 X^2, Y^2 同理, 于是 $B \setminus A \in \mathcal{G}$. 再设 $A_n \uparrow A, A_n \in \mathcal{G}$, 则

$$\begin{aligned} \int_{X^{-1}(A)} XP(d\omega) &= \int_{\Omega} XI_{X^{-1}(\cup_{n=1}^{\infty} A_n)} P(d\omega) = \int_{\Omega} XI_{\cup_{n=1}^{\infty} X^{-1}(A_n)} P(d\omega) \\ &= \int_{\Omega} \lim_{n \rightarrow \infty} XI_{X^{-1}(A_n)} P(d\omega) \stackrel{\text{DCT}}{=} \lim_{n \rightarrow \infty} \int_{\Omega} XI_{X^{-1}(A_n)} P(d\omega) \\ &= \lim_{n \rightarrow \infty} \int_{\Omega} YI_{X^{-1}(A_n)} P(d\omega) \stackrel{\text{DCT}}{=} \int_{X^{-1}(A)} YP(d\omega) \end{aligned}$$

对 X^2, Y^2 同理, 于是 $A \in \mathcal{G}$. 故 $\mathcal{G}$ 为 λ 类. 由单调类定理即知

$$\mathcal{B}^1 = \sigma(\mathcal{K}) = \lambda(\mathcal{K}) \subset \mathcal{G} \implies \mathcal{G} = \mathcal{B}^1$$

从而对 X 生成的 σ 代数 $\sigma(X) = \sigma(X^{-1}(\mathcal{B}^1))$, 可知对任意 $A \in \sigma(X)$, X, Y 与 X^2, Y^2 在 A 上的积分值分别相等, 由条件期望的唯一性即知

$$E(X|X) = E(Y|X), E(X^2|X) = E(Y^2|X)$$

又由 X, X^2 分别为 $\sigma(X)$ 上的可测函数即知

$$X = E(X|X) = E(Y|X), X^2 = E(X^2|X) = E(Y^2|X)$$

从而由 9.1.10 即知 $X \stackrel{\text{a.s.}}{=} Y$.

K-Problem 9.1.13

Let S be an r.v. such that $P(S > t) = e^{-t}, t > 0$. Compute $E(S|S \wedge t)$ and $E(S|S \vee t)$ for each $t > 0$.

Solution: 由定义知 $S \wedge t$ 在 $[0, t]$ 取值, 并且对 $x \in [0, t), S \wedge t = x \iff S = x$. 故 $E(S|S \wedge t = x) = x$, 对 $x = t, S \wedge t = x \iff S \geq t$, 于是

$$E(S|S \wedge t = t) = \frac{1}{P(S \geq t)} \int_t^\infty x e^{-x} dx = \frac{(t+1)e^{-t}}{e^{-t}} = t+1$$

由定义即知 $S \vee t$ 在 $[t, \infty)$ 取值, 并且对 $x \in (t, \infty), S \vee t = x \iff S = x$, 故 $E(S|S \vee t = x) = x$, 对 $x = t, S \vee t = x \iff S \leq t$, 于是

$$E(S|S \vee t = t) = \frac{1}{P(S \leq t)} \int_0^t x e^{-x} dx = \frac{1 - (t+1)e^{-t}}{1 - e^{-t}}$$

从而

$$E(S|S \wedge t) = \begin{cases} S, & S < t \\ t+1, & S \geq t \end{cases} \quad E(S|S \vee t) = \begin{cases} S, & S > t \\ \frac{1-(t+1)e^{-t}}{1-e^{-t}}, & S \leq t \end{cases}$$

M-Problem 4.6.1

设 $\Omega = \{0, 1, 2\}, \mathcal{A} = \sigma(\{0\}, \{1\}, \{2\})$, 且

$$P(\{0\}) = \frac{1}{2}, \quad P(\{1\}) = \frac{1}{3}, \quad P(\{2\}) = \frac{1}{6}$$

$$\mathcal{C}_1 = \{\emptyset, \{0, 1\}, \{2\}, \Omega\}, \quad \mathcal{C}_2 = \{\emptyset, \{0, 2\}, \{1\}, \Omega\}$$

取随机变量 $Y = I_{\{2\}}(\omega)$, 证明:

$$E(E(Y|\mathcal{C}_1)|\mathcal{C}_2) \neq E(E(Y|\mathcal{C}_2)|\mathcal{C}_1)$$

证明: 由于 $\mathcal{C}_1, \mathcal{C}_2$ 均为 Ω 的分割生成的 σ 代数, 因此

$$E(Y|\mathcal{C}_1) = E(Y|\{0, 1\})I_{\{0,1\}} + E(Y|\{2\})I_{\{2\}} = I_{\{2\}} = Y$$

$$E(E(Y|\mathcal{C}_1)|\mathcal{C}_2) = E(Y|\mathcal{C}_2) = E(Y|\{0, 2\})I_{\{0,2\}} + E(Y|\{1\})I_{\{1\}} = \frac{1}{4}I_{\{0,2\}}$$

$$E(E(Y|\mathcal{C}_2)|\mathcal{C}_1) = \frac{1}{4}E(I_{\{0,2\}}|\{0, 1\})I_{\{0,1\}} + \frac{1}{4}E(I_{\{0,2\}}|\{2\})I_{\{2\}} = \frac{3}{20}I_{\{0,1\}} + \frac{1}{4}I_{\{2\}}$$

于是可见上述两个随机变量在 0, 1 处取值均不相同, 故二者不等.

M-Problem 4.6.15

(1) 若 $X \in L_2$, Y 为随机变量, 满足 $E(X|Y) \stackrel{\text{a.s.}}{=} Y$ 及 $E(Y|X) \stackrel{\text{a.s.}}{=} X$, 证明: $X \stackrel{\text{a.s.}}{=} Y$;

(2) 若 $\mathcal{C}_1, \mathcal{C}_2$ 为 $\mathcal{A}$ 的子 σ 代数, $X \in L_1$, 设 $X_1 = E(X|\mathcal{C}_1)$, $X_2 = E(X_1|\mathcal{C}_2)$, 若 $X \stackrel{\text{a.s.}}{=} X_2$, 证明: $X_1 \stackrel{\text{a.s.}}{=} X_2$;

(3) 若 $X, Y \in L_1$ 且 $E(X|Y) \stackrel{\text{a.s.}}{=} Y$ 以及 $E(Y|X) \stackrel{\text{a.s.}}{=} X$, 证明: $X \stackrel{\text{a.s.}}{=} Y$.

事实上, 由课堂的如下例题可直接推出 (1), (2), (3):

Example: Assume that $\xi, \eta \in L^1(\Omega, \mathcal{F}, P)$, $\mathcal{G}_1, \mathcal{G}_2 \subset \mathcal{F}$ are σ -algebras. If $\xi = E(\eta|\mathcal{G}_1)$, $\eta = E(\xi|\mathcal{G}_2)$, then $\xi \stackrel{\text{a.s.}}{=} \eta$.

Proof: Consider a bounded, strictly increasing function f , it's trivial that

$$\square := E(\xi - \eta)(f(\xi) - f(\eta)) = 0 \iff \xi \stackrel{\text{a.s.}}{=} \eta$$

In fact, we have

$$\begin{aligned} \square &= E(\xi f(\xi)) + E(\eta f(\eta)) - E(\xi f(\eta)) - E(\eta f(\xi)) \\ E(\xi f(\eta)) &= E(E(\xi f(\eta)|\mathcal{G}_2)) = E(f(\eta)E(\xi|\mathcal{G}_2)) = E(\eta f(\eta)) \\ E(\eta f(\xi)) &= E(E(\eta f(\xi)|\mathcal{G}_1)) = E(f(\xi)E(\eta|\mathcal{G}_1)) = E(\xi f(\xi)) \end{aligned}$$

Then $\square = 0$, $\xi \stackrel{\text{a.s.}}{=} \eta$.

M-Problem 4.6.17

设 X, Y 为有界随机变量, $\mathcal{G}$ 为 $\mathcal{A}$ 的子 σ 代数, 证明:

$$E(XE(Y|\mathcal{G})) = E(YE(X|\mathcal{G}))$$

证明: 这就是钟开菜的习题 9.1.3.

Y-Problem 3.4.5.2

求 Γ 分布 $\Gamma(\lambda, r)$ 的特征函数, 并用特征函数求其 k 阶原点矩 m_k .

解: 直接计算, 可知

$$\begin{aligned} f_{\xi}(t) &= \int_0^{\infty} e^{itx} \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x} dx = \frac{\lambda^r}{\Gamma(r)} \int_0^{\infty} \sum_{n=0}^{\infty} \frac{(itx)^n}{n!} x^{r-1} e^{-\lambda x} dx \\ &= \frac{\lambda^r}{\Gamma(r)} \sum_{n=0}^{\infty} \frac{(it)^n}{n!} \int_0^{\infty} x^{r+n-1} e^{-\lambda x} dx = \frac{\lambda^r}{\Gamma(r)} \sum_{n=0}^{\infty} \frac{(it)^n}{n! \lambda^{r+n}} \Gamma(r+n) \\ &= \sum_{n=0}^{\infty} \frac{(r+n-1) \cdots (r+1)r}{n!} \left(\frac{it}{\lambda}\right)^n = \sum_{n=0}^{\infty} \binom{r+n-1}{n} \left(\frac{it}{\lambda}\right)^n \end{aligned}$$

由 Taylor 展开式

$$(1+x)^r = \sum_{n=0}^{\infty} \binom{r}{n} x^n$$

即知

$$\begin{aligned} f_{\xi}(t) &= \sum_{n=0}^{\infty} \frac{(-r)(-r-1)\cdots(-r-n+1)}{n!} \left(-\frac{it}{\lambda}\right)^n = \sum_{n=0}^{\infty} \binom{-r}{n} \left(-\frac{it}{\lambda}\right)^n \\ &= \left(1 - \frac{it}{\lambda}\right)^{-r} \end{aligned}$$

从特征函数的定义出发, 可知

$$f_{\xi}^{(k)}(t) = i^k \int_{\mathbb{R}} x^k e^{itx} dF_{\xi}(x) \implies E(\xi^k) = \frac{f_{\xi}^{(k)}(0)}{i^k}$$

于 $\Gamma(\lambda, r)$ 而言, 有

$$m_k = \frac{1}{i^k} \left(\frac{-i}{\lambda}\right)^k (-r)(-r-1)\cdots(-r-k+1) = \frac{r(r+1)\cdots(r+k-1)}{\lambda^k}$$

Y-Problem 3.4.5.3

判断下列函数哪些是特征函数, 并指出其相应的概率分布.

(1) $\cos^2 t$;

解: 是特征函数. 考虑离散型随机变量 ξ , 其服从

$$\xi \sim \begin{pmatrix} 2 & 0 & -2 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix}$$

于是

$$f_{\xi}(t) = \frac{1}{4}e^{2it} + \frac{1}{4}e^{-2it} + \frac{1}{2} = \frac{1}{2}(1 + \cos 2t) = \cos^2 t$$

(3) $\frac{1-t}{1+t^2}$;

解: 不是特征函数. 注意这是一个实函数, 由 3.4.5.4 知实特征函数必为偶函数, 但这不是一个偶函数.

(5) $\frac{1}{1+it}$;

解: 设 $\xi \sim \Gamma(1, 1)$, 于是 $f_{\xi}(t) = \frac{1}{1-it}$, 故 $f_{-\xi}(t) = \overline{f_{\xi}(t)} = \frac{1}{1+it}$.

(7) $\frac{1}{1+t^2}$.

解: 是特征函数. 易知该函数在 $\mathbb{R}$ 上可积, 故其对应一个连续型随机变量, 密度为

$$\begin{aligned} p(x) &= \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} f(t) dt = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{\cos|x|t}{1+t^2} dt = \operatorname{Re} \left(\frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{i|x|t}}{1+t^2} dt \right) \\ &= \operatorname{Re} \left(i \operatorname{Res} \left(\frac{e^{i|x|z}}{1+z^2}, i \right) \right) = \frac{1}{2} e^{-|x|} \end{aligned}$$

这是 Laplace 分布.

Y-Problem 3.4.5.4

若对一切 x 有 $F(x) + F(-x+0) = 1$, 则称分布函数 F 是对称的. 求证分布函数对称的充要条件是对应的特征函数为实的偶函数.

证明: 首先, 分布函数可写成

$$f(t) = \int_{\mathbb{R}} \cos tx dF(x) + i \int_{\mathbb{R}} \sin tx dF(x)$$

因此 f 为实函数 $\iff \int_{\mathbb{R}} \sin tx dF(x) = 0, \forall t \in \mathbb{R} \iff f(t) = f(-t), \forall t \in \mathbb{R}$.

必要性: 由 $F(x) + F(-x+0) = 1$ 可知

$$P(\xi < x) + P(\xi \leq -x) = 1 \implies P(\xi < x) = P(-\xi < x) \implies F_{\xi}(x) = F_{-\xi}(x)$$

这说明 ξ 与 $-\xi$ 具有相同的分布, 从而 $f_{\xi}(t) = f_{-\xi}(t) = f_{\xi}(-t) \implies f$ 为实函数.

充分性: f 为实函数 $\implies f_{\xi}(t) = f_{-\xi}(t)$, 由唯一性定理, pdf 可以由 cf 唯一地确定, 因此 ξ 与 $-\xi$ 有相同的分布函数, 由必要性中的论述即知 $F(x) + F(-x+0) = 1$.

Y-Problem 3.4.5.5

证明: 实值特征函数有性质

$$1 - f(2t) \leq 4(1 - f(t)), \quad 1 + f(2t) \geq 2f^2(t)$$

证明: 由 3.4.5.4 即知 $f(t) = \int_{\mathbb{R}} \cos tx \, dF(x)$, 从而

$$1 - f(2t) = \int_{\mathbb{R}} (1 - \cos 2tx) \, dF(x) = 2 \int_{\mathbb{R}} \sin^2 tx \, dF(x)$$

$$1 - f(t) = \int_{\mathbb{R}} (1 - \cos tx) \, dF(x) = 2 \int_{\mathbb{R}} \sin^2 \frac{tx}{2} \, dF(x)$$

从而

$$1 - f(2t) = 8 \int_{\mathbb{R}} \sin^2 \frac{tx}{2} \cos^2 \frac{tx}{2} \, dF(x) \leq 8 \int_{\mathbb{R}} \sin^2 \frac{tx}{2} \, dF(x) = 4(1 - f(t))$$

由 Cauchy 不等式可知

$$\begin{aligned} 1 + f(2t) &= \int_{\mathbb{R}} (1 + \cos 2tx) \, dF(x) = 2 \int_{\mathbb{R}} \cos^2 tx \, dF(x) \\ &= 2 \int_{\mathbb{R}} \cos^2 tx \, dF(x) \int_{\mathbb{R}} 1^2 \, dF(x) \\ &\geq 2 \left(\int_{\mathbb{R}} \cos tx \, dF(x) \right)^2 = 2f^2(t) \end{aligned}$$

Y-Problem 3.4.5.13

设 $f(t)$ 是随机变量 ξ 的特征函数, 证明下列函数都是某随机变量的特征函数.

(1) $f(-t)$;

证明: 我们有

$$f_{-\xi}(t) = E(e^{-it\xi}) = f(-t) (= \overline{f(t)})$$

(2) $|f(t)|^2$;

证明: 考察与 ξ 为 iid 的随机变量 η , 由独立性可知

$$f_{\xi-\eta}(t) = f_{\xi}(t)f_{-\eta}(t) = f(t) \cdot \overline{f(t)} = |f(t)|^2$$

(3) $e^{f(t)-1}$;

证明: 考虑一系列 iid $\{\xi_k\}$, $\xi_1 = \xi$, $N \sim \text{Poisson}(1)$, 考察随机变量 $\Gamma = \sum_{k=1}^N \xi_k$, 有

$$E(e^{it\Gamma}|N=n) = E(e^{it(\xi_1+\dots+\xi_n)}) = E^n(e^{it\xi}) \Rightarrow E(e^{it\Gamma}|N) = (E(e^{it\xi}))^N$$

故由全期望公式可知

$$f_{\Gamma}(t) = E(e^{it\Gamma}) = E(E(e^{it\Gamma}|N)) = \sum_{n=0}^{\infty} (E(e^{it\xi}))^n \frac{e^{-1}}{n!} = e^{E(e^{it\xi})-1} = e^{f(t)-1}$$

(4) $\operatorname{Re} f(t)$.

证明: 我们有

$$\operatorname{Re}(f(t)) = \frac{1}{2}(f_{\xi}(t) + \overline{f_{\xi}(t)}) = \frac{1}{2}(f_{\xi}(t) + f_{-\xi}(t))$$

$\frac{1}{2} + \frac{1}{2} = 1$, 由课上所讲即知 $\operatorname{Re} f(t)$ 为特征函数.

Y-Problme 3.4.5.14

设 ξ 与 η 相互独立, $\xi \sim \text{Poisson}(\lambda)$, $\eta \sim N(0, 1)$, 试证明 $\xi + \eta$ 是连续型随机变量.

证明: 只需证明 $f_{\xi+\eta}$ 可积, 我们有

$$f_{\xi+\eta}(t) = f_{\xi}(t)f_{\eta}(t) = e^{\lambda(e^{it}-1)}e^{-\frac{1}{2}t^2} = e^{-\frac{1}{2}t^2+\lambda \cos t-\lambda} = e^{-\frac{1}{2}t^2+\lambda \cos t-\lambda+i\lambda \sin t}$$

因此

$$\int_{\mathbb{R}} |f_{\xi+\eta}(t)| dt = \int_{\mathbb{R}} e^{-\frac{1}{2}t^2+\lambda \cos t-\lambda} dt \leq \int_{\mathbb{R}} e^{-\frac{1}{2}t^2} dt < +\infty$$

故 $\xi + \eta$ 为连续型随机变量.

Y-Problem 3.4.5.16

设随机变量 (ξ, η) 的联合密度函数为

$$p(x, y) = \frac{1}{4}[1 + cxy(x^2 - y^2)], |x| < 1, |y| < 1$$

其中参数 $c \in (-1, 1)$. 试求:

- (1) 联合特征函数 $f(s, t)$;
- (2) 边缘特征函数 $f_{\xi}(s)$ 与 $f_{\eta}(t)$;
- (3) $\xi + \eta$ 的特征函数.

解: (1) 直接计算可得

$$\begin{aligned}
f(s, t) &= \frac{1}{4} \int_{-1}^1 \int_{-1}^1 e^{-ixs} e^{-iyt} (1 + cxy(x^2 - y^2)) dx dy \\
&= \frac{1}{4} \frac{e^{-is} - e^{is}}{is} \frac{e^{-it} - e^{it}}{it} + \frac{c}{4} \int_{-1}^1 xy(x^2 - y^2) e^{-ixs} e^{-iyt} dx dy \\
&= \frac{\sin s}{s} \frac{\sin t}{t} + \frac{c}{s^4 t^2} (\sin t - t \cos t) ((3s^2 - 6) \sin s - (s^3 - 6s) \cos s) \\
&\quad - \frac{c}{s^2 t^4} (\sin s - s \cos s) ((3t^2 - 6) \sin t - (t^3 - 6t) \cos t)
\end{aligned}$$

(2) 先计算边缘密度

$$p_{\xi}(x) = \int_{-1}^1 \frac{1}{4} [1 + cxy(x^2 - y^2)] dy = \frac{1}{2} + cx \int_{-1}^1 y(x^2 - y^2) dy = \frac{1}{2}$$

同理有 $p_{\eta}(x) = \frac{1}{2}$, 因此边缘特征函数为

$$\begin{aligned}
f_{\xi}(s) &= \int_{-1}^1 \frac{1}{2} e^{-isx} dx = \frac{1}{2} \int_{-1}^1 \cos sx dx = \frac{\sin s}{s} \\
f_{\eta}(t) &= \int_{-1}^1 \frac{1}{2} e^{-itx} dx = \frac{1}{2} \int_{-1}^1 \cos tx dx = \frac{\sin t}{t}
\end{aligned}$$

(3) 做卷积即知 $p_{\xi+\eta}(x) = \frac{1}{4}(2 - |x|)$, $x \in (-2, 2)$, 于是

$$f_{\xi+\eta}(t) = \int_{-2}^2 \frac{1}{4} (2 - |x|) e^{-itx} dx = \left(\frac{\sin t}{t} \right)^2$$

K-Problem 6.1.1

If f is a cf, and G a df with $G(0-) = 0$, then the following functions are all cf's:

$$(1) \int_0^1 f(ut) du;$$

Solution: 显然 $g(0) = f(0) = 1$ 且 g 在 0 处连续, 对任意 $\zeta = (\zeta_1, \dots, \zeta_n)^T \in \mathbb{C}^n$ 以及 $A = (g(t_i - t_j))_{n \times n}$, 由于 f 为 cf, 具备半正定性质, 因此

$$\sum_{i,j=1}^n g(t_i - t_j) \zeta_i \bar{\zeta}_j = \int_0^1 \sum_{i,j=1}^n f(ut_i - ut_j) \zeta_i \bar{\zeta}_j du \geq 0$$

故 g 半正定, 从而 g 为 cf.

$$(2) \int_0^{\infty} f(ut) e^{-u} du;$$

Solution: 与 (1) 类似, $g(0) = f(0) = 1$ 且由 DCT 可知 g 在 0 处连续, 对任意 $\zeta = (\zeta_1, \dots, \zeta_n)^T \in \mathbb{C}^n$ 以及 $A = (g(t_i - t_j))_{n \times n}$, 由于 f 为 cf, 具备半正定性质, 因此

$$\sum_{i,j=1}^n g(t_i - t_j) \zeta_i \bar{\zeta}_j = \int_0^\infty e^{-u} \sum_{i,j=1}^n f(ut_i - ut_j) \zeta_i \bar{\zeta}_j du \geq 0$$

故 g 半正定, 从而 g 为 cf.

$$(3) \int_0^\infty e^{-|t|u} dG(u);$$

Solution: 由 $G(0-) = 0$ 可知 $g(0) = \int_0^\infty dG(u) = \int_{\mathbb{R}} dG(u) = 1$ 且由 DCT 可知 g 在 0 处连续. 对任意 $\zeta = (\zeta_1, \dots, \zeta_n)^T \in \mathbb{C}^n$ 以及 $A = (g(t_i - t_j))_{n \times n}$, 我们有

$$\sum_{i,j=1}^n g(t_i - t_j) \zeta_i \bar{\zeta}_j = \int_0^\infty \sum_{i,j=1}^\infty e^{-|t_i - t_j|u} \zeta_i \bar{\zeta}_j du$$

为使 g 半正定, 只需 $e^{-|t|u}$ 对任意 $u > 0$ 均关于 t 半正定, 注意到 $e^{-|t|u}$ 是 Cauchy 分布 $C(u, 0)$ 的特征函数, 从而半正定, 从而 g 半正定, 故为 cf.

$$(4) \int_0^\infty e^{-t^2 u} dG(u);$$

Solution: 与 (3) 类似, $g(0) = 1$ 且由 DCT 可知 g 在 0 处连续. 对任意 $\zeta = (\zeta_1, \dots, \zeta_n)^T \in \mathbb{C}^n$ 以及 $A = (g(t_i - t_j))_{n \times n}$, 我们有

$$\sum_{i,j=1}^n g(t_i - t_j) \zeta_i \bar{\zeta}_j = \int_0^\infty \sum_{i,j=1}^\infty e^{-(t_i - t_j)^2 u} \zeta_i \bar{\zeta}_j du$$

为使 g 半正定, 只需 $e^{-t^2 u}$ 对任意 $u > 0$ 均关于 t 半正定, 注意到 $e^{-t^2 u}$ 是正态分布 $N(0, 2u)$ 的特征函数, 从而半正定, 从而 g 半正定, 故为 cf.

$$(5) \int_0^\infty f(ut) dG(u).$$

Solution: 完全同 (2).

K-Problem 6.1.3

Find the df with the following cf's ($\alpha > 0, \beta > 0$):

$$(1) \frac{\alpha^2}{\alpha^2 + t^2};$$

Solution: 通过反演公式/查表知对应分布为 $\lambda = \frac{1}{\alpha}, \mu = 0$ 的 Laplace 分布, 密度

$$p(x) = \frac{\alpha}{2} e^{-\alpha|x|}, x \in \mathbb{R}$$

$$(2) \frac{1}{(1 - \alpha it)^\beta};$$

Solution: 这是 $\Gamma(\frac{1}{\alpha}, \beta)$ 的特征函数.

$$(3) \frac{1}{(1 + \alpha\beta - \alpha\beta e^{it})^{\frac{1}{\beta}}}.$$

Solution: 考虑参数 $r \in \mathbb{R}^+$ 的广义负二项分布 $NB(\frac{1}{\beta}, \frac{1}{1+\alpha\beta})$, 设随机变量 ξ 服从该分布, 记 $\eta = \xi - \frac{1}{\beta}$, 于是

$$f_{\eta}(t) = e^{-\frac{it}{\beta}} f_{\xi}(t) = e^{-\frac{it}{\beta}} \left(\frac{\frac{1}{1+\alpha\beta} e^{it}}{1 - \frac{\alpha\beta}{1+\alpha\beta} e^{it}} \right)^{\frac{1}{\beta}} = \frac{1}{(1 + \alpha\beta - \alpha\beta e^{it})^{\frac{1}{\beta}}}$$

K-Problem 6.1.5

If F_1 and F_2 are df's such that

$$F_1 = \sum_j b_j \delta_{a_j}$$

and F_2 has density p , show that $F_1 * F_2$ has a density and find it.

Proof: 直接验证 $F_1 * F_2$ 具有密度函数

$$\tilde{p}(x) = \sum_j b_j p(x - a_j)$$

我们有

$$\begin{aligned} \int_{-\infty}^x \sum_j b_j p(t - a_j) dt &\stackrel{\text{MCT}}{=} \sum_j b_j \int_{-\infty}^x p(t - a_j) dt = \sum_j b_j \int_{-\infty}^{x-a_j} p(t) dt \\ &= \sum_j b_j F(x - a_j) = F_1 * F_2(x) \end{aligned}$$

K-Problem 6.2.2

Show that for each $T > 0$:

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{(1 - \cos Tx) \cos tx}{x^2} dx = (T - |t|) \vee 0$$

Deduce from this that for each $T > 0$, the function of t given by

$$\left(1 - \frac{|t|}{T}\right) \vee 0$$

is a cf. Next, show that as a particular case of Theorem 6.2.3,

$$\frac{1 - \cos Tx}{x^2} = \frac{1}{2} \int_{-T}^T (T - |t|) e^{itx} dt$$

Finally, derive the following particularly useful relation (a case of Parseval's relation in Fourier analysis), for arbitrary a and $T > 0$:

$$\int_{-\infty}^{\infty} \frac{1 - \cos T(x-a)}{[T(x-a)]^2} dF(x) = \frac{1}{2T^2} \int_{-T}^T (T - |t|) e^{-ita} f(t) dt$$

Proof: 首先, 由积化和差公式可知

$$\begin{aligned} & \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{(1 - \cos Tx) \cos tx}{x^2} dx \\ &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\cos tx}{x^2} - \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\cos(T+t)x + \cos(T-t)x}{x^2} dx \end{aligned}$$

由重要积分结果的分部积分可知

$$\pi = \int_{-\infty}^{\infty} \frac{\sin x}{x} dx = - \int_{-\infty}^{\infty} \frac{\cos x}{x^2} dx \Rightarrow \int_{-\infty}^{\infty} \frac{\cos tx}{x^2} dx = -\pi|t|$$

于是

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{(1 - \cos Tx) \cos tx}{x^2} dx = -|t| + \frac{1}{2}(|T+t| + |T-t|) = (T - t) \vee 0$$

如上积分结果说明

$$\left(1 - \frac{|t|}{T}\right) \vee 0 = \int_{-\infty}^{\infty} \frac{1 - \cos Tx}{T\pi x^2} e^{itx} dx$$

为说明左式为特征函数, 只需说明 $\frac{1 - \cos Tx}{T\pi x^2}$ 为密度函数, 事实上

$$\int_{-\infty}^{\infty} \frac{1 - \cos Tx}{T\pi x^2} dx = \frac{2}{T\pi} \int_0^{\infty} \frac{1 - \cos Tx}{x^2} dx = 1$$

对第三个积分, 直接计算即知

$$\begin{aligned} \frac{1}{2} \int_{-T}^T (T - |t|) e^{itx} dt &= \int_0^T (T - t) \cos tx dt = \frac{T \sin Tx}{x} - \int_0^T t \cos tx dt \\ &= \frac{T \sin Tx}{x} - \left(\frac{T \sin Tx}{x} + \frac{\cos Tx}{x^2} - \frac{1}{x^2} \right) = \frac{1 - \cos Tx}{x^2} \end{aligned}$$

由以上结果可知

$$\begin{aligned}
\int_{-\infty}^{\infty} \frac{1 - \cos T(x-a)}{[T(x-a)]^2} dF(x) &= \frac{1}{2T^2} \int_{-\infty}^{\infty} \int_{-T}^T (T - |t|) e^{it(x-a)} dt dF(x) \\
&= \frac{1}{2T^2} \int_{-T}^T (T - |t|) \int_{-\infty}^{\infty} e^{it(x-a)} dF(x) dt \\
&= \frac{1}{2T^2} \int_{-T}^T (T - |t|) e^{-ita} f(t) dt
\end{aligned}$$

K-Problem 6.2.3

Prove that for each $\alpha > 0$:

$$\int_0^{\alpha} [F(x+u) - F(x-u)] du = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{1 - \cos \alpha t}{t^2} e^{-itx} f(t) dt$$

As a sort of reciprocal, we have

$$\frac{1}{2} \int_0^{\alpha} du \int_{-u}^u f(t) dt = \int_{-\infty}^{\infty} \frac{1 - \cos \alpha x}{x^2} dF(x)$$

Proof: 由于 F 的不连续点构成零测集, 因此对左侧的被积函数可以运用反演公式

$$\begin{aligned}
\int_0^{\alpha} [F(x+u) - F(x-u)] du &= \frac{1}{2\pi} \int_0^{\alpha} \int_{-\infty}^{\infty} \frac{e^{-it(x-u)} - e^{-it(x+u)}}{it} f(t) dt du \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{f(t)}{it} e^{-itx} \int_0^{\alpha} e^{itu} - e^{-itu} du dt \\
&= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{f(t)}{t} e^{-itx} \int_0^{\alpha} \sin tu du dt \\
&= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{1 - \cos \alpha t}{t^2} e^{-itx} f(t) dt
\end{aligned}$$

利用 6.2.2 最后的积分结果, 我们有

$$\begin{aligned}
\frac{1}{2} \int_0^{\alpha} \int_{-u}^u f(t) dt du &= \frac{1}{2} \int_{-\alpha}^{\alpha} \int_{|t|}^{\alpha} f(t) du dt \\
&= \frac{1}{2} \int_{-\alpha}^{\alpha} (\alpha - |t|) f(t) dt = \int_{-\infty}^{\infty} \frac{1 - \cos \alpha x}{x^2} dF(x)
\end{aligned}$$

K-Problem 6.2.10

Prove the following form of the inversion formula (due to Gil-Palaez)

$$\frac{1}{2}\{F(x+) + F(x-)\} = \frac{1}{2} + \lim_{\delta \downarrow 0, T \uparrow \infty} \int_{\delta}^T \frac{e^{itx}f(-t) - e^{-itx}f(t)}{2\pi it} dt$$

Proof: 先不考虑右式的极限, 我们有

$$\begin{aligned} \text{RHS} - \frac{1}{2} &= \int_{\delta}^T \left(\frac{e^{itx}}{2\pi it} \int_{\mathbb{R}} e^{-ity} dF(y) - \frac{e^{-itx}}{2\pi it} \int_{\mathbb{R}} e^{ity} dF(y) \right) \\ &= \int_{\delta}^T \int_{\mathbb{R}} \frac{e^{it(x-y)} - e^{it(x+y)}}{2\pi it} dF(y) dt \\ &= \int_{\delta}^T \int_{\mathbb{R}} \frac{\sin t(x-y)}{\pi t} dF(y) dt \\ &= \int_{\mathbb{R}} \int_{\delta}^T \frac{\sin t(x-y)}{\pi t} dt dF(y) \end{aligned}$$

由 DCT 可知极限可以与 $\int_{\mathbb{R}}$ 交换, 从而

$$\begin{aligned} \lim_{\delta \downarrow 0, T \uparrow \infty} \left(\text{RHS} - \frac{1}{2} \right) &= \int_{\mathbb{R}} \left(\lim_{\delta \downarrow 0, T \uparrow \infty} \int_{\delta}^T \frac{\sin t(x-y)}{\pi t} dt \right) dF(y) \\ &= \frac{1}{2} \int_{\mathbb{R}} \text{sgn}(x-y) dF(y) \\ &= \frac{1}{2} \left(\int_{-\infty}^x dF(y) - \int_x^{\infty} dF(y) \right) \\ &= \frac{1}{2}F(x+) - \frac{1}{2}(1 - F(x-)) \\ &= \text{LHS} - \frac{1}{2} \end{aligned}$$

K-Problem 6.5.8

Suppose $f(t, u)$ is a function on $\mathbb{R}^2$ such that for each u , $f(\cdot, u)$ is a cf, and for each t , $f(t, \cdot)$ is continuous. Then for any df G , $\exp\left(\int_{-\infty}^{\infty} [f(t, u) - 1] dG(u)\right)$ is a cf.

Proof: 由 $f(t, u)$ 关于 t 为特征函数可知 $f(0, u) = 1$, 记待证函数为 $\varphi(t)$, 于是 $\varphi(0) = 1$. 由 f 的连续性可知 φ 在 $t = 0$ 处连续. 再记

$$\varphi_n(t) = \exp\left(\sum_{k=-n2^n}^{n2^n} \left[f\left(t, \frac{k}{2^n}\right) - 1\right] \Delta G\left(\frac{k}{2^n}\right)\right)$$

其中 $\Delta G\left(\frac{k}{2^n}\right) = G\left(\frac{k}{2^n}\right) - G\left(\frac{k-1}{2^n}\right)$. 由 Thm 6.5.6 知对任意 λ, u , $\exp(\lambda[f(\cdot, u) - 1])$ 为分布函数, 进而

$$\varphi_n(t) = \prod_{k=-n2^n}^{n2^n} \exp\left(\Delta G\left(\frac{k}{2^n}\right) \left[f\left(t, \frac{k}{2^n}\right) - 1\right]\right)$$

为分布函数. 再由 Thm 6.3.2, 由 φ 在 $t = 0$ 处连续知 $\varphi = \lim_{n \rightarrow \infty} \varphi_n$ 为分布函数.

Y-Theorem 3.5.11

设 $\xi_1, \dots, \xi_n$ 为 iid, 服从 $N(a, \sigma^2)$ 分布, 则有

- (1) 样本均值 $\bar{\xi} := \frac{1}{n} \sum_{k=1}^n \xi_k$ 与样本方差 $S_n^2 := \frac{1}{n} \sum_{k=1}^n (\xi_k - \bar{\xi})^2$ 相互独立;
- (2) $\bar{\xi}$ 服从正态分布 $N\left(a, \frac{\sigma^2}{n}\right)$;
- (3) $\frac{nS_n^2}{\sigma^2}$ 服从 $n-1$ 个自由度的卡方分布 $\chi^2(n-1)$.

证明: (1) 易知 $\vec{\xi} = (\xi_1, \dots, \xi_n)^T \sim N((a, \dots, a)^T, \sigma^2 I)$. 考虑 n 阶正交阵

$$C = \begin{pmatrix} \frac{1}{\sqrt{1 \cdot 2}} & \frac{-1}{\sqrt{1 \cdot 2}} & 0 & 0 & \cdots & 0 & 0 \\ \frac{1}{\sqrt{2 \cdot 3}} & \frac{1}{\sqrt{2 \cdot 3}} & \frac{-2}{\sqrt{2 \cdot 3}} & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{1}{\sqrt{(n-1)n}} & \frac{1}{\sqrt{(n-1)n}} & \frac{1}{\sqrt{(n-1)n}} & \frac{1}{\sqrt{(n-1)n}} & \cdots & \frac{1}{\sqrt{(n-1)n}} & \frac{-(n-1)}{\sqrt{(n-1)n}} \\ \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \cdots & \frac{1}{\sqrt{n}} & \frac{-1}{\sqrt{n}} \end{pmatrix}$$

记 $\vec{\eta} = C\vec{\xi}$ 服从正态分布, 协方差阵 $C\sigma^2 IC^T = \sigma^2 I$, $\vec{\eta}$ 各分量独立且方差为 σ^2 , 且

$$E(\eta_1) = \cdots = E(\eta_{n-1}) = 0, E(\eta_n) = \sqrt{n}a$$

由正交变换的保范性可知

$$\sum_{k=1}^n \eta_k^2 = \|\vec{\eta}\|^2 = \|\vec{\xi}\|^2 = \sum_{k=1}^n \xi_k^2$$

注意到 $\eta_n = \frac{1}{\sqrt{n}} \sum_{k=1}^n \xi_k = \sqrt{n} \bar{\xi}$, 而

$$nS_n^2 = \sum_{j=1}^n (\xi_k - \bar{\xi})^2 = \sum_{k=1}^n \xi_k^2 - n\bar{\xi}^2 = \sum_{k=1}^n \eta_k^2 - \eta_n^2 = \sum_{k=1}^{n-1} \eta_k^2$$

由 $\eta_1, \dots, \eta_n$ 相互独立即知 $\bar{\xi} = \frac{\eta_n}{\sqrt{n}}$ 与 $S_n^2 = \frac{1}{n} \sum_{k=1}^{n-1} \eta_k^2$ 相互独立.

(2) 由 $\bar{\xi} = (\frac{1}{n}, \dots, \frac{1}{n}) \cdot \vec{\xi}$ 即知 $\bar{\xi} \sim N(a, \frac{\sigma^2}{n})$.

(3) 我们有 $\frac{nS_n^2}{\sigma^2} = \sum_{k=1}^{n-1} \left(\frac{\eta_k}{\sigma}\right)^2$, 其中 $\frac{\eta_k}{\sigma} \sim N(0, 1)$ 且互相独立, 由之前作业的计算可知 $\frac{\eta_k}{\sigma}$ 为 iid 且服从 $\Gamma\left(\frac{1}{2}, \frac{1}{2}\right)$, 于是其和 $\frac{nS_n^2}{\sigma^2}$ 服从 $\Gamma\left(\frac{1}{2}, \frac{n-1}{2}\right) = \chi^2(n-1)$.

Y-Problem 3.4.5.15

证明: $\xi_1, \dots, \xi_n$ 相互独立的充要条件是其联合特征函数等于各边缘特征函数之积.

证明: 必要性: 设 $\xi_1, \dots, \xi_n$ 独立, 则

$$f_{(\xi_1, \dots, \xi_n)}(\vec{t}) = E(e^{it_1 \xi_1} \dots e^{it_n \xi_n}) = E(e^{it_1 \xi_1}) \dots E(e^{it_n \xi_n}) = f_{\xi_1}(t_1) \dots f_{\xi_n}(t_n)$$

充分性: 设 $\xi_1, \dots, \xi_n$ 的分布为 $F_1, \dots, F_n$, 于是 $F(\vec{t}) := \prod_{k=1}^n F_k(t_k)$ 也为分布, 且

$$f_{\xi_1}(t_1) \dots f_{\xi_n}(t_n) = \prod_{k=1}^n \int_{\mathbb{R}} e^{it_k x} dF_k(x) = \int \dots \int_{\mathbb{R}^n} e^{i\vec{t}^T \vec{x}} dF(\vec{x})$$

故可将 $f_{\xi_1}(t_1) \dots f_{\xi_n}(t_n)$ 视作 n 维特征函数, 其对应分布函数 F , 而 $f_{(\xi_1, \dots, \xi_n)}(\vec{t}) = f_{\xi_1}(t_1) \dots f_{\xi_n}(t_n)$, 前者对应分布函数 $F_{(\xi_1, \dots, \xi_n)}$, 由唯一性定理即知

$$F_{(\xi_1, \dots, \xi_n)}(\vec{t}) = F(\vec{t}) = \prod_{k=1}^n F_k(t_k)$$

即 $\xi_1, \dots, \xi_n$ 独立.

Y-Problem 3.5.5.2

若 $\vec{\xi} = (\xi_1, \dots, \xi_n)^T$ 服从 n 元正态分布. 证明: 对任何常数 $b_0, b_1, \dots, b_n$ 有

$$P(b_1 \xi_1 + \dots + b_n \xi_n = b_0) = 0 \quad \text{或} \quad 1$$

证明: 若 $(b_0, \dots, b_n) = 0$, 则所求概率为 0, 否则 $\mathbb{R}^n$ 中曲面 $b_1 x_1 + \dots + b_n x_n = b_0$ 的维数是 $n-1$, 由实分析的知识可知其为 $\mathbb{R}^n$ 中的零测集, 其上任何有限函数的积分均为 0. 而 $\vec{\xi}$ 为 n 维连续型随机变量, 故

$$P(b_1 \xi_1 + \dots + b_n \xi_n = b_0) = \int \dots \int_{b_1 x_1 + \dots + b_n x_n = b_0} p_{\vec{\xi}}(\vec{x}) dx_1 \dots dx_n = 0$$

Y-Problem 3.5.5.4

设 $(\xi_1, \xi_2)^T$ 服从二元正态分布, 对 $k, j = 1, 2$ 有 $E(\xi_k) = 0$ 与 $E(\xi_j \xi_k) = \sigma^2 > 0$. 试求 $\eta = \sigma^{-2} \xi_1 \xi_2$ 的密度函数.

证明: 可知 $\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2 & \sigma^2 \\ \sigma^2 & \sigma^2 \end{pmatrix}\right)$, 其中 B 为半正定阵, 可见其为退化的二元正态分布. 考虑如下线性变换

$$\begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2\sigma^2 & 0 \\ 0 & 0 \end{pmatrix}\right)$$

可见 η_1, η_2 相互独立, 且 $\eta_1 \sim N(0, 2\sigma^2)$, η_2 为退化的一维正态分布, 即 $\eta_2 \equiv 0$. 由 $\eta_1 = \frac{1}{\sqrt{2}}(\xi_1 + \xi_2)$, $\eta_2 = \frac{1}{\sqrt{2}}(\xi_1 - \xi_2)$ 即知 $\eta = \sigma^{-2} \xi_1 \xi_2 = \frac{1}{2\sigma^2}(\eta_1^2 - \eta_2^2) = \frac{\eta_1^2}{2\sigma^2}$, 故

$$\begin{aligned} F_\eta(x) &= P\left(\frac{\eta_1^2}{2\sigma^2} < x\right) = P(-\sqrt{2x}\sigma < \eta_1 < \sqrt{2x}\sigma) = \int_{-\sqrt{2x}\sigma}^{\sqrt{2x}\sigma} \frac{1}{2\sqrt{\pi}\sigma} e^{-\frac{t^2}{4\sigma^2}} dt \\ \Rightarrow p_\eta(x) &= F'_\eta(x) = 2 \cdot \frac{\sigma}{\sqrt{2x}} \frac{1}{2\sqrt{\pi}\sigma} e^{-\frac{x}{2}} = \frac{1}{\sqrt{2\pi x}} e^{-\frac{x}{2}} \end{aligned}$$

Y-Problem 3.5.5.5

设随机变量 ξ, η 独立, 服从 $N(0, 1)$, 令

$$\zeta = \begin{cases} |\eta|, & \xi > 0 \\ -|\eta|, & \xi < 0 \end{cases}$$

由例题可知 $\zeta \sim N(0, 1)$. 求 $(\zeta, \xi)^T$ 的协方差阵, 并证明它不服从二元正态分布.

解: 由正态分布即知 $\text{cov}(\zeta, \zeta) = D(\zeta) = 1$, $\text{cov}(\xi, \xi) = 1$. 只需计算 $\text{cov}(\zeta, \xi)$:

$$\begin{aligned} \text{cov}(\zeta, \xi) &= E(\zeta \xi) - E(\zeta)E(\xi) = E(\zeta \xi) \\ &= \frac{1}{2}E(|\eta|\xi \mid \xi > 0) + \frac{1}{2}E(-|\eta|\xi \mid \xi < 0) \\ &\stackrel{\text{indep}}{=} \frac{1}{2}E(|\eta|)E(\xi \mid \xi > 0) - \frac{1}{2}E(|\eta|)E(\xi \mid \xi < 0) \\ &= \frac{1}{4}(E(\eta \mid \eta > 0) + E(-\eta \mid \eta < 0))(E(\xi \mid \xi > 0) - E(\xi \mid \xi < 0)) \\ &= E^2(\xi \mid \xi > 0) = 4 \left(\int_0^\infty \frac{1}{\sqrt{2\pi}} x e^{-\frac{x^2}{2}} dx \right)^2 = \frac{2}{\pi} \end{aligned}$$

故协方差阵为 $\begin{pmatrix} 1 & \frac{2}{\pi} \\ \frac{2}{\pi} & 1 \end{pmatrix}$. 由 $P(\xi > 0, \zeta < 0) = 0$ 即知 $(\zeta, \xi)^T$ 不服从二元正态分布.

Y-Problem 3.5.5.6

设 $(\xi, \eta)^T$ 服从二元正态分布, 试求 $\xi + \eta$ 与 $\xi - \eta$ 相互独立的充要条件.

解: 设 $(\xi, \eta)^T \sim N\left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & r\sigma_1\sigma_2 \\ r\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}\right), \sigma_1, \sigma_2 > 0, -1 < r < 1$, 计算得

$$\begin{pmatrix} \xi + \eta \\ \xi - \eta \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix} \\ \sim N\left(\begin{pmatrix} a_1 + a_2 \\ a_1 - a_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 + 2r\sigma_1\sigma_2 + \sigma_2^2 & \sigma_1^2 - \sigma_2^2 \\ \sigma_1^2 - \sigma_2^2 & \sigma_1^2 - 2r\sigma_1\sigma_2 + \sigma_2^2 \end{pmatrix}\right)$$

于是 $\xi + \eta$ 与 $\xi - \eta$ 独立等价于

$$\sigma_1^2 - \sigma_2^2 = 0 \iff \sigma_1 = \sigma_2$$

L-Problem 1

Assume $\vec{\xi} = (\xi_1, \xi_2, \xi_3)^T \sim N(\vec{a}, B)$, where

$$\vec{a} = (1, 0, -1)^T, B = \begin{pmatrix} 4 & 2 & -2 \\ 2 & 10 & 2 \\ -2 & 2 & 5 \end{pmatrix}$$

Define $X = \xi_1 - 2\xi_2 + \frac{1}{4}\xi_3, Y = 2\xi_2 - \xi_2 + 5\xi_3$.

求: (1) $F_{X,Y}$;

(2) $E(X|Y), E(Y|X)$;

(3) $D(X|Y)$.

解: 我们有 $\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} 1 & -2 & \frac{1}{4} \\ 2 & -1 & 5 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix}$, 于是计算可知

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N\left(\begin{pmatrix} \frac{3}{4} \\ -3 \end{pmatrix}, \begin{pmatrix} \frac{533}{16} & -\frac{29}{4} \\ -\frac{29}{4} & 83 \end{pmatrix}\right) = N\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & r\sigma_1\sigma_2 \\ r\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}\right)$$

其中

$$\mu_1 = \frac{3}{4}, \mu_2 = -3, \sigma_1 = \frac{\sqrt{533}}{4}, \sigma_2 = \sqrt{83}, r = -\frac{29}{\sqrt{44239}}$$

故我们有密度函数

$$\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-r^2}} \exp\left(-\frac{1}{2(1-r^2)} \left[\frac{(x-\mu_1)^2}{\sigma_1^2} - \frac{2r(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2} \right]\right)$$

于是分布函数即为 $F_{(X,Y)}(x,y) = \int_{-\infty}^x \int_{-\infty}^y p(s,t) dt ds$.

(2) 对于如上二元正态分布, 利用熟知的计算结果

$$E(X|Y) = \mu_1 + r \frac{\sigma_1}{\sigma_2}(Y - \mu_2), E(Y|X) = \mu_2 + r \frac{\sigma_2}{\sigma_1}(X - \mu_1)$$

可知

$$E(X|Y) = \frac{3}{4} - \frac{29}{332}(Y + 3) = -\frac{29}{332}Y + \frac{81}{166}$$
$$E(Y|X) = -3 - \frac{116}{533}\left(X - \frac{3}{4}\right) = -\frac{116}{533}X - \frac{1512}{533}$$

(3) 利用熟知的计算结果

$$D(X|Y) = \sigma_1^2(1 - r^2) = \frac{21699}{664}$$

K-Problem 4.1.2

If $0 \leq X_n \leq X \in L^1$ and $X_n \xrightarrow{\mathbb{P}} X$, then $X_n \xrightarrow{L^1} X$

Proof: 存在子列 $\{X_{n_k}\}$ 使得 $\lim_{k \rightarrow \infty} E(X_{n_k}) = \lim_n E(X_n)$, 对该子列, 仍存在子列 $\{X_{n'_k}\}$ 几乎处处收敛到 X , 于是由 Fatou 引理知

$$E(X) \leq \lim_{k \rightarrow \infty} E(X_{n'_k}) = \lim_{k \rightarrow \infty} E(X_{n_k}) = \lim_n E(X_n)$$

对 $-X \leq -X_n \leq 0$ 重复上述证明, 可得

$$E(-X) \leq \lim_n E(-X_n) \implies E(X) \geq \overline{\lim}_n E(X_n)$$

于是得到 $\lim_n E(X_n) = E(X)$, 由课上所讲, 该条件结合依概率收敛的 L^1 收敛.

K-Problem 4.1.3

If $X_n \xrightarrow{\mathbb{P}} X, Y_n \xrightarrow{\mathbb{P}} Y$, then $X_n \pm Y_n \xrightarrow{\mathbb{P}} X \pm Y, X_n Y_n \xrightarrow{\mathbb{P}} XY$.

Proof: $X_n + Y_n \xrightarrow{\mathbb{P}} X + Y$: 只需证明对任意子列 $\{X_{n_k} + Y_{n_k}\}$, 存在子列的子列 $\{X_{n'_k} + Y_{n'_k}\}$ 收敛到 $X + Y$. 由条件知存在 $\{X_{n'_k}\} \subset \{X_{n_k}\}$ 收敛到 X , 于是对子列 $\{Y_{n'_k}\}$, 存在子列 $\{Y_{n''_k}\}$ 收敛到 Y , 从而 $\{X_{n''_k} + Y_{n''_k}\}$ 收敛到 $X + Y$. 这对 $\{X_n - Y_n\}, \{X_n Y_n\}$ 完全同理.

K-Problem 4.1.6

If $X_n \xrightarrow{L^p} X, Y_n \xrightarrow{L^p} Y$, then $X_n \pm Y_n \xrightarrow{L^p} X \pm Y$. If $X_n \xrightarrow{L^p} X, Y_n \xrightarrow{L^q} Y$, where we have $p > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$, the $X_n Y_n \xrightarrow{L^1} XY$.

Proof: 由 C_p 不等式可知

$$E|(X_n + Y_n) - (X + Y)|^p \leq C_p(E|X_n - X|^p + E|Y_n - Y|^p)$$

令 $n \rightarrow \infty$ 即知 $X_n \pm Y_n \xrightarrow{L^p} X \pm Y$. 再利用 Hölder 不等式, 有

$$\begin{aligned} E|X_n Y_n - XY| &\leq E|X_n||Y_n - Y| + E|Y||X_n - X| \\ &\leq \|X_n\|_p \|Y_n - Y\|_q + \|Y\|_q \|X_n - X\|_p \rightarrow 0 (n \rightarrow \infty) \end{aligned}$$

注意最后一步趋于 0 需使用 $X_n \xrightarrow{L^p} X \implies E|X_n|^p \rightarrow E|X|^p$, 因此 $X_n Y_n \xrightarrow{L^1} XY$.

K-Problem 4.1.9

Give an example in which $E(X_n) \rightarrow 0$ but there does not exist any subsequence $\{X_{n_k}\}$ such that $X_{n_k} \xrightarrow{\mathbb{P}} 0$.

Solution: 考虑取 $\Omega = [0, 1)$, $\mathcal{F} = \mathcal{B}(\Omega)$, $dP = dx$ 为 Lebesgue 测度. 将 $[0, 1)$ 均分为 n 份, X_n 满足:

$$X_n = \begin{cases} 1, & x \in [\frac{i}{n}, \frac{i+1}{n}), i \text{ 为偶数} \\ -1, & x \in [\frac{i}{n}, \frac{i+1}{n}), i \text{ 为奇数} \end{cases}, i = 0, 1, \dots, n$$

易见此时 $E(X_n) = 0$, n 为偶数; $E(X_n) = \frac{1}{n}$, n 为奇数. 于是 $E(X_n) \rightarrow 0$, 但对任何 X_n 与 $0 < \varepsilon < 1$, 有

$$P(\{\omega : |X_n| > \varepsilon\}) = P([0, 1)) = 1 > 0$$

于是不存在子列依概率收敛到 0.

K-Problem 4.1.18

If $X_n \downarrow X$ a.s., each X_n is integrable and $\inf_n E(X_n) > -\infty$, then $X_n \xrightarrow{L^1} X$.

Proof: 考虑 $\{-X_n\}$, 故只需证明对偶命题: 设 $X_n \uparrow X$ a.s., X_n 可积且 $\sup_n E(X_n) < +\infty$, 则 $X_n \xrightarrow{L^1} X$. 可知 $\{X_n - X_1\}$ 为非负单增趋于 $X - X_1$ 的随机变量列, 故由 MCT 可知

$$E(X) - E(X_1) = \lim_n E(X_n) - E(X_1) \implies E(X) = \lim_n E(X_n) < +\infty$$

于是 X 可积, 故 $|X_n - X| \leq |X_n| + |X| < 2|X|$, 由 DCT 即知

$$\lim_n E|X_n - X| = E\left(\lim_n |X_n - X|\right) = 0$$

即 $X_n \xrightarrow{L^1} X$

K-Problem 4.2.4

For any sequence of rv's $\{X_n\}$ there exists a sequence of constants $\{A_n\}$ s.t. $\frac{X_n}{A_n} \xrightarrow{\text{a.s.}} 0$.

Proof: 由于

$$\lim_{n \rightarrow \infty} P(\{|X| > n\}) = P\left(\bigcap_{n=1}^{\infty} \{|X| > n\}\right) = 0$$

对任意 X 成立. 故存在 $\{a_n\}$ 使 $P(|X_n| > a_n) < \frac{1}{n^2}$. 记 $A_n = na_n$, 固定 $m \in \mathbb{N}^*$, 则

$$\begin{aligned} \sum_{n=1}^{\infty} P\left(\left\{\omega : \left|\frac{X_n}{A_n}\right| > \frac{1}{m}\right\}\right) &= \sum_{n=1}^{\infty} P\left(\left\{\omega : |X_n| > \frac{n}{m}a_n\right\}\right) \\ &< \sum_{n=1}^m P\left(\left\{\omega : |X_n| > \frac{n}{m}a_n\right\}\right) + \sum_{n \geq m} P(\{\omega : |X_n| > a_n\}) < +\infty \end{aligned}$$

故由 Borel-Cantelli 引理可知 $P(\{\omega : |X_n/A_n| > \frac{1}{m}\} \text{ i.o.}) = 0$. 上式对任意 $m \geq 1$ 成立, 故由几乎处处收敛的等价定义即知 $X_n/A_n \xrightarrow{\text{a.s.}} 0$.

K-Problem 4.2.6

Cauchy convergence of $\{X_n\}$ in pr (or in L^p) implies the existence of an X (finite a.e.), s.t. X_n converges to X in pr (or in L^p).

Proof: 先证明存在子列 $\{X_{n_k}\}$ 使得

$$P\left(\left\{\omega : |X_{n_{k+1}} - X_{n_k}| > \frac{1}{2^k}\right\}\right) < \frac{1}{2^k}$$

取 N_1 使得对任意 $i, j > N_1$ 有 $P(\{\omega : |X_i - X_j| > \frac{1}{2}\}) < \frac{1}{2}$, 取 $n_1 > N_1$. 再取 N_2 使得对任意 $i, j > N_2$ 有 $P(\{\omega : |X_i - X_j| > \frac{1}{2^2}\}) < \frac{1}{2^2}$, 再取 $n_2 > \max(n_1, N_2)$. 重复以上操作, 取 N_k 使得对任意 $i, j > N_k$ 有 $P(\{\omega : |X_i - X_j| > \frac{1}{2^k}\}) < \frac{1}{2^k}$, 再取 $n_k > \max(N_k, n_{k-1})$. 这样得到的子列 $\{X_{n_k}\}$ 满足要求.

再证明存在 X 使得 $X_{n_k} \xrightarrow{\text{a.s.}} X$, 令

$$F_k = \left\{\omega : |X_{n_{k+1}} - X_{n_k}| > \frac{1}{2^k}\right\}$$

则有

$$P\left(\bigcup_{k=p}^{\infty} F_k\right) \leq \sum_{k=p}^{\infty} P(F_k) < \sum_{k=p}^{\infty} \frac{1}{2^k} \rightarrow 0 (p \rightarrow \infty) \implies P\left(\overline{\lim}_k F_k\right) = 0$$

对任意 $\omega \in \Omega \setminus \overline{\lim}_k F_k$, 存在 K 使得对任意 $k > K$ 有 $|X_{n_{k+1}} - X_{n_k}| \leq \frac{1}{2^k}$, 于是由数列收敛的 Cauchy 定理即知 $\{X_{n_k}(\omega)\}$ 收敛. 故 $\{X_{n_k}\}$ 几乎处处收敛到某个 X .

最后证明 $X_n \xrightarrow{\mathbb{P}} X$. 固定 $\varepsilon_1, \varepsilon_2 > 0$, 于是存在 N, K 使得对任意的 $i, j > N$ 成立 $P(\{\omega : |X_i - X_j| > \frac{\varepsilon_1}{2}\}) < \frac{\varepsilon_2}{2}$, 对任意 $n_k > K$, $P(\{\omega : |X_{n_k} - X| > \frac{\varepsilon_1}{2}\}) < \frac{\varepsilon_2}{2}$. 故对任意的 $n > \max(N, K)$, 存在 $n_k > n$, 从而

$$P(\{|X_n - X| > \varepsilon_1\}) \leq P\left(\left\{|X_n - X_{n_k}| > \frac{\varepsilon_1}{2}\right\}\right) + P\left(\left\{|X_{n_k} - X| > \frac{\varepsilon_1}{2}\right\}\right) < \varepsilon_2$$

于是 $\lim_n P(\{\omega : |X_n - X| > \varepsilon_1\}) = 0$, 即 $X_n \xrightarrow{\mathbb{P}} X$.

K-Problem 4.2.7

$\{X_n\}$ converges in pr to X if and only if every subsequence $\{X_{n_k}\}$ contains a further subsequence that converges a.e. to X .

Proof: $\implies$: It's easy to see that for any $\{\xi_{n'}\} \subset \{\xi_n\}$, $\xi_{n'} \xrightarrow{\mathbb{P}} \xi$. Thus without loss of generality, we only need to prove that there exists $\{\xi_{n_k}\} \subset \{\xi_n\}$, s.t. $\xi_{n_k} \xrightarrow{\text{a.s.}} \xi$. Actually it's Riesz thm. From definition, we can choose a subsequence $\{\xi_n^1\}$ s.t.

$$P(\{\omega : |\xi_n^1 - \xi| > 1\}) < \frac{1}{2^n}, \forall n \geq 1$$

Next, from $\{\xi_n^1\}$, we extract a further subsequence $\{\xi_n^2\}$ such that

$$P\left(\left\{\omega : |\xi_n^2 - \xi| > \frac{1}{2}\right\}\right) < \frac{1}{2^n}, \forall n \geq 1$$

Proceeding inductively, we get a subsequence $\{\xi_n^n\}$. Observe that for any $j > i$, the term ξ_j^j belongs to $\{\xi_n^i\}$ at least from index j onward. Therefore,

$$\begin{aligned} P\left(\left\{\omega : |\xi_j^j - \xi| > \frac{1}{i}\right\}\right) < \frac{1}{2^j} &\Rightarrow P\left(\bigcup_{p=j}^{\infty} \left\{\omega : |\xi_p^j - \xi| > \frac{1}{i}\right\}\right) < \sum_{p=j}^{\infty} \frac{1}{2^p} \\ \Rightarrow P\left(\bigcap_{j=1}^{\infty} \bigcup_{p=j}^{\infty} \left\{\omega : |\xi_p^j - \xi| > \frac{1}{i}\right\}\right) &= \lim_{j \rightarrow \infty} P\left(\bigcup_{p=j}^{\infty} \left\{\omega : |\xi_p^j - \xi| > \frac{1}{i}\right\}\right) = 0 \end{aligned}$$

Hence

$$P\left(\bigcup_{i=1}^{\infty} \bigcap_{j=1}^{\infty} \bigcup_{p=j}^{\infty} \left\{\omega : |\xi_p^j - \xi| > \frac{1}{i}\right\}\right) = 0$$

which implies that $\xi_n^n \xrightarrow{\text{a.s.}} \xi$.

$\Leftarrow$: If $\xi_n \xrightarrow{\mathbb{P}} \xi$ doesn't hold, then there exists ε_0 and $\delta > 0$, s.t.

$$\overline{\lim}_n P(\{\omega : |\xi_n - \xi| > \varepsilon_0\}) > \delta > 0$$

Thus there exists $\{\xi_{n_k}\}$ s.t. $\lim_k P(\{\omega : |\xi_{n_k} - \xi| > \varepsilon_0\}) > \delta > 0$. Then there is no $\{\xi_{n'_k}\} \subset \{\xi_{n_k}\}$ s.t. $\xi_{n'_k} \xrightarrow{\text{a.s.}} \xi$ due to

$$\xi_{n'_k} \xrightarrow{\text{a.s.}} \xi \Rightarrow \xi_{n'_k} \xrightarrow{\mathbb{P}} \xi \Rightarrow \lim_k P(\{\omega : |\xi_{n'_k} - \xi| > \varepsilon_0\}) = 0$$

It's contradicts with the condition.

K-Problem 4.2.17

If $E(X_n) = 1$ and $E(X_n^2)$ is bounded in n , then

$$P\left(\left\{\overline{\lim}_{n \rightarrow \infty} X_n \geq 1\right\}\right) > 0$$

Proof: 对 $A > 0$, 考虑 $\{X_n\}$ 的截断函数

$$X_n^A = \begin{cases} A, & X_n > A \\ X_n, & X_n \leq A \end{cases}$$

于是 $|X_n^A| \leq |X_n|$, 故 X_n^A 可积. 我们有

$$\begin{aligned} E(X_n - X_n^A) &= \int_{\{X_n > A\}} (X_n - A) dP \\ &= \int_{\{X_n > A\}} X_n dP - AP(\{X_n > A\}) \leq \int_{\{X_n > A\}} X_n dP \end{aligned}$$

设 $\sup_n E(X_n^2) < M$, 于是有

$$\begin{aligned} \int_{\{X_n > A\}} X_n dP &< \frac{1}{A} \int_{\{X_n > A\}} X_n^2 dP < \frac{M}{A} \\ \Rightarrow E(X_n^A) &> 1 - \frac{M}{A}, \forall n \geq 1 \end{aligned}$$

注意 $\{A - X_n^A\}$ 为非负随机变量, 故有 Fatou 引理知

$$\begin{aligned} \int_{\Omega} \liminf_n (A - X_n^A) dP &\leq \liminf_n \int_{\Omega} (A - X_n^A) dP \\ \Rightarrow E\left(\overline{\lim}_n X_n^A\right) &\geq \overline{\lim}_n E(X_n^A) \geq 1 - \frac{M}{A} \end{aligned}$$

而 $X_n^A \leq X_n$, 故 $\overline{\lim}_n X_n^A \leq \overline{\lim}_n X_n$, 于是 $E(\overline{\lim}_n X_n) \geq 1 - \frac{M}{A}$, 由 A 的任意性, 令 $A \rightarrow \infty$ 即得 $E(\overline{\lim}_n X_n) \geq 1$. 从而若 $P(\{\overline{\lim}_n X_n \geq 1\}) = 0$, 则

$$1 \leq E(\overline{\lim}_n X_n) = \int_{\{\overline{\lim}_n X_n < 1\}} \overline{\lim}_n X_n dP < P(\{\overline{\lim}_n X_n < 1\})$$

矛盾! 因此有 $P(\{\overline{\lim}_n X_n \geq 1\}) > 0$.

K-Problem 4.2.18

Let $\{E_n\}$ be events and $\{I_n\}$ be their indicators. Prove the inequality

$$P\left(\bigcup_{k=1}^n E_k\right) \geq \frac{\left(E\left(\sum_{k=1}^n I_k\right)\right)^2}{E\left(\left(\sum_{k=1}^n I_k\right)^2\right)}$$

Deduce from this that if (i) $\sum_n P(E_n) = \infty$ and (ii) there exists $c > 0$ s.t. we have

$$\forall m < n : P(E_m E_n) \leq cP(E_m)P(E_{n-m})$$

then

$$P\left(\overline{\lim}_n E_n\right) > 0$$

Proof: (1) 考虑随机变量 $S := \sum_{k=1}^n I_k$, 注意到

$$\omega \in \bigcup_{k=1}^n E_k \iff \omega \in \{\omega : S \geq 1\}$$

这说明

$$I_{\bigcup_{k=1}^n E_k} = I_{\{\omega: S(\omega) \geq 1\}} \implies P\left(\bigcup_{k=1}^n E_k\right) = E(I_{\{\omega: S \geq 1\}})$$

于是由 Cauchy 不等式

$$P\left(\bigcup_{k=1}^n E_k\right) E\left(\left(\sum_{k=1}^n I_k\right)^2\right) = E(I_{\{\omega: S \geq 1\}}) E(S^2) \geq E^2(S \cdot I_{\{\omega: S \geq 1\}}) = E^2(S)$$

(2) 对任意 $k > m$, 有

$$\begin{aligned} P\left(\bigcup_{n=k}^m E_n\right) &\geq \frac{E^2\left(\sum_{n=k}^m I_n\right)}{E\left(\left(\sum_{n=k}^m I_n\right)^2\right)} = \frac{\left(\sum_{n=k}^m P(E_n)\right)^2}{\sum_{n=k}^m P(E_n) + 2 \sum_{k \leq i < j \leq m} P(E_i E_j)} \\ &\geq \frac{\left(\sum_{n=k}^m P(E_n)\right)^2}{\sum_{n=k}^m P(E_n) + 2c \sum_{i=k}^m \sum_{j=1}^{m-i} P(E_i) P(E_j)} \\ &\geq \frac{\left(\sum_{n=k}^m P(E_n)\right)^2}{\sum_{n=k}^m P(E_n) + 2c \sum_{i=1}^m \sum_{j=1}^m P(E_i) P(E_j)} \\ &= \frac{1}{\left(\sum_{n=k}^m P(E_n)\right)^{-1} + 2c \left(\frac{\sum_{n=1}^{k-1} P(E_n)}{\sum_{n=k}^m P(E_n)} + 1\right)^2} \end{aligned}$$

两侧令 $m \rightarrow \infty$, 由概率的上连续性以及 $\sum_{n=k}^{\infty} P(E_n) = \infty$ 可知

$$P\left(\bigcup_{n=k}^{\infty} E_n\right) \geq \frac{1}{2c} \implies P\left(\overline{\lim}_n E_n\right) = \lim_{k \rightarrow \infty} P\left(\bigcup_{n=k}^{\infty} E_n\right) \geq \frac{1}{2c} > 0$$

K-Problem 4.2.19

If $\sum_n P(E_n) = \infty$ and

$$\lim_n \frac{\sum_{j=1}^n \sum_{k=1}^n P(E_j E_k)}{\left(\sum_{k=1}^n P(E_k)\right)^2} = 1$$

then $P\left(\overline{\lim}_n E_n\right) = 1$.

Proof: 我们证明如下 Kochen-Stone Lemma, 若

$$\lim_n \frac{\sum_{j=1}^n \sum_{k=1}^n P(E_j E_k)}{\left(\sum_{k=1}^n P(E_k)\right)^2} = c > 0$$

则 $P(\overline{\lim}_n E_n) \geq \frac{1}{c}$.

由 18 题的不等式, 我们有如下估计

$$\begin{aligned} P\left(\bigcup_{k=l}^n E_k\right) &\geq \frac{\left(E\left(\sum_{k=l}^n I_k\right)\right)^2}{E\left(\left(\sum_{k=l}^n I_k\right)^2\right)} = \frac{\left(\sum_{k=l}^n P(E_k)\right)^2}{\sum_{i=l}^n \sum_{j=l}^n P(E_i E_j)} \\ &\geq \frac{\left(\sum_{k=1}^n P(E_k) - \sum_{k=1}^{l-1} P(E_k)\right)^2}{\sum_{i=1}^n \sum_{j=1}^n P(E_i E_j)} \\ &= \left(1 - \frac{\sum_{k=1}^{l-1} P(E_k)}{\sum_{k=1}^n P(E_k)}\right)^2 \left(\frac{\sum_{i=1}^n \sum_{j=1}^n P(E_i E_j)}{\left(\sum_{k=1}^n P(E_k)\right)^2}\right)^{-1} \end{aligned}$$

固定 l , 令 $n \rightarrow \infty$ 并取上极限, 由 $\sum_{k=1}^{\infty} P(E_k) = \infty$ 以及条件即知

$$P\left(\bigcup_{k=l}^{\infty} E_k\right) \geq \frac{1}{c}$$

上式对任意 l 成立, 于是

$$P\left(\overline{\lim}_n E_n\right) = \lim_{l \rightarrow \infty} P\left(\bigcup_{k=l}^{\infty} E_k\right) \geq \frac{1}{c}$$

令 $c = 1$, 立得原题结论.

Y-Problem 4.1.4.1

取 $\Omega = [0, 1)$, $\mathcal{F} = \mathcal{B}(\Omega)$, $dP = dx$, 定义

$$\eta_{ki} = I_{\left[\frac{i-1}{k}, \frac{i}{k}\right)}, k \geq 1, i = 1, 2, \dots, k$$

$\xi_1 = \eta_{11}, \xi_2 = \eta_{21}, \xi_3 = \eta_{22}, \xi_4 = \eta_{31}, \dots$, 证明: $\xi_n \xrightarrow{L^p} 0$.

证明: 易见

$$E|\eta_{ki}|^p = \frac{1}{k} \rightarrow 0 (k \rightarrow \infty)$$

而 ξ_n 的指标 $n \rightarrow \infty$ 时, 对应 η_{ki} 的指标 $k \rightarrow \infty$, 故 $E|\xi_n|^p \rightarrow 0$.

Y-Problem 4.1.4.4

试证 $\xi_n \xrightarrow{\mathbb{P}} \xi$ 的充要条件是

$$E\left(\frac{|\xi_n - \xi|}{1 + |\xi_n - \xi|}\right) \rightarrow 0$$

证明: $\xi_n \xrightarrow{\mathbb{P}} \xi \iff \xi_n - \xi \xrightarrow{\mathbb{P}} 0$, 因此不妨设 $\xi \equiv 0$.

必要性: 固定 $\varepsilon > 0$, 于是

$$\begin{aligned} E\left(\frac{|\xi_n|}{1 + |\xi_n|}\right) &= \int_{\{\omega: |\xi_n| > \varepsilon\}} \frac{|\xi_n|}{1 + |\xi_n|} dP + \int_{\{\omega: |\xi_n| < \varepsilon\}} \frac{|\xi_n|}{1 + |\xi_n|} dP \\ &\leq P(\{\omega: |\xi_n| > \varepsilon\}) + \frac{\varepsilon}{1 + \varepsilon} \end{aligned}$$

令 $n \rightarrow \infty$, 得到 $\overline{\lim}_n \int_{\Omega} \frac{|\xi_n|}{1 + |\xi_n|} dP \leq \frac{\varepsilon}{1 + \varepsilon}$, 由 ε 任意性, 再令 $\varepsilon \rightarrow 0$ 即知.

充分性: L^1 收敛蕴涵依概率收敛, 因此对任意 $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} P\left(\left\{\omega: \frac{|\xi_n|}{1 + |\xi_n|} > \varepsilon\right\}\right) = 0 \implies \lim_{n \rightarrow \infty} P\left(\left\{\omega: |\xi_n| > \frac{\varepsilon}{1 - \varepsilon}\right\}\right) = 0$$

故对任意 $\varepsilon' > 0$, 将上式的 ε 代入 $\frac{\varepsilon'}{1 + \varepsilon'}$ 即得 $\lim_n P(\{\omega: |\xi_n| > \varepsilon'\}) = 0$.

Y-Problem 4.1.6

设 $\{\xi_n\}$ 为非负随机变量的单调下降序列, 且 $\xi_n \xrightarrow{\mathbb{P}} 0$, 证明: $\xi_n \xrightarrow{\text{a.s.}} 0$.

证明: 只需证对任意 $\varepsilon > 0$, $P(\{\omega: |\xi_n| > \varepsilon\} \text{ i.o.}) = 0$. 事实上, 由单调性即知

$$P(\{\omega: |\xi_n| > \varepsilon\} \text{ i.o.}) = \lim_{k \rightarrow \infty} P\left(\bigcup_{n=k}^{\infty} \{\omega: \xi_n > \varepsilon\}\right) = \lim_{k \rightarrow \infty} P(\{\omega: \xi_k > \varepsilon\}) = 0$$

Y-Problem 4.1.7

设对于一切自然数 $n \geq 1$, ξ_n 服从 Cauchy 分布, 密度函数为 $p_n(x) = \frac{n}{\pi(1 + n^2 x^2)}$,

证明: $\xi_n \xrightarrow{\mathbb{P}} 0$.

证明: 直接计算即知

$$\begin{aligned} P(\{\omega: |\xi_n| > \varepsilon\}) &= P(\{\omega: \xi_n < -\varepsilon\}) + P(\{\omega: \xi_n > \varepsilon\}) \\ &= 2 \int_{-\infty}^{-\varepsilon} \frac{n}{\pi(1 + n^2 x^2)} dx = \frac{2}{\pi} \left(-\arctan n\varepsilon + \frac{\pi}{2} \right) \end{aligned}$$

令 $n \rightarrow \infty$ 可得 $\lim_n P(\{\omega: |\xi_n| > \varepsilon\}) = 0$, 即 $\xi_n \xrightarrow{\mathbb{P}} 0$.

Y-Problem 4.1.4.9

设正态分布列 $\{F_n(x)\}$ 弱收敛到分布函数 $F(x)$, 求证: $F(x)$ 也是正态分布函数, 且相应的期望与方差也收敛.

证明: 设这列正态分布为 $\{N(a_n, \sigma_n^2)\}$, 于是特征函数列

$$f_n(t) = e^{ia_n t - \frac{1}{2}\sigma_n^2 t^2} \rightarrow f(t), \forall t$$

于是 $ia_n t - \frac{1}{2}\sigma_n^2 t^2 \rightarrow \log f(t)$, 由任意性可知 $\{a_n\}$ 与 $\{\sigma_n\}$ 存在极限 a, σ , 于是

$$f(t) = \lim_{n \rightarrow \infty} f_n(t) = e^{iat - \frac{1}{2}\sigma^2 t^2}$$

故 $F(x)$ 为 $N(a, \sigma^2)$ 的正态分布函数.

Y-Problem 4.1.4.10

设 $\{\xi_k\}$ 为独立随机变量序列, 它们都服从 -1 与 1 两点上的等可能分布. 试证随机变量列 $\eta_n = \sum_{k=1}^n \frac{\xi_k}{2^k}$ 依分布收敛到区间 $(-1, 1)$ 上的均匀分布随机变量.

证明: 容易求得 ξ_k 的特征函数为 $f_{\xi_k}(t) = \frac{1}{2}(e^{it} + e^{-it}) = \cos t$. 由独立性可知

$$f_{\eta_n}(t) = \prod_{k=1}^n E\left(e^{it \frac{\xi_k}{2^k}}\right) = \prod_{k=1}^n f_{\xi_k}\left(\frac{t}{2^k}\right) = \prod_{k=1}^n \cos\left(\frac{t}{2^k}\right) = \frac{\sin t}{2^n \sin \frac{t}{2^n}}$$

对任意 $t \in \mathbb{R}$, 令 $n \rightarrow \infty$ 可得

$$\lim_{n \rightarrow \infty} f_{\eta_n}(t) = \frac{\sin t}{t} = \frac{e^{it} - e^{-it}}{2it}$$

后者正是 $(-1, 1)$ 上均匀分布随机变量的特征函数, 因此 η_n 依分布收敛与该分布.

Y-Problem 4.1.4.11

设 $\xi_n \xrightarrow{d} \xi, \eta_n \xrightarrow{d} a \in \mathbb{R}$. 证明: $\xi_n + \eta_n \xrightarrow{d} \xi + a$.

证明: 我们先证明, $X_n \xrightarrow{\mathbb{P}} a \iff X_n \xrightarrow{d} a, a$ 为常数. 必要性已证, 只需证充分性, 我们有

$$\begin{aligned} P(\{|X_n - a| > \varepsilon\}) &= P(\{X_n > a + \varepsilon\}) + P(\{X_n < a - \varepsilon\}) \\ &= 1 + P(\{X_n < a - \varepsilon\}) - P(\{X_n \leq a + \varepsilon\}) \\ &\leq 1 + P(\{X_n < a - \varepsilon\}) - P(\{X_n < a + \varepsilon\}) \end{aligned}$$

令 $n \rightarrow \infty$, 由依分布收敛的定义即知右式趋于 $1 + F_a(a - \varepsilon) - F_a(a + \varepsilon) = 0$. 于是有 $\lim_n P(\{|X_n - a| > \varepsilon\}) = 0, \forall \varepsilon > 0$, 即 $X_n \xrightarrow{\mathbb{P}} a$.

回到本题可知 $\eta_n \xrightarrow{\mathbb{P}} a$. 于是有如下估计

$$P(\xi_n + \eta_n < x) \leq P(\xi_n + \eta_n < x, |\eta_n - a| < \varepsilon) + P(|\eta_n - a| > \varepsilon)$$

对事件 $\{\xi_n + \eta_n < x, |\eta_n - a| < \varepsilon\}$, 有

$$\{\xi_n + \eta_n < x, |\eta_n - a| < \varepsilon\} \subset \{\xi_n < x - a + \varepsilon\}$$

于是取上极限可得

$$\varlimsup_n P(\xi_n + \eta_n < x) \leq \lim_n P(\xi_n < x - a + \varepsilon) = P(\xi < x - a + \varepsilon)$$

对于下界, 我们有关系

$$\{\xi_n \leq x - a - \varepsilon\} \subset \{|\eta_n - a| > \varepsilon\} \cup \{\xi_n + \eta_n < x\}$$

因此

$$P(\xi_n + \eta_n < x) \geq P(\xi_n < x - a - \varepsilon) - P(|\eta_n - a| > \varepsilon)$$

取下极限可得

$$\varliminf_n P(\xi_n + \eta_n < x) \geq \lim_n P(\xi_n < x - a - \varepsilon) = P(\xi < x - a - \varepsilon)$$

取 $x \in C_{\xi+a}$, 于是 $x - a \in C_\xi$, 令上述两个式子中 $\varepsilon \rightarrow 0$, 即得

$$\lim_n P(\xi_n + \eta_n < x) = P(\xi + a < x)$$

也即 $\xi_n + \eta_n \xrightarrow{d} \xi + a$.

Y-Problem 4.1.4.12

设 $\xi_n \xrightarrow{d} \xi, \eta_n \xrightarrow{d} 0$. 证明: $\xi_n \eta_n \xrightarrow{d} 0$.

证明: 首先有 $\eta_n \xrightarrow{\mathbb{P}} 0$, 要证 $\xi_n \eta_n \xrightarrow{d} 0$, 只需证 $\xi_n \eta_n \xrightarrow{\mathbb{P}} 0$. 每个 ξ_n 以及 ξ 的不连续点是可列的, 因此这些不连续点集的并也是可列的. 因此对固定的 $\varepsilon > 0$, 可取充分大的 $A, -A \in C_{F_\xi} \cup \left(\bigcup_{n=1}^{\infty} C_{F_{\xi_n}} \right)$, 且 $P(\{|\xi| > A\}) < \varepsilon$. 于是有如下估计

$$\begin{aligned} P(\{|\xi_n \eta_n| > \delta\}) &\leq P(\{|\xi_n| > A\}) + P\left(\left\{|\eta_n| \geq \frac{\delta}{A}\right\}\right) \\ &= 1 - F_{\xi_n}(A + 0) + F_{\xi_n}(-A) + P\left(\left\{|\eta_n| \geq \frac{\delta}{A}\right\}\right) \\ &= 1 - F_{\xi_n}(A) + F_{\xi_n}(-A) + P\left(\left\{|\eta_n| \geq \frac{\delta}{A}\right\}\right) \\ &\xrightarrow{n \rightarrow \infty} 1 - F_\xi(A) + F_\xi(-A) = P(\{|\xi| > A\}) < \varepsilon \end{aligned}$$

令 $\varepsilon \rightarrow 0$ 即得 $\lim_n P(\{|\xi_n \eta_n| > \delta\}) = 0$, 综上命题得证.

Y-Problem 4.1.4.13

设 $\xi_n \xrightarrow{d} \xi$, 数列 $a_n \rightarrow a, b_n \rightarrow b$. 证明: $a_n \xi_n + b_n \xrightarrow{d} a\xi + b$.

证明: 只需考察分布函数

$$f_{a_n \xi_n + b_n}(t) = e^{itb_n} f_{\xi_n}(a_n t)$$

我们有 $f_{\xi_n}(t) \rightarrow f_{\xi}(t), \forall t$, 再由

$$|f_{\xi_n}(a_n t) - f_{\xi}(at)| \leq |f_{\xi_n}(a_n t) - f_{\xi_n}(at)| + |f_{\xi_n}(at) - f_{\xi}(at)|$$

令 $n \rightarrow \infty$, 由分布函数的连续性即知 RHS 趋于 0, 从而 $f_{\xi_n}(a_n t) \rightarrow f_{\xi}(at)$, 进而有 $e^{itb_n} f_{\xi_n}(a_n t) \rightarrow e^{itb} f_{\xi}(at)$, 即 $f_{a_n \xi_n + b_n}(t) \rightarrow f_{a\xi + b}(t)$, 即 $a_n \xi_n + b_n \xrightarrow{d} a\xi + b$.

Y-Problem 4.1.4.14

设 $\xi_n \xrightarrow{\mathbb{P}} \xi$, 证明: 对任意连续函数 g , 随机变量序列 $g(\xi_n) \xrightarrow{\mathbb{P}} g(\xi)$. 再证明将依概率收敛改为依分布收敛后命题仍成立.

证明: 固定 $\varepsilon, M > 0$. g 在 $[-M-1, M+1]$ 上一致连续, 存在 $0 < \delta < 1$ 使得 $|x_1 - x_2| < \delta \implies |g(x_1) - g(x_2)| < \varepsilon$. 因此我们有

$$\begin{aligned} \{|g(\xi_n) - g(\xi)| > \varepsilon\} &\subset \{|\xi| > M\} \cup \{|\xi_n - \xi| \geq \delta\} \\ \implies P(|g(\xi_n) - g(\xi)| > \varepsilon) &\leq P(|\xi| > M) + P(|\xi_n - \xi| \geq \delta) \\ \implies \overline{\lim}_n P(|g(\xi_n) - g(\xi)| > \varepsilon) &\leq P(|\xi| > M) \end{aligned}$$

令 $M \rightarrow \infty$ 即得 $\lim_n P(|g(\xi_n) - g(\xi)| > \varepsilon) = 0$, 即 $g(\xi_n) \xrightarrow{\mathbb{P}} g(\xi)$.

对于依分布收敛的情形, 使用 Helly 第二定理: $\xi_n \xrightarrow{d} \xi \iff$ 对任意有界连续函数 f 有 $E(f(\xi_n)) \rightarrow E(f(\xi))$. 于是只需证对任意有界连续函数 f 有 $E(f \circ g(\xi_n)) \rightarrow E(f \circ g(\xi))$, 由 g 的连续性知 $f \circ g$ 为有界连续函数, 从而对 $\xi_n \xrightarrow{d} \xi$ 使用定理即可.

Y-Problem 4.1.4.15

设分布函数列 $\{F_n(x)\}$ 弱收敛到连续的分布函数 $F(x)$, 求证该收敛对 $x \in \mathbb{R}$ 一致.

证明:

对任意 $\varepsilon > 0$, 存在 $M > 0$, 使得

$$F(-M) < \frac{\varepsilon}{3}, F(M) > 1 - \frac{\varepsilon}{3}$$

由弱收敛, 对充分大的 n , 有

$$F_n(-M) < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \frac{2\varepsilon}{3}, F_n(M) > 1 - \frac{\varepsilon}{3} - \frac{\varepsilon}{3} = 1 - \frac{2\varepsilon}{3}$$

由于 F 在 $[-M, M]$ 上一致连续, 故存在 $\delta > 0$ 使得 $|x - y| < \delta \implies |F(x) - F(y)| < \varepsilon$. 将 $[-M, M]$ 划分成有限个小区间, 分点为 $x_0 = -M, x_1, \dots, x_k = M$, 使得每个小区间的长度小于 δ . 对每个分点 x_i , 存在 N_i 使得 $n > N_i \implies |F_n(x_i) - F(x_i)| < \frac{\varepsilon}{3}$, 取 $N = \max N_i$, 则 $n \geq N$ 时上述不等式对所有分点成立.

对任意 $x \in [-M, M]$, 存在 $x_i \leq x \leq x_{i+1}$. 由 F 的单调性, 有

$$F_n(x_i) \leq F_n(x) \leq F_n(x_{i+1})$$

结合分点处的收敛性和 $F(x)$ 的一致连续性, 对 $n > N$

$$F_n(x) \leq F_n(x_{i+1}) \leq F(x_{i+1}) + \frac{\varepsilon}{3} \leq F(x) + \frac{2\varepsilon}{3}$$

$$F_n(x) \geq F_n(x_i) \geq F(x_i) - \frac{\varepsilon}{3} \geq F(x) - \frac{2\varepsilon}{3}$$

因此在区间 $[-M, M]$ 上有

$$|F_n(x) - F(x)| < \frac{2\varepsilon}{3} < \varepsilon$$

对尾部区域, 当 $x > M$ 时, $1 \geq F(x) > 1 - \frac{\varepsilon}{3}$, $1 \geq F_n(x) \geq F_n(M) \geq 1 - \frac{2\varepsilon}{3}$, 故

$$|F_n(x) - F(x)| \leq \frac{2\varepsilon}{3} < \varepsilon$$

当 $x < -M$ 时同理. 综上, 当 $n > N$ 时, 对任意 $x \in \mathbb{R}$ 有 $|F_n(x) - F(x)| < \varepsilon$, 也即该收敛是一致的.

K-Problem 4.5.4

Let $\{F_n\}, 1 \leq n \leq \infty$ be df's such that $F_n \xrightarrow{\omega} F$. Let 0 be uniformly distributed on $[0, 1]$ and put $X_n = F_n^{-1}(\theta), 1 \leq n \leq \infty$, where $F_n^{-1}(y) = \sup\{x : F_n(x) < y\}$. Then X_n has df F_n and $X_n \xrightarrow{\mathbb{P}} X_\infty$.

Proof: 事实上, 课上的引理 2 证明了更强的: X_n 逐点收敛到 X_∞ .

Step1: Let $\Omega = (0, 1], \mathcal{F} = \mathcal{B}(\Omega), dP = dx$ define

$$\tilde{\xi}_n(\omega) := \inf\{x : F_n(x) > \omega\}, \tilde{\xi}(\omega) = \inf\{x : F(x) > \omega\}$$

Then we have

$$(a) : \omega < F_n(x) \iff \tilde{\xi}_n(\omega) < x; \omega < F(x) \iff \tilde{\xi}(\omega) < x$$

类似地

$$(b) : F_{\tilde{\xi}_n}(x) = P(\{\omega : \tilde{\xi}_n(\omega) < x\}) = P(\{\omega : \omega < F_n(x)\}) = F_n(x); F_{\tilde{\xi}}(x) = F(x)$$

$\tilde{\xi}_n(\omega)$ and $\tilde{\xi}(\omega)$ are right-continuous and non-decreasing function wrt ω .

Define: $g(\omega) = \underline{\lim}_n \tilde{\xi}_n(\omega)$. Then $g(\omega)$ is non-decreasing function. Thus $P(\tilde{\Omega}) = 1$, where $\tilde{\Omega} = C_g := \{\omega : g \text{ is continuous at } \omega\}$.

We show that

$$(*) : \lim \tilde{\xi}_n(\omega) = \tilde{\xi}(\omega), \forall \omega \in \tilde{\Omega}$$

$$\text{i.e. } \tilde{\xi}(\omega) \stackrel{(1)}{\leq} \underline{\lim}_n \tilde{\xi}_n(\omega) \leq \overline{\lim}_n \tilde{\xi}_n(\omega) \stackrel{(2)}{\leq} \tilde{\xi}(\omega)$$

To show (1) as follow: $\forall \omega \in \tilde{\Omega}, \forall \varepsilon > 0$, as $C_F = \mathbb{R}, \exists x \in C_F$ s.t. $\tilde{\xi}(\omega) - \varepsilon < x \leq \tilde{\xi}(\omega) \xrightarrow{(a)} F(x) \leq \omega < \omega + \varepsilon_k$ ($\forall \varepsilon_k \downarrow$). Using $F_n(x) \rightarrow F(x)$ we have

$$F_n(x) < \omega + \varepsilon_k \text{ for large } n \implies x \leq \tilde{\xi}_n(\omega + \varepsilon_k)$$

$$\implies \tilde{\xi}(\omega) - \varepsilon < x \leq \tilde{\xi}_n(\omega + \varepsilon_k) \implies \tilde{\xi}(\omega) \leq \underline{\lim}_n \tilde{\xi}_n(\omega + \varepsilon_k) = g(\omega + \varepsilon_k) \sim (3)$$

as $\omega \in C_g$, letting $\varepsilon_k \downarrow 0$, we have $\tilde{\xi}(\omega) \leq \underline{\lim}_n \tilde{\xi}_n(\omega)$ i.e. (1) holds.

To show (2) as follows: Taking $\omega \in \Omega, \forall \varepsilon > 0, \exists x \in C_F$ s.t. $\tilde{\xi}(\omega) < x < \tilde{\xi}(\omega) + \varepsilon$.

$$\implies \omega < F(x) = \lim_n F_n(x) \left(\text{as } F_n \xrightarrow{\omega} F \right) \implies \omega < F_n(x) \text{ for large } n$$

$$\implies \tilde{\xi}_n(\omega) \stackrel{(a)}{<} x < \tilde{\xi}(\omega) + \varepsilon \implies \overline{\lim}_n \tilde{\xi}_n(\omega) \leq \tilde{\xi}(\omega) + \varepsilon \implies \overline{\lim}_n \tilde{\xi}_n(\omega) \leq \tilde{\xi}(\omega)$$

step3: Define

$$\xi_n(\omega) := \begin{cases} \tilde{\xi}_n(\omega), & \omega \in \tilde{\Omega} \\ 0, & \omega \notin \tilde{\Omega} \end{cases}, \quad \xi(\omega) := \begin{cases} \tilde{\xi}(\omega), & \omega \in \tilde{\Omega} \\ 0, & \omega \notin \tilde{\Omega} \end{cases}$$

Then using (*), we have

$$\xi_n(\omega) \rightarrow \xi(\omega), \forall \omega \in \Omega = (0, 1]$$

and $P(\{\omega : \xi_n(\omega) \neq \tilde{\xi}_n(\omega)\}) = P(\tilde{\Omega}^c) = 0, P(\{\omega : \xi(\omega) \neq \tilde{\xi}(\omega)\}) = P(\tilde{\Omega}^c) = 0$.

Using (b), we get $F_{\xi_n} = F_{\tilde{\xi}_n} = F_n, F_{\xi} = F_{\tilde{\xi}} = F$. Thus, we end the proof.

K-Problem 4.5.6

If $\{X_t\}$ and $\{Y_t\}$ are uniformly integrable, then so is $\{X_t + Y_1\}$ and $\{X_t + Y_t\}$.

Proof: 我们有

$$E|X_t + Y_1| \leq E|X_t| + E|Y_1|, \quad E|X_t + Y_t| \leq E|X_t| + E|Y_t|$$

$$\sup_{t \in T} \int_A |X_t + Y_1| dP \leq \sup_{t \in T} \int_A |X_t| dP + \int_A |Y_1| dP$$

$$\sup_{t \in T} \int_A |X_t + Y_t| dP \leq \sup_{t \in T} \int_A |X_t| dP + \sup_{t \in T} \int_A |Y_t| dP$$

从而由一致可积的如下等价定义: $\{\xi_t, t \in T\}$ is UI if and only if

$$(a) \tilde{A} := \sup_{t \in T} E(|\xi_t|) < +\infty;$$

$$(b) \forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } P(A) < \delta \implies \sup_{t \in T} \int_A |\xi_t| dP < \varepsilon$$

以及 Y_1 积分的一致连续性, 立得 $\{X_t + Y_1\}$ 与 $\{X_t + Y_t\}$ 一致可积.

K-Problem 4.5.7

If $\{X_n\}$ is dominated by some Y in L^1 or if it is identically distributed with finite mean, then it is uniformly integrable.

Proof: 若 $\{X_n\}$ 被 $Y \in L^1$ 控制, 则

$$E|X_n| \leq E|Y| < +\infty, \quad \sup_{n \geq 1} \int_A |X_n| dP \leq \int_A |Y| dP$$

于是由积分的一致连续性立得一致可积.

若 $\{X_n\}$ 同分布且期望有限, 记这些分布为 F , 由积分变换公式可知

$$\begin{aligned} \sup_{n \geq 1} \int_{\{|X_n| > c\}} |X_n| dP &= \sup_{n \geq 1} \int_c^\infty x dF_{X_n}(x) = \int_c^\infty x dF(x) \\ \implies \lim_{c \rightarrow \infty} \sup_{n \geq 1} \int_{\{|X_n| > c\}} |X_n| dP &= \lim_{c \rightarrow \infty} \int_c^\infty x dF(x) = 0 \end{aligned}$$

由定义即知 $\{X_n\}$ 一致可积.

K-Problem 4.5.8

If $\sup_n E(|X_n|^p) < \infty$ for some $p > 1$, then $\{X_n\}$ is uniformly integrable.

Proof: 取 p 的共轭指标 q . 对任意事件 A , 由 Hölder 不等式知

$$\int_A |X_n| dP \leq \left(\int_A |X_n|^p dP \right)^{\frac{1}{p}} \left(\int_A 1^q dP \right)^{\frac{1}{q}} \leq \left(\sup_n E|X_n|^p \right)^{\frac{1}{p}} P(A)^{\frac{1}{q}}$$

与 $A = \Omega$ 可得一致可积性要求的 (a), 令 $P(A) \rightarrow 0$ 可得 (b), 故 $\{X_n\}$ 一致可积.

K-Problem 4.5.9

If $\{X_n\}$ is uniformly integrable, then $\left\{\frac{1}{n} \sum_{j=1}^n X_j, n \geq 1\right\}$ is uniformly integrable.

Proof: 记 $Y_n = \frac{1}{n} \sum_{j=1}^n X_j$, 于是

$$\int_A |Y_n| dP \leq \frac{1}{n} \sum_{j=1}^n \int_A |X_j| dP \leq \sup_{1 \leq j \leq n} \int_A |X_j| dP$$

故

$$\sup_{n \geq 1} \int_A |Y_n| dP \leq \sup_{n \geq 1} \int_A |X_n| dP$$

由 $\{X_n\}$ 一致可积立得 $\{Y_n\}$ 一致可积.

Y-Problem 4.2.5.1

假定 $\{\xi_n\}$ 为独立随机变量列, 有如下概率分布, 验证 $\{\xi_n\}$ 是否满足 Markov 条件.

(1) $P(\{\xi_n = \pm 2^n\}) = \frac{1}{2};$

解: 易知 $D(\xi_n) = 2^{2n}$, 由独立性可知

$$\frac{1}{n^2} D\left(\sum_{k=1}^n \xi_k\right) = \frac{1}{n^2} \sum_{k=1}^n D(\xi_k) = \frac{4^{n+1} - 4}{3n^2} \rightarrow \infty$$

因此不满足 Markov 条件.

(2) $P(\{\xi_n = 0\}) = 1 - 2^{-2n}, P(\{\xi_n = \pm 2^n\}) = 2^{-2n-1};$

解: 易知 $D(\xi_n) = \frac{1}{2}$, 由独立性可知

$$\frac{1}{n^2} D\left(\sum_{k=1}^n \xi_k\right) = \frac{1}{n^2} \sum_{k=1}^n D(\xi_k) = \frac{1}{2n} \rightarrow 0$$

因此满足 Markov 条件.

(4) $P(\{\xi_n = 0\}) = 1 - n^{-\frac{1}{2}}, P(\{\xi_n = \pm n\}) = (4n)^{-\frac{1}{2}};$

解: 易知 $D(\xi_n) = \frac{1}{2}n^{\frac{3}{2}}$, 由独立性可知

$$\frac{1}{n^2} D\left(\sum_{k=1}^n \xi_k\right) = \frac{1}{n^2} \sum_{k=1}^n D(\xi_k) = \frac{\sum_{k=1}^n k^{\frac{3}{2}}}{2n^2} \xrightarrow[n \rightarrow \infty]{\text{Stolz}} \lim_{n \rightarrow \infty} \frac{n^{\frac{3}{2}}}{2n^2 - 2(n-1)^2} = \infty$$

因此不满足 Markov 条件.

(5) $P(\{\xi_n = \pm \sqrt{\ln n}\}) = \frac{1}{2}.$

解: 易知 $D(\xi_n) = \ln n$, 由独立性可知

$$\frac{1}{n^2} D\left(\sum_{k=1}^n \xi_k\right) = \frac{1}{n^2} \sum_{k=1}^n D(\xi_k) = \frac{\ln n!}{n^2} \xrightarrow[n \rightarrow \infty]{\text{Stolz}} \lim_{n \rightarrow \infty} \frac{\ln n}{2n - 1} = 0$$

因此满足 Markov 条件.

Y-Problem 4.2.5.2

参加集会的 n 个人将他们的帽子混放在一起, 会后每人任选一顶戴上. 以 S_n 表示他们中戴对自己帽子的人的个数, 求证

$$\frac{1}{n}(S_n - E(S_n)) \xrightarrow{\mathbb{P}} 0$$

证明: 设随机变量 ξ_k 表示第 k 个人是否戴对了帽子 (戴对则 $\xi_k = 1$, 否则为 0), 于是 $S_n = \sum_{k=1}^n \xi_k$. 只需证 $\{\xi_n\}$ 满足 WLLN. 易见

$$P(\xi_k = 1) = \frac{1}{n}, P(\xi_k = 0) = 1 - \frac{1}{n} \implies E(\xi_k) = \frac{1}{n}, D(\xi_k) = \frac{1}{n} - \frac{1}{n^2}$$

$$P(\xi_i \xi_j = 1) = \frac{1}{n(n-1)} \implies E(\xi_i \xi_j) = \frac{1}{n(n-1)}, \text{cov}(\xi_i, \xi_j) = \frac{1}{n^2(n-1)}, \forall i \neq j$$

于是

$$D(S_n) = \sum_{i=1}^n D(\xi_i) + \sum_{i \neq j} \text{cov}(\xi_i, \xi_j) = n \left(\frac{1}{n} - \frac{1}{n^2} \right) + (n^2 - n) \frac{1}{n^2(n-1)} = 1$$

从而 $\frac{1}{n^2} D(S_n) \rightarrow 0$, WLLN 成立.

Y-Problem 4.2.5.3

设 $\{\xi_n\}$ 的方差有界, 且当 $|i - j| \rightarrow \infty$ 时 ξ_i 与 ξ_j 的相关系数 $r_{ij} \rightarrow 0$. 试证对 $\{\xi_n\}$ 成立大数定律.

证明: 设 $D(\xi_n) < C$. 取 $\varepsilon > 0$, 存在 M 使得 $|i - j| > M \implies |r_{ij}| < \varepsilon$. 于是

$$\begin{aligned} \frac{1}{n^2} D(S_n) &= \frac{1}{n^2} \left| \left(\sum_{k=1}^n D(\xi_k) + \sum_{1 \leq i \neq j \leq n} \text{cov}(\xi_i, \xi_j) \right) \right| \\ &\leq \frac{1}{n^2} \left(nC + \sum_{1 \leq i \neq j \leq n} \sqrt{D(\xi_i)D(\xi_j)} |r_{ij}| \right) \\ &\leq \frac{1}{n^2} \left(nC + C \sum_{i=1}^n \left(\sum_{\substack{1 \leq j \neq i \leq n \\ |j-i| \leq M}} |r_{ij}| + \sum_{\substack{1 \leq j \neq i \leq n \\ |j-i| > M}} |r_{ij}| \right) \right) \\ &\leq \frac{1}{n^2} \left(nC + C \sum_{i=1}^n (2M + n\varepsilon) \right) \\ &= \frac{(2M+1)C}{n} + C\varepsilon \end{aligned}$$

令 $n \rightarrow \infty$ 得 $\overline{\lim}_n \frac{1}{n^2} D(S_n) \leq C\varepsilon$, 再令 $\varepsilon \rightarrow 0$, 得 $\frac{1}{n^2} D(S_n) \rightarrow 0$, 于是 WLLN 成立.

Y-Problem 4.2.5.4

设 $\{\xi_n\}$ 同分布, 每个 ξ_n 方差有限, 当 $|k - l| \geq 2$ 时 ξ_k 与 ξ_l 独立. 试证对 $\{\xi_n\}$ 大数定律成立.

证明: 由同分布, 可设所有随机变量的方差都是 C , 方差为 E , 于是有

$$\begin{aligned}
\frac{1}{n^2}D(S_n) &= \frac{1}{n^2} \left(nC + \sum_{1 \leq i \neq j \leq n} \text{cov}(\xi_i, \xi_j) \right) = \frac{1}{n^2} \left(nC + 2 \sum_{i=1}^{n-1} \text{cov}(\xi_i, \xi_{i+1}) \right) \\
&= \frac{1}{n^2} \left(nC + 2 \sum_{i=1}^{n-1} (E(\xi_i \xi_{i+1}) - E^2) \right) \\
&\leq \frac{1}{n^2} \left(nC + 2 \sum_{i=1}^{n-1} \left(\sqrt{E(\xi_i^2)E(\xi_{i+1}^2)} - E^2 \right) \right) \\
&= \frac{1}{n^2} (nC + 2(n-1)C) = \frac{3n-2}{n^2}C
\end{aligned}$$

令 $n \rightarrow \infty$ 得 $\frac{1}{n^2}D(S_n) \rightarrow 0$, 于是 WLLN 成立.

Y-Problem 4.2.5.5

设 $\{\xi_k\}$ 为 iid, 服从 $(0, 1)$ 上的均匀分布, 求证 $(\prod_{k=1}^n \xi_k)^{\frac{1}{n}} \xrightarrow{\mathbb{P}} c$, 并求 c .

证明: 由连续函数保依概率收敛, 因此只需要考虑 $\frac{1}{n} \sum_{i=1}^n \ln \xi_k$. $\{\ln \xi_k\}$ 为 iid, 且

$$E(|\ln \xi_1|) = \int_0^1 |\ln x| dx = 1 < \infty, E(\ln \xi_1) = -1$$

于是由 Kolmogorov 大数定律即知

$$\frac{1}{n} \sum_{i=1}^n \ln \xi_i \xrightarrow{\text{a.s.}} E(\ln \xi_1) = -1$$

于是 $(\prod_{k=1}^n \xi_k)^{\frac{1}{n}} \xrightarrow{\mathbb{P}} \frac{1}{e} = c$.

K-Problem 5.1.2

If $\{X_j\}$ are uncorrelated and their second moments have a common bound, then

$$\frac{S_n - E(S_n)}{n^\alpha} \xrightarrow{\text{a.s.}} 0, \forall \alpha > \frac{3}{4}$$

Proof: 通过变换 $X'_j = X_j - E(X_j)$, 可不妨设 $E(X_j) = 0$. 设 $E(X_j^2) < M$, 则

$$E(S_n^2) = \sum_{i=1}^n E(X_i^2) + 2 \sum_{1 \leq i < j \leq n} E(X_i X_j) \leq nM$$

由 Chebyshev 不等式, 对任意 $\varepsilon > 0$ 有

$$P(\{|S_n| > n^\alpha \varepsilon\}) \leq \frac{M}{n^{2\alpha-1} \varepsilon^2} \implies P(\{|S_{n^2}| > n^{2\alpha} \varepsilon\}) \leq \frac{M}{n^{4\alpha-2} \varepsilon^2}$$

由 $4\alpha - 2 > 1$ 知右式对 n 求和收敛, 故由 B-C 引理知 $P\left(\overline{\lim}_n \left\{ \left| \frac{S_{n^2}}{n^{2\alpha}} \right| > \varepsilon \right\}\right) = 0$.

由此即知 $\frac{S_{n^2}}{n^{2\alpha}} \xrightarrow{\text{a.s.}} 0$. 对任意 $n \geq 1$, 考虑

$$D_n = \max_{n^2 \leq k < (n+1)^2} |S_k - S_{n^2}|$$

于是有

$$E(D_n^2) \leq 2nE\left(|S_{(n+1)^2} - S_{n^2}|^2\right) = 2n \sum_{j=n^2+1}^{(n+1)^2} E^2(X_j) \leq 4n^2 M$$

于是由 Chebyshev 不等式知

$$P(D_n > n^{2\alpha}\varepsilon) \leq \frac{4M}{n^{4\alpha-2}\varepsilon^2} \xrightarrow{\text{B-C}} P\left(\overline{\lim}_n \left\{ \frac{D_n}{n^{2\alpha}} > \varepsilon \right\}\right) = 0, \forall \varepsilon > 0$$

故 $\frac{D_n}{n^{2\alpha}} \xrightarrow{\text{a.s.}} 0$. 因此对 $n^2 \leq k < (n+1)^2$, 有

$$\frac{|S_k|}{k^\alpha} \leq \frac{D_n + |S_{n^2}|}{n^{2\alpha}} \xrightarrow{\text{a.s.}} 0$$

于是命题得证.

K-Problem 5.1.4

If $\{X_n\}$ are independent rv's such that the fourth moments $E(X_n^4)$ have a common bound, then SLLN holds.

Proof: 由 Cauchy 不等式

$$\begin{aligned} E^2(X_n^2) &\leq E(X_n^4) \cdot E(1) \leq M \\ \Rightarrow E(X_n^2) &\leq \sqrt{M} \\ \Rightarrow \sum_{n=1}^{\infty} \frac{E(X_n^2)}{n^2} &\leq \sum_{n=1}^{\infty} \frac{E(X_n^2)}{n^2} \leq \sqrt{M} \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty \end{aligned}$$

于是由 Kolmogorov 大数定律即知 SLLN 成立.

K-Problem 5.2.1

For any sequence of rv's $\{X_n\}$, and any $p > 1$:

$$X_n \xrightarrow{\text{a.s.}} 0 \Rightarrow \frac{S_n}{n} \xrightarrow{\text{a.s.}} 0, \quad X_n \xrightarrow{L^p} 0 \Rightarrow \frac{S_n}{n} \xrightarrow{L^p} 0$$

The second result is false if $p < 1$.

Proof: a.s. 收敛到 0 由 Kronecker 引理立得. 当 $p > 1$ 时, 由 Minkovski 不等式

$$\left\| \frac{S_n}{n} \right\|_p \leq \frac{1}{n} \left(\sum_{i=1}^n \|X_i\|_p \right)$$

$\|X_n\|_p \rightarrow 0$, 由 Kronecker 引理, $n \rightarrow \infty$ 时右式趋于 0, 从而 $\frac{S_n}{n} \xrightarrow{L_p} 0$.

当 $0 < p < 1$ 时 Minkovski 不等式需要反向, 考虑随机变量 $\{X_n\}$, 满足

$$X_n \sim \begin{pmatrix} n & 0 \\ n^{-\delta} & 1 - n^{-\delta} \end{pmatrix}, 0 < p < \delta < 1$$

于是

$$E(X_n^p) = n^{p-\delta} \rightarrow 0 (n \rightarrow \infty)$$

但是

$$\left\| \frac{S_n}{n} \right\|_p \geq \frac{1}{n} \left(\sum_{i=1}^n \|X_i\|_p \right) = \frac{1}{n} \left(\sum_{k=1}^n k^{1-\frac{\delta}{p}} \right)$$

令 $n \rightarrow \infty$, 由 Stolz 定理知右式趋于 $\lim_n n^{1-\frac{\delta}{p}} = \infty$, 从而命题不再成立.

K-Problem 5.2.2

Even for a sequence of independent rv's $\{X_j\}$

$$X_n \xrightarrow{\mathbb{P}} 0 \not\Rightarrow \frac{S_n}{n} \xrightarrow{\mathbb{P}} 0$$

Proof: 构造 iid $\{X_n\}$ 满足

$$X_n \sim \begin{pmatrix} 2^n & 0 \\ \frac{1}{n} & 1 - \frac{1}{n} \end{pmatrix}$$

易见 $X_n \xrightarrow{\mathbb{P}} 0$. 我们估计 $\left| \frac{S_n}{n} \right| > 1$ 的概率, 由 $S_n = \sum_{k=1}^n X_k$, 若存在 $k > \log_2 n$ 使得 $X_k = 2^k$, 则必有 $S_n > n$, 这给出

$$\begin{aligned} P\left(\left\{\frac{S_n}{n} > 1\right\}\right) &\geq 1 - \prod_{k=\lceil \log_2 n \rceil}^n \left(1 - \frac{1}{k}\right) = 1 - \exp\left(\sum_{k=\lceil \log_2 n \rceil}^n \ln\left(1 - \frac{1}{k}\right)\right) \\ &\geq 1 - \exp\left(-\sum_{k=\lceil \log_2 n \rceil}^n \frac{1}{k}\right) \sim 1 - \exp(-(\ln n - \ln \ln n)) \end{aligned}$$

取 $n \rightarrow \infty$, 于是 $\lim_n P\left(\left\{\frac{S_n}{n} > 1\right\}\right) \geq 1$, 故 $\frac{S_n}{n} \not\xrightarrow{\mathbb{P}} 0$.

Y-Problem 4.2.5.10

设 $\{\xi_n\}$ 独立, 满足 $E(\xi_n) \rightarrow 0$ 且 $\sum_{n=1}^{\infty} \frac{E(\xi_n^2)}{n^2} < \infty$, 证明: $\frac{1}{n} \sum_{k=1}^n \xi_k \xrightarrow{\text{a.s.}} 0$.

证明: 易见

$$\sum_{n=1}^{\infty} \frac{D(\xi_n)}{n^2} \leq \sum_{n=1}^{\infty} \frac{E(\xi_n^2)}{n^2} < \infty$$

于是由 Kolmogorov 大数定律知 $\frac{1}{n} \left(\sum_{k=1}^n \xi_k - \sum_{k=1}^n E(\xi_k) \right) \xrightarrow{\text{a.s.}} 0$. 由 $E(\xi_n) \rightarrow 0$ 与 Kronecker 引理可得 $\frac{1}{n} \sum_{k=1}^n E(\xi_k) \rightarrow 0$, 从而 $\frac{1}{n} \sum_{k=1}^n \xi_k \xrightarrow{\text{a.s.}} 0$.

K-Problem 5.4.1

Assume that $\{X_n\}$ are iid. If $E(X_1^+) = \infty$, $E(X_1^-) < \infty$, then $\frac{S_n}{n} \xrightarrow{\text{a.s.}} \infty$.

Proof: 考虑截断

$$X_{nk} = X_n I_{(-\infty, k]}(X_n)$$

则 $E(X_{nk}^+) < \infty$, $E(X_{nk}^-) < \infty$ 且 $\{X_{nk}\}$ 也为独立同分布, SLLN 成立. 我们有

$$\lim_{n \rightarrow \infty} \frac{S_{nk}}{n} = E(X_{1k}), \quad \forall k, \frac{S_n}{n} > \frac{S_{kn}}{n} \implies \lim_{n \rightarrow \infty} \frac{S_n}{n} > \lim_{n \rightarrow \infty} \frac{S_{kn}}{n} = E(X_{1k})$$

从而取 $\{k_i\}$ 使得 $E(X_{1k_i}) > i$, 令 $i \rightarrow \infty$ 即证.

K-Problem 5.4.2

Let $\{X_n\}$ be a sequence of independent rv's with $E(X_n) = 0$, $\forall n$, and $0 < a_n \uparrow \infty$. If φ is a positive, even and continuous function on $\mathbb{R}^1$ such that as $|x|$ increases

$$\varphi(x) \uparrow, \frac{\varphi(x)}{|x|} \downarrow$$

Then

$$\sum_n \frac{E(\varphi(X_n))}{\varphi(a_n)} < \infty$$

implies

$$\sum_n \frac{X_n}{a_n} \text{ a.s. converges}$$

Proof: 由 $\frac{\varphi(x)}{|x|} \downarrow$ 可知 $\frac{\varphi(x)}{x^2} \downarrow$, 令 $Y_n = X_n I_{[-a_n, a_n]}(X_n)$, 则

$$\sum_{n=1}^{\infty} D\left(\frac{Y_n}{a_n}\right) \leq \sum_{n=1}^{\infty} E\left(\frac{Y_n^2}{a_n^2}\right) \leq \sum_{n=1}^{\infty} \int_{|x|>a_n} \frac{\varphi(x)}{\varphi(a_n)} dF_{X_n}(x) = \sum_{n=1}^{\infty} \frac{E(\varphi(X_n))}{\varphi(a_n)} < \infty$$

于是 $\sum_n \frac{1}{a_n}(Y_n - E(Y_n))$ 收敛. 我们有

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{E(Y_n)}{a_n} &= \sum_{n=1}^{\infty} \int_{|x|>a_n} \frac{|x|}{a_n} dF_{X_n}(x) \\ &\leq \sum_{n=1}^{\infty} \int_{|x|>a_n} \frac{\varphi(x)}{\varphi(a_n)} dF_{X_n}(x) \leq \sum_{n=1}^{\infty} \frac{E(\varphi(X_n))}{\varphi(a_n)} < \infty \end{aligned}$$

于是

$$\sum_{n=1}^{\infty} P(\{X_n \neq Y_n\}) = \sum_{n=1}^{\infty} \int_{|x|>a_n} dF_{X_n}(x) \leq \sum_{n=1}^{\infty} \frac{E(\varphi(X_n))}{\varphi(a_n)} < \infty$$

于是 $\sum_{n=1}^{\infty} (X_n - Y_n)$ a.s. 收敛, 从而 $\sum_{n=1}^{\infty} \frac{X_n}{a_n}$ a.s. 收敛.

K-Problem 5.4.6

Assume that $\{X_n\}$ are iid. Let $E(X_1) = 0$ and $\{c_n\}$ be a bounded sequence of real numbers. Then

$$\frac{1}{n} \sum_{j=1}^n c_j X_j \xrightarrow{\text{a.s.}} 0$$

Proof: 设 $|c_k| < M$. 令 $Y_n := X_n \cdot I_{[-n,n]}(X_n)$ (对 X_n 作截断), 于是 $|Y_n| \leq n$, 记

$$S_n = \sum_{k=1}^n c_k X_k, \quad S_n^* = \sum_{k=1}^n c_k Y_k$$

于是

$$\frac{1}{n} S_n = \frac{1}{n} (S_n^* - E(S_n^*)) + \frac{1}{n} (S_n - S_n^*) + \frac{1}{n} E(S_n^*)$$

• 对于 $\frac{1}{n} E(S_n^*) = \frac{1}{n} \sum_{k=1}^n c_k E(Y_k)$, 我们有

$$\lim_{k \rightarrow \infty} E(Y_k) = \lim_{k \rightarrow \infty} \int_{\mathbb{R}} x \cdot I_{[-k,k]}(x) dF_{X_1}(x) \stackrel{\text{DCT}}{=} \int_{\mathbb{R}} x dF_{X_1}(x) = E(X_1) = 0$$

由有界性立得 $\lim_{k \rightarrow \infty} c_k E(Y_k) = 0$, 由 Kronecker's 引理得 $\frac{1}{n} E(S_n^*) \rightarrow 0$.

• 为证 $\frac{1}{n} (S_n^* - E(S_n^*)) \xrightarrow{\text{a.s.}} 0$, 由 Kolmogorov 大数定律知只需证

$$\sum_{n=1}^{\infty} \frac{c_n^2 D(Y_n)}{n^2} < +\infty$$

我们有

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{c_n^2 D(Y_n)}{n^2} &\leq \sum_{n=1}^{\infty} \frac{c_n^2 E(Y_n^2)}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} \int_{\mathbb{R}} c_n^2 x^2 I_{[-n,n]}(x) dF_{X_1}(x) \\
&= \sum_{n=1}^{\infty} \frac{c_n^2}{n^2} \sum_{j=1}^n \int_{\{j-1 \leq |x| \leq j\}} x^2 dF_{X_1}(x) \\
(\text{Fubini}) &= \sum_{j=1}^{\infty} \left(\int_{\{j-1 \leq |x| \leq j\}} x^2 dF_{X_1}(x) \sum_{n=j}^{\infty} \frac{c_n^2}{n^2} \right) \\
&\leq \sum_{j=1}^{\infty} \frac{C'}{j} \int_{\{j-1 \leq |x| \leq j\}} x^2 dF_{X_1}(x) \\
&\leq C' \sum_{j=1}^{\infty} \int_{\{j-1 \leq |x| \leq j\}} |x| dF_{X_1}(x) \\
&= C' E(|\xi_1|) < +\infty
\end{aligned}$$

- 为证 $\frac{1}{n}(S_n - S_n^*) \xrightarrow{\text{a.s.}} 0$, 我们有

$$\begin{aligned}
\sum_{n=1}^{\infty} P(\{X_n \neq Y_n\}) &= \sum_{n=1}^{\infty} P(\{\omega : |X_n| \geq n\}) \leq E|X_n| < +\infty \\
\Rightarrow \sum_{n=1}^{\infty} (X_n - Y_n) \text{ a.s. converges} &\Rightarrow X_n - Y_n \xrightarrow{\text{a.s.}} 0 \\
\Rightarrow c_k(X_k - Y_k) \xrightarrow{\text{a.s.}} 0 &\Rightarrow \frac{1}{n}(S_n - S_n^*) \xrightarrow{\text{a.s.}} 0
\end{aligned}$$

综上, 我们得到 $\frac{1}{n} \sum_{k=1}^{\infty} c_k X_k \xrightarrow{\text{a.s.}} 0$.

K-Problem 5.4.8

Assume that $\{X_n\}$ are iid. If $E(|X_1|) < \infty$, then the sequence $\left\{\frac{S_n}{n}\right\}$ is uniformly integrable and $\frac{S_n}{n} \rightarrow E(X_1)$ in L^1 as well as a.s.

Proof: 最后的 $\frac{S_n}{n} \xrightarrow{\text{a.s.}} E(X_1)$ 就是大数定律. 对一致可积, 我们有

- 由 iid 可得 $E(|X_i|)$ 的值不随 i 的变化而变化, 于是

$$\sup_n \left| \frac{S_n}{n} \right| \leq \sup_n \frac{1}{n} \left(\sum_{i=1}^n E(|X_i|) \right) = E(|X_1|) < \infty$$

- 对任意事件 A , 由 iid 可得

$$\begin{aligned}\int_A \left| \frac{S_n}{n} \right| dP &\leq \frac{1}{n} \sum_{i=1}^n \int_A |X_i| dP \leq \sup_{1 \leq i \leq n} \int_A |X_i| dP \\ &\Rightarrow \sup_n \int_A \left| \frac{S_n}{n} \right| dP \leq \sup_n \int_A |X_n| dP\end{aligned}$$

对 $\{X_n\}$, 注意

$$\begin{aligned}\sup_n \int_{|X_n| \geq c} |X_n| dP &= \sup_n \int_c^\infty |x| dF_{X_n}(x) = \int_c^\infty |x| dF_{X_1}(x) \\ &\Rightarrow \lim_{c \rightarrow \infty} \sup_n \int_{|X_n| \geq c} |X_n| dP = \lim_{c \rightarrow \infty} \int_c^\infty |x| dF_{X_1}(x) = 0\end{aligned}$$

于是 $\{X_n\}$ 一致可积, 对 $\varepsilon > 0$, 存在 $\delta > 0$ 使 $P(A) < \delta \Rightarrow \sup_n \int_A |X_n| dP < \varepsilon$, 于是 $\frac{S_n}{n}$ 也满足这一性质, 综上 $\left\{ \frac{S_n}{n} \right\}$ 一致可积.

最后, 由 $\frac{S_n}{n} \xrightarrow{\mathbb{P}} E(X_1)$ 与一致可积立得 $\frac{S_n}{n} \xrightarrow{L^1} E(X_1)$.

Y-Problem 4.3.6.1

反复抛一枚硬币, 为保证正面出现的频率在 0.4 至 0.6 之间的概率不小于 90%, 用下述方法确定至少要抛多少次: (1) Chebyshev 不等式; (2) 二项分布的正态近似.

解: (1) 以 ξ_k 表第 k 次出现正面的次数. 记 $S_n = \sum_{k=1}^n \xi_k$, 则 $\frac{S_n}{n}$ 表示正面出现的频率. “正面出现的频率在 0.4 至 0.6 之间的概率不小于 90%” 等价于

$$P\left\{\left|\frac{S_n}{n} - \frac{1}{2}\right| \leq 0.1\right\} \geq 90\% \iff P\left\{\left|\frac{S_n}{n} - \frac{1}{2}\right| > 0.1\right\} \leq 0.1$$

易见 $E(S_n) = \frac{1}{2}n$, 由 Chebyshev 不等式知

$$P\left\{\left|\frac{S_n}{n} - \frac{1}{2}\right| > 0.1\right\} \leq \frac{D(S_n)}{(0.1n)^2} = \frac{1}{0.04n}$$

因此若 $n \geq 250$, 则 $\frac{1}{0.04n} \leq 0.1$, 故 n 至少是 250.

(2) 易见 $S_n \sim B(n, \frac{1}{2})$, $E(S_n) = \frac{1}{2}n$, $D(S_n) = \frac{1}{4}n$, 于是对充分大的 n , 由 CLT 知

$$\begin{aligned} P\left\{\frac{S_n - \frac{1}{2}n}{\frac{1}{2}\sqrt{n}} < x\right\} &\approx \Phi(x) \implies P\left\{\frac{S_n - \frac{1}{2}n}{n} < x\right\} \approx \Phi(2\sqrt{n}x) \\ &\implies P\left\{\left|\frac{S_n}{n} - \frac{1}{2}\right| > 0.1\right\} \approx 2\Phi(-0.2\sqrt{n}) \end{aligned}$$

因此只需 $2\Phi(-0.2\sqrt{n}) \leq 0.1$, 查表可知只需 $n \geq 69$, 故 n 至少是 69.

Y-Problem 4.3.6.5

设独立随机变量列 $\{\xi_n\}$ 有如下分布:

(1) $P\{\xi_n = \pm\sqrt{n}\} = \frac{1}{2}$;

(2) $P\{\xi_n = 0\} = \frac{1}{3}$, $P\{\xi_n = \pm n^\alpha\} = \frac{1}{3}$ ($\alpha > 0$).

试验证 Lyapunov 条件是否成立.

解: (1) 取 $\delta > 0$, $E(\xi_k) = 0$, $E(|\xi_k|^{2+\delta}) = k^{1+\frac{\delta}{2}}$, $D(\xi_k) = k$, $B_n^2 = \frac{1}{2}n(n+1)$, 故

$$\begin{aligned} \frac{1}{B_n^{2+\delta}} \sum_{k=1}^n E|\xi_k - E(\xi_k)|^{2+\delta} &= \frac{1}{\left(\frac{1}{2}n(n+1)\right)^{1+\frac{\delta}{2}}} \sum_{k=1}^n k^{1+\frac{\delta}{2}} \\ &\approx \frac{n^{2+\frac{\delta}{2}}}{\left(\frac{1}{2}n(n+1)\right)^{1+\frac{\delta}{2}}} \rightarrow 0 (n \rightarrow \infty), \forall \delta > 0 \end{aligned}$$

从而 $\{\xi_n\}$ 成立 Lyapunov 条件.

(2) 取 $\delta > 0$, $E(\xi_k) = 0$, $E(|\xi_k|^{2+\delta}) = \frac{2}{3}k^{\alpha(2+\delta)}$, $D(\xi_k) = \frac{2}{3}k^{2\alpha}$, $B_n^2 = \frac{2}{3}\sum_{k=1}^n k^{2\alpha}$

$$\frac{1}{B_n^{2+\delta}} \sum_{k=1}^n E|\xi_k - E(\xi_k)|^{2+\delta} = \frac{\frac{2}{3}\sum_{k=1}^n k^{\alpha(2+\delta)}}{\left(\frac{2}{3}\sum_{k=1}^n k^{2\alpha}\right)^{1+\frac{\delta}{2}}} \approx \frac{n^{2\alpha+\delta\alpha+1}}{n^{2\alpha+\delta\alpha+1+\frac{\delta}{2}}} \rightarrow 0 (n \rightarrow \infty)$$

从而 $\{\xi_n\}$ 成立 Lyapunov 条件.

Y-Problem 4.3.6.6

用中心极限定理证明: $n \rightarrow \infty$ 时, $e^{-n} \sum_{k=0}^n \frac{n^k}{k!} \rightarrow \frac{1}{2}$.

证明: 考虑 iid $\{\xi_n\}$ 满足 $\xi_1 \sim \text{Poisson}(1)$, 由结论, $S_n = \sum_{k=1}^n \xi_k \sim \text{Poisson}(n)$. 于是 $E(S_n) = n = D(S_n)$. 由 iid 得 $\{\xi_n\}$ 满足 CLT, 于是

$$P\left\{\frac{S_n - E(S_n)}{\sqrt{D(S_n)}} < 0\right\} = P\{S_n < n\} = e^{-n} \sum_{k=0}^{n-1} \frac{n^k}{k!}$$

$$\xrightarrow{\text{CLT}} \lim_n e^{-n} \sum_{k=0}^{n-1} \frac{n^k}{k!} = \Phi(0) = \frac{1}{2} \xrightarrow{\text{Stirling}} \lim_n e^{-n} \sum_{k=0}^n \frac{n^k}{k!} = \frac{1}{2}$$

Y-Problem 4.3.6.7

设 $\eta = 0.\xi_1\xi_2\cdots$ 是区间 $[0, 1]$ 上均匀分布的随机变量, 试证 $\{\xi_n\}$ 满足 CLT.

证明: 易见

$$\xi_1 = [10\eta], \xi_2 = [100\eta] - 10[10\eta], \cdots, \xi_n = [10^n\eta] - 10[10^{n-1}\eta]$$

易知 $10^n\eta$ 服从 $[0, 10^n]$ 上的均匀分布, 而 ξ_n 表示 $10^n\eta$ 的个位数字, 从而 ξ_n 服从 $0, 1, \cdots, 9$ 上的均匀分布. 由均匀性知 $\{\xi_n\}$ 为 iid, 期望方差均有限, 从而满足 CLT.

Y-Problem 4.3.6.8

设独立随机变量序列 $\{\xi_n\}$ 满足 CLT, 试证 $\{\xi_n\}$ 满足大数定律的充要条件是

$$\frac{1}{n^2} \sum_{k=1}^n D(\xi_k) \rightarrow 0$$

证明: 由独立可知 $\sum_{k=1}^n D(\xi_k) = D(S_n)$, 于是充分性由 WLLN 的充分条件立得. 下证必要性: 只需证 $\lim_n \frac{\sqrt{D(S_n)}}{n} = 0$. 反证, 设存在 $\varepsilon > 0$ 使得存在一列 $\{n_k\}$ 满足 $\frac{\sqrt{D(S_{n_k})}}{n_k} > \varepsilon$, 此时有

$$P\left\{\left|\frac{S_n - E(S_n)}{n}\right| > \varepsilon\right\} = P\left\{\left|\frac{S_n - E(S_n)}{\sqrt{D(S_n)}}\right| > \varepsilon \frac{n}{\sqrt{D(S_n)}}\right\}$$

由于 $\varepsilon \frac{n_k}{\sqrt{D(S_{n_k})}} < 1$, 因此

$$P\left\{\left|\frac{S_{n_k} - E(S_{n_k})}{\sqrt{D(S_{n_k})}}\right| > 1\right\} \leq P\left\{\left|\frac{S_{n_k} - E(S_{n_k})}{n_k}\right| > \varepsilon\right\}$$

令两侧的 $k \rightarrow \infty$, 由 $\{\xi_n\}$ 同时满足 CLT 和 WLLN 可得 $2\Phi(-1) \leq 0$, 矛盾!

Y-Problem 4.3.6.9

设 $\{\xi_n\}$ 为独立随机变量序列, 每个 ξ_n 有分布 $P\{\xi_n = \pm n^\alpha\} = \frac{1}{2}$. 试证:

(1) 当 $\alpha > -\frac{1}{2}$ 时, $\{\xi_n\}$ 满足 CLT;

(2) 当 $\alpha > 0$ 时, $\{\xi_n\}$ 服从大数定律的充要条件是 $\alpha < \frac{1}{2}$.

证明: (1) 易见 $E(\xi_k) = 0$, $D(\xi_k) = n^{2\alpha}$, $D(S_n) = \sum_{k=1}^n k^{2\alpha}$, 从而由 $2\alpha > -1$, 选取 $\delta > 0$ 使得 $\alpha(2 + \delta) \neq -1$, 于是

$$\frac{1}{B_n^{2+\delta}} \sum_{k=1}^n E(|\xi_k|^{2+\delta}) = \frac{\sum_{k=1}^n k^{\alpha(2+\delta)}}{(\sum_{k=1}^n k^{2\alpha})^{1+\frac{\delta}{2}}} \approx \frac{n^{2\alpha+\alpha\delta+1}}{n^{2\alpha+\alpha\delta+1+\frac{\delta}{2}}} \rightarrow 0 (n \rightarrow \infty)$$

从而 $\{\xi_n\}$ 满足 Lyapunov 条件, 故满足 CLT.

(2) 由(1), 此时 $\{\xi_n\}$ 为独立的满足 CLT 的随机变量列, 故由 4.3.6.8 可知 $\{\xi_n\}$ 满足大数定律当且仅当 $\frac{1}{n^2} D(S_n) = \frac{1}{n^2} \sum_{k=1}^n k^{2\alpha} \approx n^{2\alpha-1} \rightarrow 0$, 显然这等价于 $\alpha < \frac{1}{2}$.

Y-Problem 4.3.6.10

设 $\{\xi_n\}$ 为 iid, $E(\xi_n) = 0$, $D(\xi_n) = 1$. 试证如下随机变量列为渐近正态分布的

$$(1) \eta_n = \sqrt{n} \frac{\xi_1 + \cdots + \xi_n}{\xi_1^2 + \cdots + \xi_n^2} \quad (2) \zeta_n = \frac{\xi_1 + \cdots + \xi_n}{\sqrt{\xi_1^2 + \cdots + \xi_n^2}}$$

证明: (1) 记 $S_n = \sum_{k=1}^n \xi_k$, $T_n = \sum_{k=1}^n \xi_k^2$, 则 $E(S_n) = 0$, $D(S_n) = E(T_n) = n$. 再由 $\{\xi_n\}, \{\xi_n^2\}$ 均为 iid, 且 ξ_1 期望方差有限, ξ_1^2 期望有限, 故 $\{\xi_n\}$ 满足 CLT, $\{\xi_n^2\}$ 满足 SLLN, 特别地, 其满足 WLLN, 即

$$\frac{S_n}{\sqrt{n}} \xrightarrow{d} N(0, 1), \quad \frac{T_n}{n} \xrightarrow{\mathbb{P}} 1 \implies \frac{n}{T_n} \xrightarrow{\mathbb{P}} 1 \implies \eta_n = \frac{S_n}{\sqrt{n}} \frac{n}{T_n} \xrightarrow{d} N(0, 1)$$

(2) 同理

$$\frac{S_n}{\sqrt{n}} \xrightarrow{d} N(0, 1), \quad \sqrt{\frac{n}{T_n}} \xrightarrow{\mathbb{P}} 1 \implies \zeta_n = \frac{S_n}{\sqrt{n}} \sqrt{\frac{n}{T_n}} \xrightarrow{d} N(0, 1)$$

Y-Problem 4.3.6.11

设独立随机变量列 $\{\xi_n\}$ 有分布

$$P\{\xi_n = \pm n\} = \frac{1}{2n^2}, \quad P\{\xi_n = 0\} = 1 - \frac{1}{n^2}$$

求证: $\{\xi_n\}$ 不满足 CLT.

证明: 易见 $E(\xi_k) = 0, D(\xi_k) = 1, D(S_n) = n \Rightarrow \lim_n \max_{1 \leq k \leq n} \frac{\sigma_k^2}{B_n^2} = 0$. 从而 $\{\xi_n\}$ 满足 Feller 条件, 若其满足 CLT, 则必然满足 Linderberg 条件. 我们计算

$$\begin{aligned} \frac{1}{B_n^2} \sum_{k=1}^n \int_{\frac{|x-a_k|}{B_n} > \tau} (x-a_k)^2 dF_{\xi_k}(x) &= \frac{1}{B_n^2} \sum_{k=1}^n E(\xi_k^2 I_{|\xi_k| > \tau B_n}) \\ &= \frac{1}{n} \sum_{k=\lceil \tau n \rceil}^n 1 = 1 + \frac{1}{n} - \frac{\lceil \tau n \rceil}{n} \\ &\approx 1 - \tau + \frac{1}{n} \end{aligned}$$

于是令 $n \rightarrow \infty$, 上式趋于 $1 - \tau \neq 0$ ($\tau \neq 1$ 时), 矛盾! 从而 $\{\xi_n\}$ 不满足 CLT.

Y-Problem 4.3.6.13

设 $\{\xi_n\}$ 为独立随机变量列, $\xi_n \sim N(0, \sigma_n^2)$, 其中 $\sigma_1^2 = 1, \sigma_n^2 = nB_{n-1}^2$. 试证 $\{\xi_n\}$ 满足 CLT 但不满足 Feller 条件.

证明: 我们有 $B_n^2 = \sigma_n^2 + B_{n-1}^2 = (n+1)B_{n-1}^2 = \dots = \frac{1}{2}(n+1)!, \sigma_n^2 = \frac{n}{2}n!$, 从而

$$\max_{1 \leq k \leq n} \frac{\sigma_k^2}{B_n^2} = \max_{1 \leq k \leq n} \frac{k \cdot k!}{(n+1)!} = \frac{n}{n+1} \Rightarrow \lim_n \max_{1 \leq k \leq n} \frac{\sigma_k^2}{B_n^2} = 1$$

故 $\{\xi_n\}$ 不满足 Feller 条件. 我们有

$$S_n \sim N(0, B_n^2) \Rightarrow \frac{S_n - E(S_n)}{\sqrt{D(S_n)}} \sim N(0, 1)$$

从而 $\{\xi_n\}$ 满足 CLT.

L-Problem 1

Assume that $\{\xi_{nk}, 1 \leq k \leq n\}$ are indep rv's on $(\Omega, \mathcal{F}, P)$ and $\xi_{nk} \sim \begin{pmatrix} 0 & 1 \\ 1-p_{nk} & p_{nk} \end{pmatrix}$. Denote $S_n = \sum_{k=1}^n \xi_k$. show that

(1) If $\frac{1}{n} \sum_{k=1}^n p_{nk} \rightarrow p (n \rightarrow \infty)$, then $\frac{1}{n} S_n \xrightarrow{\text{a.s.}} p$;

(2) If $\max_{1 \leq k \leq n} p_{nk} \rightarrow 0 (n \rightarrow \infty)$ and $\sum_{k=1}^n p_{nk} \rightarrow \lambda > 0 (n \rightarrow \infty)$, then $S_n \xrightarrow{d} S$, where $S \sim \text{Poisson}(\lambda)$.

Proof: (1) 易见 $E(\xi_{nk}) = p_{nk}$, $D(\xi_{nk}) = p_{nk} - p_{nk}^2$. 对 $\varepsilon > 0$, 由 Chebyshev's Inq

$$\begin{aligned} P\left\{\left|\frac{S_n - E(S_n)}{n}\right| \geq \varepsilon\right\} &\leq \frac{E(S_n - E(S_n))^4}{n^4 \varepsilon^4} = \frac{1}{n^4 \varepsilon^4} E\left[\sum_{k=1}^n (\xi_{nk} - E(\xi_{nk}))\right]^4 \\ &= \frac{1}{n^4 \varepsilon^4} \left(\sum_{k=1}^n E(\xi_{nk} - E(\xi_{nk}))^4 + 6 \sum_{1 \leq i < j \leq n} E[(\xi_{ni} - E(\xi_{ni}))^2 (\xi_{nj} - E(\xi_{nj}))^2] \right) \end{aligned}$$

上述最后一个展开需要使用 $\{\xi_{nk}\}$ 的独立性. 由于 $|\xi_{nk} - E(\xi_{nk})| \in [0, 1]$, 故有

$$P\left\{\left|\frac{S_n - E(S_n)}{n}\right| > \varepsilon\right\} \leq \frac{1}{n^4 \varepsilon^4} \left(n + 6 \cdot \frac{n(n-1)}{2} \right) \approx \frac{1}{n^2}$$

从而由 $\sum \frac{1}{n^2} < \infty$ 以及 Borel-Cantelli 引理即知

$$P\left\{\overline{\lim}_n \left\{\left|\frac{S_n - E(S_n)}{n}\right| > \varepsilon\right\}\right\} = 0 \Rightarrow \frac{S_n - E(S_n)}{n} \xrightarrow{\text{a.s.}} 0$$

$$\text{于是 } \frac{E(S_n)}{n} = \frac{1}{n} \sum_{k=1}^n p_{nk} \rightarrow p \Rightarrow \frac{S_n}{n} \xrightarrow{\text{a.s.}} p.$$

(2) 由独立性可知

$$f_{S_n}(t) = \prod_{k=1}^n E(e^{it\xi_{nk}}) = \prod_{k=1}^n (1 - p_{nk} + p_{nk}e^{it}), \quad f_S(t) = e^{\lambda(e^{it}-1)}$$

根据证明 CLT 时用到的引理, 对任意 t 如下不等式成立, 令 $n \rightarrow \infty$ 即知命题成立.

$$\begin{aligned} |f_{S_n}(t) - f_S(t)| &= \left| \prod_{k=1}^n (1 - p_{nk} + p_{nk}e^{it}) - e^{\lambda(e^{it}-1)} \right| \\ &= \left| \prod_{k=1}^n (1 - p_{nk} + p_{nk}e^{it}) - e^{\sum_{k=1}^n p_{nk}(e^{it}-1)} + e^{(e^{it}-1)\sum_{k=1}^n p_{nk}} - e^{\lambda(e^{it}-1)} \right| \\ &\leq \left| \prod_{k=1}^n (1 - p_{nk} + p_{nk}e^{it}) - \prod_{k=1}^n e^{p_{nk}(e^{it}-1)} \right| + \left| e^{(e^{it}-1)\sum_{k=1}^n p_{nk}} - e^{\lambda(e^{it}-1)} \right| \\ &\leq \sum_{k=1}^n |e^{p_{nk}(e^{it}-1)} - 1 - p_{nk}(e^{it}-1)| + \left| e^{(e^{it}-1)\sum_{k=1}^n p_{nk}} - e^{\lambda(e^{it}-1)} \right| \\ &\leq \sum_{k=1}^n p_{nk}^2 |e^{it} - 1|^2 + \left| e^{(e^{it}-1)\sum_{k=1}^n p_{nk}} - e^{\lambda(e^{it}-1)} \right| \\ &\leq 4 \max_{1 \leq k \leq n} p_{nk} \sum_{k=1}^n p_{nk} + \left| e^{(e^{it}-1)\sum_{k=1}^n p_{nk}} - e^{\lambda(e^{it}-1)} \right| \rightarrow 0 \quad (n \rightarrow \infty) \end{aligned}$$

L-Problem 2

$\{\xi_n\}$ are iid with $E(\xi_1) = 0, D(\xi_1) = 1$. v_n is rv and $v_n \in \mathbb{Z}^+, \frac{v_n}{n} \xrightarrow{\mathbb{P}} c \in (0, \infty)$, then

$$\frac{S_{v_n}}{\sqrt{v_n}} \xrightarrow{d} \xi \sim N(0, 1) \text{ where } S_n = \sum_{k=1}^n \xi_k$$

证明: 利用分解

$$\frac{S_{v_n}}{\sqrt{v_n}} = \left(\frac{S_{[nc]}}{\sqrt{[nc]}} + \frac{S_{v_n} - S_{[nc]}}{\sqrt{[nc]}} \right) \sqrt{\frac{[nc]}{v_n}}$$

由经典的 CLT 以及依概率收敛的相关性质, 我们有

$$\frac{S_{[nc]}}{\sqrt{[nc]}} \xrightarrow{d} \xi, \quad \frac{[nc]}{v_n} = \frac{nc}{v_n} \frac{[nc]}{nc} \xrightarrow{\mathbb{P}} 1 \left(\Leftrightarrow \xrightarrow{d} 1 \right) \Rightarrow \sqrt{\frac{[nc]}{v_n}} \xrightarrow{\mathbb{P}} 1 \left(\Leftrightarrow \xrightarrow{d} 1 \right)$$

如果有

$$\frac{S_{v_n} - S_{[nc]}}{\sqrt{[nc]}} \xrightarrow{\mathbb{P}} 0 \left(\Leftrightarrow \xrightarrow{d} 0 \right) \sim (1)$$

于是由中文教材作业题 4.1.4.11 和 4.1.4.12 知

$$\frac{S_{[nc]}}{\sqrt{[nc]}} + \frac{S_{v_n} - S_{[nc]}}{\sqrt{[nc]}} \xrightarrow{d} \xi \Rightarrow \frac{S_{v_n}}{\sqrt{v_n}} \xrightarrow{d} \xi$$

下证 (1) 成立. 固定 $0 < \varepsilon < 1$, 由于 $\frac{v_n}{[cn]} \xrightarrow{\mathbb{P}} 1$, 故存在 N 使得对 $n > N$ 有

$$P \left\{ \left| \frac{v_n}{[cn]} - 1 \right| > \varepsilon^3 \right\} < \varepsilon$$

记 $a_n = \lfloor (1 - \varepsilon^3)[cn] \rfloor, b_n = \lfloor (1 + \varepsilon^3)[cn] \rfloor$, 记 $\Lambda := \{\omega : a_n \leq v_n(\omega) \leq b_n\}$, 若 $\omega \in \Lambda$, 则 $S_{v_n(\omega)}(\omega)$ 是某个和式 S_j , 其中 $a_n \leq j \leq b_n$. 对 $[cn] < j \leq b_n$, 有

$$S_j - S_{[cn]} = X_{[cn]+1} + X_{[cn]+2} + \cdots + X_j$$

于是由 Kolmogorov 不等式可知

$$P \left\{ \max_{[cn] \leq j \leq b_n} |S_j - S_{[cn]}| > \varepsilon [cn] \right\} \leq \frac{D(S_{b_n} - S_{[cn]})}{\varepsilon^2 cn} = \frac{b_n - [cn]}{\varepsilon^2 cn} \leq \frac{\varepsilon^3 [cn]}{\varepsilon^2 cn} \leq \varepsilon$$

对 $a_n \leq j < [cn]$, 同理可知

$$P\left\{\max_{a_n \leq j < \lfloor cn \rfloor} |S_j - S_{\lfloor cn \rfloor}| > \varepsilon \lfloor cn \rfloor\right\} \leq \frac{D(S_{\lfloor cn \rfloor} - S_{a_n})}{\varepsilon^2 cn} \leq \varepsilon$$

结合以上两个不等式可知

$$P\left\{\max_{a_n \leq j \leq b_n} |S_j - S_{\lfloor cn \rfloor}| > \varepsilon \lfloor cn \rfloor\right\} \leq 2\varepsilon$$

故对 $n > N$, 我们有

$$\begin{aligned} P\left\{\left|\frac{S_{v_n} - \lfloor cn \rfloor}{\sqrt{\lfloor cn \rfloor}}\right| > \varepsilon\right\} &= \sum_{j=1}^{\infty} P\left\{v_n = j; \left|\frac{S_{v_n} - S_{\lfloor cn \rfloor}}{\sqrt{\lfloor cn \rfloor}}\right| > \varepsilon\right\} \\ &\leq \sum_{a_n \leq j \leq b_n} P\left\{v_n = j; \max_{a_n \leq j \leq b_n} |(S_j - S_{\lfloor cn \rfloor})| > \varepsilon \sqrt{\lfloor cn \rfloor}\right\} + \sum_{j \notin [a_n, b_n]} P\{v_n = j\} \\ &\leq P\left\{\max_{a_n \leq j \leq b_n} |S_j - S_{\lfloor cn \rfloor}| > \varepsilon \sqrt{\lfloor cn \rfloor}\right\} + P\{v_n \notin [a_n, b_n]\} \leq 2\varepsilon + \varepsilon = 3\varepsilon \end{aligned}$$

由 ε 的任意性可知 (1) 成立, 从而原命题成立.

L-Problem 3

构造 Feller 和 Linderberg 条件不成立但 CLT 成立的例子.

解: 取中文教材 4.3.6.13 的例子即可, 已证其满足 CLT 但不满足 Feller 条件, 从而其也不满足 Linderberg 条件.